Solver Specification and Verification Report:

`highOrderManufacturedFDAImplicitPETScDistributed`

Manufactured monodomain-like PDE–ODE solver with high-order finite volumes, Picard, diagonal-I_{ion}, and JFNK nonlinear coupling on a distributed PETSc Krylov backend

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Abstract

This report documents the numerical formulation implemented in:

`highOrderManufacturedFDAImplicitPETScDistributed`.

The solver solves a monodomain-like reaction–diffusion PDE for the transmembrane potential V_{m}, evolves local state ODEs for (u_1, u_2, u_3), evaluates a nonlinear ionic-current surrogate $I_{\text{ion}}(V_{\text{m}}, u_1, u_2, u_3)$, and allows the same nonlinear coupling strategies used in the physiological solver: `Picard`, `diagonalIion` and `JFNK`.

- The diffusion operator is discretised with the OpenFOAM cell-centred finite-volume framework, optionally enhanced to a high-order Local Regression Estimators (LRE) reconstruction;
- The ionic source is evaluated either at cell centres or at the cell quadrature points (required for high order);
- There are different ways to integrate the reaction ODE equations:
  (i) at each quadrature point used to evaluate the ionic source;
  (ii) only at the cell centres, reconstructing them afterwards wherever the ionic source has to be evaluated;
- The ODE uses the solver's own integrator (adaptive `RKF45` by default, `RK4` or `Euler`) with configurable tolerances, driven by a polynomial in time of V_{m} across the step;
- Time integration follows a θ scheme (backward Euler or Crank–Nicolson) or BDF1–4 with an ESDIRK3 startup;
- The mass operator can be `lumped` or high-order `consistent`;
- Linear voltage systems are solved through a thin PETSc wrapper that *caches* the KSP, the matrix and the ILU(T) symbolic factorisation across nonlinear iterations and time steps; an Eigen fallback (`SparseLU/BiCGSTAB`) is also available. JFNK uses a persistent matrix-free `MatShell`+KSP pair, defaulting to restarted GMRES, and an optional preconditioner derived from $\mathbf{A}$ with a local I_{ion} linearisation (`diagonalIion`).

The sparse-matrix assemblies of $\mathbf{M}$ and $\mathbf{K}$ are split into *chunks* over cells/faces to bound the transient triplet-vector peak on large 3D unstructured meshes. The current implementation also exposes an automatic memory-optimisation mode with adaptive stiffness triplet reservation, compact consistent-mass assembly, conditional allocation of ionic quadrature-point fields, and allocator trimming around the large setup phases. The results section is data driven: tables are generated from the current tutorial outputs and fitted directly from the aggregated `errors_vs_N` files.

This `...Distributed` variant additionally makes the linear algebra *genuinely distributed*: the mesh is decomposed with `decomposePar`, the operators are assembled with local rows and global columns and become PETSc `MPIAIJ` matrices on `PETSC_COMM_WORLD`, and every matrix–vector product goes through `MatMult` rather than Eigen. The formulation is unchanged. Section 11 documents what the decomposition requires — including the exchange that carries a neighbouring cell's reconstruction row across a partition, without which the face-jump stabilisation is silently absent along every cut — how a parallel operator is verified entry by entry, and the inconsistency

that was measured in the stencil selection of the high-order library (Section 11.4); in 2-D it disappeared once the `Nn` semantics were corrected (Section 4.7).

Status as of 2026-10-18. Since October the code this solver shares with the electro solver lives in `src/monodomainCommon/`, and the solver gained a nonlinear stopping test common to the three methods (with a floor exit), right preconditioning of the KSP, the V_m history at the Gauss points, and the fix of the stage step on the `gaussPointODE` path. Section 12 collects those changes and the new M02 results; the earlier sections have been corrected where they had become false. The result tables of Section 13 **predate** those changes.

Contents

1 Purpose and notation **3**

2 Governing equations **4**
2.1 Electrophysiology motivation . . . 4
2.2 Manufactured PDE–ODE system . . . 4
2.3 Exact solution and the role of β . . . 4

3 Spatial discretisation **5**
3.1 Finite-volume integral form . . . 5
3.2 Standard cell-centred operator . . . 5
3.3 High-order LRE operator . . . 6
3.4 Lumped and consistent mass matrices . . . 8
3.5 High-order averaging of I_{ion} and states . . . 9

4 Time integration **9**
4.1 The θ method . . . 9
4.2 Backward differentiation formulae . . . 10
4.3 Starting values and the ESDIRK startup . . . 11
4.4 ODE states inside one PDE time step . . . 13
4.5 The full triad grid, and the cost of each combination . . . 15
4.6 Measured temporal orders . . . 16
4.7 Two things the convergence study measures against, and how they were wrong . . . 16

5 Nonlinear coupled solve **18**
5.1 Residual and diagnostics . . . 18
5.2 Picard fixed-point method . . . 19
5.3 Diagonal-I_{ion} linearisation . . . 20
5.4 Jacobian-free Newton–Krylov . . . 20
5.5 The top-of-iteration exit, and the defect it hid . . . 22
5.6 Armijo line search and rollback . . . 22
5.7 Method comparison . . . 23

6 Linear-algebra backend and implementation notes **25**
6.1 JFNK preconditioner: `diagonalIion` ILU(T) . . . 26
6.2 Chunked assembly of **M** and **K** . . . 26
6.3 Memory layout in the time loop . . . 27

7 Theoretical accuracy and practical error floors **28**

8 Solver controls **28**

9 Output diagnostics **31**

10 Error Measures and Convergence Rates **31**

11 Distributed-memory parallelisation **32**
11.1 Matrix layout: local rows, global columns . . . 32
11.2 Stabilisation across processor faces . . . 33
11.3 Verifying a parallel operator . . . 34
11.4 Partition dependence of the MLS stencils . . . 34
11.5 Verification cases: what was run and where it lives . . . 36

12 October 2026 changes and new results **37**
12.1 Code shared with the electro solver . . . 37
12.2 One stopping test, with a floor exit . . . 37
12.3 The V_{m} history at the Gauss points . . . 38
12.4 New results: M02 with GP . . . 38

13 Results **38**
13.1 Rollback audit and nonlinear robustness . . . 39
13.2 Two-dimensional manufactured problem . . . 40
13.2.1 `Picard` . . . 40
13.2.2 `diagonalIion` . . . 41
13.2.3 `JFNK` . . . 41
13.3 Discussion of the 2D results . . . 42
13.4 Three-dimensional manufactured problem . . . 47
13.4.1 `Picard` . . . 47
13.4.2 `diagonalIion` . . . 53
13.4.3 `JFNK` . . . 58
13.5 Discussion of the 3D results . . . 64

1 Purpose and notation

The MMS problem is not a physiological ionic model. Its role is to provide an exact solution for a PDE–ODE–source structure that is deliberately close to the electro-activation solver. If the same finite-volume operator, same source quadrature, same state ODE integration, same θ time discretisation, and same nonlinear coupling are used here and in the electro solver, then the observed MMS convergence gives evidence for the numerical order of those shared building blocks.

Symbol	Meaning	Location in solver
$\Omega \subset \mathbb{R}^d$	computational domain, $d = 1, 2, 3$	OpenFOAM mesh
K_i	finite-volume cell	`mesh.cells()`
$\lvert K_i \rvert$	cell volume	`mesh.V()`
$V_{\mathrm{m}}(\mathbf{x}, t)$	manufactured transmembrane potential	`Vm`
$\mathbf{u} = (u_1, u_2, u_3)$	manufactured state variables	`u1,u2,u3`
$I_{\mathrm{ion}}(V_{\mathrm{m}}, \mathbf{u})$	nonlinear source surrogate	`Iion`
χ and C_m	monodomain scaling constants	`cardiacProperties`
$\mathbf{D}$	conductivity tensor	`conductivity`
$\mathbf{M}$	cell-average mass operator	`M` (moved into `BImplicit` after $\mathbf{A}/\mathbf{B}$
$\mathbf{K}$	volume-normalised diffusion/stiffness operator	`K`
$\mathbf{L} = (\chi C_m)^{-1}\mathbf{K}$	scaled diffusion operator	absorbed into $\mathbf{A}/\mathbf{B}$ (no separate st
$\mathbf{A}, \mathbf{B}$	implicit θ-scheme matrices	`AImplicit,BImplicit`
$\mathsf{F}(\mathsf{x})$	nonlinear residual for voltage candidate x	`residualFor`
$\mathbf{P}$	persistent JFNK preconditioner matrix $\approx \mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d})$	`jfnkP`

The cell-centred vector of voltages at time t^n is denoted $\mathsf{V}^n \in \mathbb{R}^{N_c}$. A nonlinear iteration inside one time step uses the superscript k, e.g. V^k is a candidate for V^{n+1}. Quadrature point q in cell K_i is denoted $\mathbf{x}_{iq}$ with positive weight ω_{iq}.

2 Governing equations

2.1 Electrophysiology motivation

The physiological monodomain equation can be written, in one common convention, as

$$\chi C_m \frac{\partial V_{\mathrm{m}}}{\partial t} = \nabla\cdot(\mathbf{D}\nabla V_{\mathrm{m}}) - \chi C_m\, I_{\mathrm{ion}}(V_{\mathrm{m}}, \mathbf{s}) + I_{\mathrm{ext}}, \tag{1}$$

where $\mathbf{s}$ are ionic states and I_{ion} is often stored as a voltage rate with units $\mathrm{V\,s^{-1}}$. Dividing by χC_m gives

$$\frac{\partial V_{\mathrm{m}}}{\partial t} = \frac{1}{\chi C_m}\nabla\cdot(\mathbf{D}\nabla V_{\mathrm{m}}) - I_{\mathrm{ion}}(V_{\mathrm{m}}, \mathbf{s}) + \frac{I_{\mathrm{ext}}}{\chi C_m}. \tag{2}$$

The manufactured solver replaces the physiological states by three smooth MMS states and replaces the cell-model ionic current by a constructed nonlinear function. The purpose is not to reproduce cellular electrophysiology, but to exercise the same numerical responsibilities: diffusion, local ODEs, nonlinear source, quadrature, implicit time stepping, and nonlinear iteration.

2.2 Manufactured PDE–ODE system

The solver advances

$$\frac{\partial V_{\mathrm{m}}}{\partial t} = \frac{1}{\chi C_m}\nabla\cdot(\mathbf{D}\nabla V_{\mathrm{m}}) - I_{\mathrm{ion}}(V_{\mathrm{m}}, u_1, u_2, u_3), \tag{3}$$

$$\dot{u}_1 = (u_1 + u_3 - V_{\mathrm{m}})^2 u_2^2 + \frac{1}{2}(u_1 + u_3 - V_{\mathrm{m}})u_2^2(V_{\mathrm{m}} - u_3), \tag{4}$$

$$\dot{u}_2 = -(u_1 + u_3 - V_{\mathrm{m}})u_2^3, \tag{5}$$

$$\dot{u}_3 = 0. \tag{6}$$

The nonlinear ionic-current surrogate is

$$I_{\mathrm{ion}}(V_{\mathrm{m}}, u_1, u_2, u_3) = -\frac{1}{2}(u_1 + u_3 - V_{\mathrm{m}})u_2^2(V_{\mathrm{m}} - u_3) + \frac{\beta}{\chi C_m}(V_{\mathrm{m}} - u_3). \tag{7}$$

The PDE source used in the matrix right-hand side is

$$S(V_{\mathrm{m}}, \mathbf{u}) = -I_{\mathrm{ion}}(V_{\mathrm{m}}, \mathbf{u}). \tag{8}$$

Thus the term called `Iion` in the output is the nonlinear manufactured current, while the term passed into the voltage equation is its negative.

2.3 Exact solution and the role of β

The exact fields are built from two smooth spatial functions F and G:

$$V_{\mathrm{m}}^\star(\mathbf{x}, t) = \sqrt{1+t}\, F(\mathbf{x}), \tag{9}$$

$$u_1^\star(\mathbf{x}, t) = (1+t)G(\mathbf{x}) + V_{\mathrm{m}}^\star(\mathbf{x}, t), \tag{10}$$

$$u_2^\star(\mathbf{x}, t) = \frac{1}{(1+t)\sqrt{G(\mathbf{x})}}, \tag{11}$$

$$u_3^\star(\mathbf{x}, t) = 0. \tag{12}$$

The solver uses

$$F = \begin{cases} \cos(\pi x), & d = 1, \\ \cos(\pi x)\cos(2\pi y), & d = 2, \\ \cos(\pi x)\cos(2\pi y)\cos(3\pi z), & d = 3, \end{cases} \qquad G = \begin{cases} 1 + x, & d = 1, \\ 1 + xy^2, & d = 2, \\ 1 + xy^2 z^3, & d = 3. \end{cases} \tag{13}$$

The constant β closes the analytical balance between the Laplacian of F and the conductivity tensor. For a diagonal constant conductivity sampled from `conductivity[0]`,

$$\beta = \begin{cases} -\pi^2 D_{xx}, & d = 1, \\ -\pi^2 (D_{xx} + 4D_{yy}), & d = 2, \\ -\pi^2 (D_{xx} + 4D_{yy} + 9D_{zz}), & d = 3. \end{cases} \tag{14}$$

This choice makes the manufactured source compatible with the exact trigonometric mode under the selected conductivity tensor. Boundary values for V_m, u_1, u_2, and u_3 are overwritten with the exact solution where appropriate, so the verification error is a discretisation error rather than a boundary-data error.

In the case of this report V_m uses `zeroGradient`, a condition that the manufactured solution satisfies exactly, so that the overwriting mechanism and the vector $\mathsf{b}(t)$ are inactive for V_m (active only for the states $\mathbf{u}$).

3 Spatial discretisation

3.1 Finite-volume integral form

Integrating (3) over a control volume K_i and dividing by the cell volume gives the cell-average balance used by the implementation:

$$\frac{1}{|K_i|}\int_{K_i} \frac{\partial V_\mathrm{m}}{\partial t}\, dV = \frac{1}{\chi C_m |K_i|}\int_{\partial K_i} \mathbf{n}\cdot\mathbf{D}\nabla V_\mathrm{m}\, dA + \frac{1}{|K_i|}\int_{K_i} S(V_\mathrm{m}, \mathbf{u})\, dV. \tag{15}$$

After spatial discretisation the semi-discrete cell-average problem has the matrix form

$$\mathbf{M}\frac{d\mathsf{V}}{dt} = \mathbf{L}\mathsf{V} + \mathsf{S}(\mathsf{V}, \mathbf{u}), \qquad \mathbf{L} = \frac{1}{\chi C_m}\mathbf{K}. \tag{16}$$

Here $\mathbf{K}\mathsf{V}$ approximates the volume-normalised divergence $(1/|K_i|)\int_{\partial K_i} \mathbf{n}\cdot\mathbf{D}\nabla V_\mathrm{m}\, dA$, and S_i is a cell-average source rate, not a volume-integrated source. Consequently the lumped finite-volume mass used by the code is the identity matrix.

3.2 Standard cell-centred operator

When `useHighOrder_Vm=false`, the solver assembles the standard cell-centred finite-volume diffusion operator. For an internal face f between cells P and N, the orthogonal part is of the form

$$\frac{1}{|K_P|}\int_f \mathbf{n}\cdot\mathbf{D}\nabla V_\mathrm{m}\, dA \approx \frac{a_f}{|K_P|}(V_N - V_P), \tag{17}$$

where a_f contains the face area, face-normal direction, conductivity tensor, and distance between cell centres. Boundary faces use the OpenFOAM patch type. `zeroGradient` and `empty` patches give no-flux contributions; fixed value patches contribute known boundary terms.

The non-orthogonal correction, and why the two-point flux alone is not the standard operator. Equation (17) is only the part of the flux that lies along the line joining the two cell centres. Writing $\mathbf{e} = \mathbf{d}/|\mathbf{d}|$ for that direction and $\mathbf{g} = \mathbf{D}\mathbf{n}$, the exact face flux splits exactly as

$$|S_f|\,\mathbf{n}\cdot\mathbf{D}\nabla V_{\mathrm{m}} = \underbrace{|S_f|\,(\mathbf{g}\cdot\mathbf{e})\,(\nabla V_{\mathrm{m}}\cdot\mathbf{e})}_{\approx\, a_f(V_N - V_P)} + \underbrace{|S_f|\,\nabla V_{\mathrm{m}}\cdot\mathbf{w}}_{\text{dropped}}, \qquad \mathbf{w} = \mathbf{g} - (\mathbf{g}\cdot\mathbf{e})\,\mathbf{e}. \tag{18}$$

The second term is the tangential remainder. It vanishes identically when $\mathbf{e} \parallel \mathbf{n}$ and $\mathbf{D}$ is isotropic, so on a regular hexahedral mesh the two-point flux is complete. On a mesh where the centre-to-centre line is not parallel to the face normal it does not vanish, and dropping it makes the operator *inconsistent*: on the manufactured solution, suite test `M01` fitted order 0.81 on unstructured triangles against 2.00 on hexahedra, with an error 230× larger at the same resolution.

The solver therefore assembles the second term as well, and **not optionally** — there is no dictionary key to switch it off. On an orthogonal mesh with isotropic conductivity it is identically zero and costs nothing; on a non-orthogonal mesh it is what makes the operator consistent. There is no configuration in which omitting it is the wanted behaviour. With it, the unstructured runs fit 2.49–2.74 at order 2 nominal. The correction is assembled *implicitly*, through the reconstruction rows, rather than deferred to the right-hand side.

This is the same decomposition OpenFOAM makes: `gaussLaplacianScheme` uses `nonOrthDeltaCoeffs` for the implicit part and adds an explicit `correction()` built from `nonOrthCorrectionVectors`, which vanish identically on an orthogonal mesh.

Two implementation notes, because both cost a debugging pass. The far side of a processor face computes $-\mathbf{w}$, so a *second* row exchange is needed next to the one the stabilisation jump already performs. And the test on $|\mathbf{w}|$ must be *relative* to $|\mathbf{g}|$: with the absolute `SMALL` used elsewhere, the roundoff residue of $\mathbf{w}$ on an orthogonal mesh survives into K and shifts the fitted error by 2.8×10^{-7} where the correction should be exactly zero. The relative floor of 10^{-12} sits inside a measured fourteen-order gap: the largest entry the correction adds to K is 8.0×10^{-16} of the largest entry of K on hexahedra and 2.3×10^{-2} on the unstructured mesh.

For smooth solutions on regular orthogonal meshes, this standard finite-volume operator is second-order accurate in space for diffusion. With the non-orthogonal correction it remains second order on unstructured meshes as well; without it the effective order degrades with mesh quality, as quantified above.

3.3 High-order LRE operator

When `useHighOrder_Vm=true`, the solver uses Local Regression Estimators (LRE) stencils. Around each cell centre $\mathbf{x}_i$, a polynomial reconstruction of degree N is fitted from neighbouring cell values:

$$V_{\mathrm{m}}(\mathbf{x}_i + \mathbf{r}) \approx V_i + \nabla V_i \cdot \mathbf{r} + \frac{1}{2}\mathbf{r}^T H_i \mathbf{r} + \frac{1}{6} T_i[\mathbf{r},\mathbf{r},\mathbf{r}] + \cdots . \tag{19}$$

The dictionary entries `N`, `Nn` and `k` control the polynomial degree, the stencil size and the radial weight exponent.

`Nn` is worth stating precisely, because reading it as a total silently doubles the stencil. It is the number of cells *added on top of* the minimum the degree needs, and that minimum is the number of terms in (19) — $(N{+}1)(N{+}2)/2$ in 2-D and $(N{+}1)(N{+}2)(N{+}3)/6$ in 3-D — which is the number of unknowns the local least-squares fit determines. So

$$\text{stencil size} \;=\; \text{minimum} \;+\; \texttt{Nn},$$

and `Nn` is the redundancy given to that fit. With `N 3` in 2-D the minimum is 10, so `Nn 10` gives a 20-cell stencil, two points per unknown. The sweeps in this report use $\texttt{Nn} = 10$ in 2-D and 16/30/40 for $p1/p2/p3$ in 3-D, where the minima grow faster and a tetrahedral stencil is geometrically worse conditioned; those keep the ratio at 3× or better.

`maxStencilSize` is read but inert: the adapter over `movingLeastSquares` has no equivalent and drops it.

The solver can evaluate reconstructed values, gradients, Hessians, and third-order tensors, depending on the selected polynomial order.

Boundary patches are classified automatically before the LRE objects are constructed. `fixedValue` and `fixedVoltage` patches are inserted into the LRE stencils as boundary ghost points so that the manufactured boundary value can be used. `zeroGradient` patches are passed to LRE as `mirrorPatches`, which doubles the local stencil by reflecting neighbour locations across the boundary plane and preserves the full reconstruction order at homogeneous Neumann boundaries. `empty`, processor, and unsupported patches are left out of the explicit include/mirror lists. A patch cannot be both included as a Dirichlet ghost patch and mirrored; the LRE constructor checks this and aborts if the dictionary is inconsistent.

The high-order dictionaries are read from `highOrderCoeffs/LRECoeffs_Vm` for the diffusion/mass reconstruction and from `highOrderCoeffs/LRECoeffs_Iion` for source/state quadrature. The entry `LRECoeffs_Iion` is required when high-order ionic quadrature is enabled.

The high-order stiffness matrix is assembled by evaluating reconstructed gradients at face quadrature points:

$$(\mathbf{K}\mathsf{V})_i \approx \frac{1}{|K_i|} \sum_{f \subset \partial K_i} \sigma_{if} |f| \sum_{q \in f} \omega_{fq}\, \mathbf{n}_f \cdot \mathbf{D}_{\mathrm{own}(f)} \nabla_h V_{\mathrm{m}}(\mathbf{x}_{fq}), \tag{20}$$

where σ_{if} is the owner/neighbour orientation sign. For the manufactured verification cases the conductivity is constant, so using the owner-cell tensor in this face reconstruction is equivalent to any consistent face interpolation. Fixed-value boundaries contribute known exact-boundary terms to a separate vector $\mathsf{b}(t)$. The resulting operator is sparse but has a wider stencil than the standard finite-volume Laplacian. For smooth fields, good stencils, sufficiently accurate quadrature, and non-degenerate meshes, the expected spatial consistency improves with the LRE polynomial degree. In practice the observed order is the minimum of: reconstruction order, quadrature order, time-integration order, ODE tolerance floor, nonlinear residual floor, and mesh/stencil conditioning. If the error is controlled only by the interpolation/reconstruction of V_{m}, then the behaviour expected from Taylor term counting is the usual polynomial one: linear reconstruction (`p1`) gives second-order accuracy, quadratic reconstruction (`p2`) gives third-order accuracy, and cubic reconstruction (`p3`) gives fourth-order accuracy.

That count, however, ignores stencil-symmetry cancellation, and the results of §13.3 show that the actual behaviour of this diffusion operator follows the odd/even pattern: the reconstructed face gradient has $O(h^N)$ truncation for degree N, but on approximately symmetric stencils the even-order error term cancels, so odd degrees gain one extra order and even degrees do not. The 2D sweep returns 2.00, 2.61, 2.03, and 4.32 for `NO`, `p1`, `p2`, and `p3` respectively, that is 2, 2, and 4 for the three LRE degrees. The expectation above should therefore be read as a term-counting upper bound, not as the order actually attainable with an even degree.

The boundary vector $\mathsf{b}(t)$. $\mathsf{b}(t) \in \mathbb{R}^{N_c}$ is the Dirichlet lifting vector: the part of the discrete diffusion operator that multiplies known values and therefore cannot stay inside the matrix.

The idea is standard. The discrete stencil of $\mathbf{K}$ in a cell i adjacent to the boundary involves a ghost value V_b on the boundary face. If V_b were an unknown it would go into the matrix; since it is data ($V_b = V_{\mathrm{m}}^{\star}(\mathbf{x}_b, t)$), it is moved to the right-hand side. Formally, the complete diffusion operator is split as

$$\underbrace{\mathbf{K}\,\mathsf{V}}_{\text{depends on the interior unknowns}} + \underbrace{\mathsf{b}(t)}_{\text{depends only on boundary data}} \approx \frac{1}{|K_i|} \int_{\partial K_i} \mathbf{n} \cdot \mathbf{D} \nabla V_{\mathrm{m}}\, dA.$$

This is why the diagnostic numerical Laplacian is formed as `lapNow = K*Vm + bcNow` and not as `K*Vm` alone.

- Low-order construction of $\mathsf{b}(t)$:

$$b_i = \sum_{f \subset \partial K_i \cap \Gamma_D} \frac{a_f}{|K_i|} V_{\mathrm{m}}^{\star}(\mathbf{x}_f, t) + (\text{stabilisation term if } \alpha > 0),$$

  with a_f the same orthogonal coefficient $\mathbf{S}_f \cdot \mathbf{D}\,\mathbf{d}/|\mathbf{d}|^2$ that was subtracted from the diagonal of $\mathbf{K}$.

- High-order construction of $\mathsf{b}(t)$:

$$b_i = \sum_{f \subset \partial K_i \cap \Gamma_D} \frac{|f|}{|K_i|} \sum_{q \in f} \omega_{fq} \Big(\mathbf{n}_f \cdot \mathbf{D}_{\mathrm{own}(f)}\, \mathbf{g}_{fq}^{\mathrm{ghost}} \Big)\, V_{\mathrm{m}}^{\star}(\mathbf{x}_{fq}, t),$$

  where $\mathbf{g}_{fq}^{\mathrm{ghost}}$ is the LRE gradient coefficient associated with the boundary ghost point (index `ghostID = faceStencils[globalFaceI].size()`, i.e. the last stencil entry, which is the one LRE reserves for the Dirichlet point). In other words: the column of the LRE row that corresponds to the ghost point is split off, multiplied by the exact value, and that product is b_i.

Three consequences are worth stating explicitly.

1. b depends on time because $V_{\mathrm{m}}^{\star}$ does. This is why the θ scheme carries two distinct vectors, $\mathsf{b}^n = \mathsf{b}(t^n)$ and $\mathsf{b}^{n+1} = \mathsf{b}(t^{n+1})$, weighted by $(1-\theta)$ and θ exactly like the source. Without that distinction, Crank–Nicolson would lose its second-order temporal accuracy near the boundary.

2. b is constant with respect to the unknowns x. It therefore drops out of the JFNK Jacobian–vector product: $\mathsf{F}(\mathsf{x} + \epsilon \mathbf{v}) - \mathsf{F}(\mathsf{x})$ cancels b. It only appears in the residual itself.

3. In the worked example, $\mathsf{b}(t) \equiv \mathbf{0}$ at every step, because no patch of V_{m} is fixedValue. The terms $\theta \mathsf{b}^{n+1} + (1-\theta)\mathsf{b}^n$ of (27), (28) and (38) vanish identically in the example. The machinery is there for the Dirichlet case, which other variants of the case do exercise.

3.4 Lumped and consistent mass matrices

The `lumped` mass operator is the identity:

$$\mathbf{M}_{ij}^{\mathrm{lump}} = \delta_{ij}. \tag{21}$$

This follows from the volume-normalised finite-volume form in (15): the unknowns are advanced as cell-average rates, so the volume factor has already been divided out of both the diffusion operator and the source. The identity mass is cheap, robust, positive, and coincides with the usual one-degree-of-freedom cell-centred update. If high-order V_{m} is disabled, the solver uses this diagonal mass even if the input asks for `consistent`, because there is no high-order reconstruction operator to integrate.

The `consistent` mass matrix is available when `useHighOrder_Vm=true` and is assembled with cell quadrature and the LRE reconstruction coefficients. For quadrature point $\mathbf{x}_{iq}$ in cell K_i, write the local Taylor/LRE reconstruction as

$$V_{\mathrm{m}h}(\mathbf{x}_{iq}) = \sum_j c_{ijq} V_j, \tag{22}$$

where the non-zero coefficients c_{ijq} are restricted to the cell stencil of K_i and include the self coefficient. The implemented consistent cell-average mass is then

$$\mathbf{M}_{ij}^{\mathrm{cons}} \approx \sum_{q \in K_i} \bar{\omega}_{iq}\, c_{ijq}, \qquad \bar{\omega}_{iq} = \frac{\omega_{iq}}{\sum_{q'} \omega_{iq'}}. \tag{23}$$

It is therefore a high-order averaging/projection operator induced by the LRE reconstruction, not a Galerkin mass matrix with products of basis functions. It better matches the high-order spatial

approximation, but it is sparse rather than diagonal and increases the cost and conditioning sensitivity of the implicit solve.

Note that usually (and in our `lumped` case):

$$\sum_{q'} \omega_{iq'} = 1. \tag{24}$$

3.5 High-order averaging of I_{ion} and states

The cell-average source rate can be approximated either by a cell-centred rule or by quadrature. Since $S = -I_{\text{ion}}$, when `useHighOrder_Iion=false` or when the mesh is effectively one-dimensional, the solver uses the cell-centred rate

$$\mathsf{S}_i \approx -I_{\text{ion}}(V_i, u_{1,i}, u_{2,i}, u_{3,i}). \tag{25}$$

When `useHighOrder_Iion=true` and $d > 1$, the voltage and states are evaluated at ionic quadrature points and the source rate is averaged back to the cell:

$$\mathsf{S}_i \approx -\sum_{q \in K_i} \bar{\omega}_{iq}\, I_{\text{ion}}\left(V_{\text{m}h}(\mathbf{x}_{iq}), u_{1,iq}, u_{2,iq}, u_{3,iq}\right). \tag{26}$$

These quadrature-point scalar fields are allocated only for multi-dimensional high-order ionic integration, i.e. when `useHighOrder_Iion=true` and $d > 1$. In all other cases the solver still reports the nominal number of ionic integration points for auditability, but the allocated quadrature-point storage is zero and the source/state update stays cell-centred.

There are two state-update modes for the high-order ionic path:

- `cellCentredReconstruct` (default) advances the ODE once per cell from the time-step-start states using the cell-centred old voltage and the current cell-centred voltage candidate. The updated cell-centred states are then reconstructed to ionic quadrature points with the separate `LRECoeffs_states` interpolator before evaluating I_{ion}. This is the default because it costs one ODE integration per cell.
- `gaussPointODE` reconstructs the old and candidate voltages to each ionic quadrature point and advances an independent state ODE at every quadrature point. The quadrature-point states are then averaged back to cell centres. This legacy path costs one ODE integration per quadrature point.

The state reconstruction is the routine shared with the electro solver (`reconstructCellFieldsAtIntegrationPoin` the three states in a single pass). It accepts a neighbour-range limiter, which this solver **never** requests: on a smooth field it would only clip legitimate extrema.

4 Time integration

4.1 The θ method

Let $\Delta t = t^{n+1} - t^n$. The solver maps `backwardEuler` to $\theta = 1$ and `crankNicolson` to $\theta = 1/2$. Applying the θ method to (16) gives

$$\mathbf{M}\frac{\mathsf{V}^{n+1} - \mathsf{V}^n}{\Delta t} = \mathbf{L}\left(\theta \mathsf{V}^{n+1} + (1-\theta)\mathsf{V}^n\right) + \theta \mathsf{S}^{n+1} + (1-\theta)\mathsf{S}^n + \theta \mathsf{b}^{n+1} + (1-\theta)\mathsf{b}^n. \tag{27}$$

Equivalently,

$$\mathbf{A}\mathsf{V}^{n+1} = \mathbf{B}\mathsf{V}^n + \theta \mathsf{S}^{n+1} + (1-\theta)\mathsf{S}^n + \theta \mathsf{b}^{n+1} + (1-\theta)\mathsf{b}^n, \tag{28}$$

$$\mathbf{A} = \frac{1}{\Delta t}\mathbf{M} - \theta \mathbf{L}, \qquad \mathbf{B} = \frac{1}{\Delta t}\mathbf{M} + (1-\theta)\mathbf{L}. \tag{29}$$

In the implementation, $\mathbf{L}$ is never stored as a separate sparse matrix. The solver builds $\mathbf{M}$ and $\mathbf{K}$ once (with the chunked assemblies described in §6.2) and constructs $\mathbf{A}$, $\mathbf{B}$ directly from the relation $\mathbf{L} = (\chi C_m)^{-1}\mathbf{K}$ via expression-template arithmetic on the sparse matrices: $\mathbf{A} \leftarrow \mathbf{M}/\Delta t - (\theta/\chi C_m)\mathbf{K}$ and $\mathbf{B} \leftarrow \mathbf{M}/\Delta t + ((1-\theta)/\chi C_m)\mathbf{K}$. After $\mathbf{A}$ is copied from $\mathbf{M}$, the storage of $\mathbf{M}$ is moved into $\mathbf{B}$ to avoid keeping a third copy of the mass matrix alive throughout the run.

`backwardEuler` is first order in time and strongly damping. It is usually the most robust choice for stiff or poorly converged nonlinear problems. `crankNicolson` is second order in time for smooth solutions and converged nonlinear/source terms, but is less dissipative. Its observed order can drop if the nonlinear iterations, state ODE tolerance, or spatial error dominate.

A property of Crank–Nicolson that matters on a stiff problem and is easy to overlook: it is A-stable but *not* L-stable. Its amplification factor tends to -1, not to 0, as $\lambda\Delta t \to -\infty$, so the stiffest modes are not damped — they oscillate with undiminished amplitude. `BDF2` below is also second order and *is* L-stable, which makes it the better of the two second-order options wherever a sharp front is present.

4.2 Backward differentiation formulae

The θ family cannot be pushed past second order. Dahlquist's second barrier states that no A-stable linear multistep method has order greater than two, and the trapezoidal rule ($\theta = 1/2$) is the A-stable member of order two with the smallest error constant. Higher temporal order therefore requires either giving up A-stability or leaving the multistep family altogether. The solver takes the first route for the scheme itself, and the second only to start it (§4.3).

Applying the k-step backward differentiation formula to (16) gives

$$\frac{1}{\Delta t}\sum_{j=0}^{k} a_j \,\mathbf{M}\,\mathsf{V}^{n+1-j} = \mathbf{L}\mathsf{V}^{n+1} + \mathsf{S}^{n+1} + \mathsf{b}^{n+1}, \tag{30}$$

which rearranges into exactly the same pair of operators the θ method uses:

$$\mathbf{A}\mathsf{V}^{n+1} = \mathbf{B}\,\mathsf{H}^{n} + \mathsf{S}^{n+1} + \mathsf{b}^{n+1}, \tag{31}$$

$$\mathbf{A} = \frac{a_0}{\Delta t}\mathbf{M} - \mathbf{L}, \qquad \mathbf{B} = \frac{1}{\Delta t}\mathbf{M}, \qquad \mathsf{H}^{n} = -\sum_{j=1}^{k} a_j\,\mathsf{V}^{n+1-j}. \tag{32}$$

Two consequences are what make BDF nearly free to add here. First, $\mathbf{A}$ differs from the backward-Euler operator *only* in the scalar multiplying $\mathbf{M}$: the assembly, the sparsity pattern and the cached factorisation are untouched. Second, the history enters through a *single* mass-matrix product, because the linear combination H^n is formed as a field before $\mathbf{B}$ is applied — a BDF-k step costs one matrix–vector product, not k. BDF is also fully implicit, so the source is needed at t^{n+1} only, one source evaluation per step fewer than Crank–Nicolson.

The coefficients, the theoretical order and the stability of each member are:

`implicitScheme`	$(a_0, a_1, \ldots, a_k)$	Order	Stability
`BDF1`	$(1,\ -1)$	$O(\Delta t)$	A-stable, L-stable
`BDF2`	$(\frac{3}{2},\ -2,\ \frac{1}{2})$	$O(\Delta t^2)$	A-stable, L-stable
`BDF3`	$(\frac{11}{6},\ -3,\ \frac{3}{2},\ -\frac{1}{3})$	$O(\Delta t^3)$	$A(86.03°)$
`BDF4`	$(\frac{25}{12},\ -4,\ 3,\ -\frac{4}{3},\ \frac{1}{4})$	$O(\Delta t^4)$	$A(73.35°)$

`BDF1` and `backwardEuler` are the same scheme, and the solver is built so that this is exact rather than approximate: $a_0 = 1$ and $\mathsf{H}^n = \mathsf{V}^n$ reduce (31) to the $\theta = 1$ case term by term, and the two produce byte-identical operators and byte-identical answers. That identity is used as the regression test guarding every change to this part of the solver.

`BDF5` and `BDF6` are deliberately not offered. Past BDF2 no member is A-stable, and the $A(\alpha)$ wedge closes quickly: 86.03° at BDF3 and 73.35° at BDF4, which a diffusion-dominated spectrum

stays well inside, but 51.84° at BDF5 and 17.84° at BDF6, which the reaction Jacobian of a depolarisation front cannot be assumed to respect. BDF7 and beyond are zero-unstable and do not exist as convergent methods.

4.3 Starting values and the ESDIRK startup

A k-step method needs k levels before it can take a step, and at t^0 only one exists. How the missing $k-1$ are supplied is not an implementation detail: it decides the order that will be measured, and it does so silently.

The requirement is that the starting values be accurate to $O(\Delta t^k)$ *with respect to the semi-discrete system* (16), not with respect to the PDE. That distinction is the whole difficulty. Seeding the history with the manufactured solution evaluated at $t^0 - j\Delta t$ looks exact and is not: the exact solution of the PDE does not satisfy the semi-discrete equations, it satisfies them up to the spatial truncation error τ_h, and the mismatch accumulated over the span of the history is $O(\tau_h\, \Delta t)$. That is *first* order in Δt, so it pins the global order at one, with a coefficient proportional to the spatial error.

The prediction that follows is falsifiable, and was tested — suite test `M20`, on `BDF3`, 2-D hexahedral, $p3$ triad. What is measured there is the effect of the startup in isolation,

$$e(\Delta t) = \left\| \mathsf{V}_{\text{startup}}(\Delta t) - \mathsf{V}_{\text{ESDIRK3}}(\Delta t) \right\|, \tag{33}$$

at the same Δt and the same mesh, so that BDF3's own truncation error is common to both runs and cancels. Judging this on raw self-convergence differences instead does *not* work: they mix the startup term, which falls with the mesh, with BDF3's own $O(\Delta t^3)$ term, which does not, and the mesh-refinement ratio comes out diluted.

Startup	order of e, $N = 40$	order of e, $N = 80$	$e(N{=}40)/e(N{=}80)$
`constant`	1.00	1.00	1.00
`exact`	1.01	1.13	$\mathbf{15.5} = 2^{3.95}$

Both modes are $O(\Delta t)$, as predicted, and they differ exactly where the theory says they should: `constant`'s coefficient is $\dot{V}_{\mathrm{m}}$, which does not depend on h, so refining the mesh changes nothing; `exact`'s is τ_h, so its error falls by 15.5 per refinement and the cap lifts as $\tau_h \lesssim \Delta t^{k-1}$ is approached. At $N = 40$ the two coefficients differ by a factor of 1.28×10^4.

The other way a BDF loses its order silently: the history index. A BDF-k advances with $V^n = \sum_j a_j V^{n-j}$, and the coefficients a_j assume that level j of the history array holds the solution at $t - (j+1)\Delta t$. If that array is off by one step, the formula combines the wrong instants with the right weights: the order drops and *nothing reports it* — no exception, no large residual, exit 0.

`CF_DUMP_BDF_HISTORY` prints, for the first four steps, the maximum deviation of each history level from $V_{\mathrm{m}}^{\mathrm{exact}}$ at the instant that level should hold. Suite test `M23` turns that into a criterion with no invented tolerance: if the indexing is right, level j at step n *is* the field that was the previous solution at step $n-j-1$, so its printed instant and deviation must match that step's digit for digit. The test performs 18 such exact comparisons.

It also validates the diagnostic itself, by making the three startup modes produce three predictable signatures at step 0, where the levels still hold whatever the startup seeded:

Startup	Level 0	Level 1	Level 2
`exact`	0	0	0
`constant`	6.322×10^{-3}	1.269×10^{-2}	1.909×10^{-2}
`ESDIRK3`	0.978	0.972	0.966

`exact` seeds each level from the exact solution at its own instant, so the deviations must be *exactly* zero. `constant` repeats the initial field, so level j deviates by $|V_{\mathrm{m}}(0) - V_{\mathrm{m}}^{\mathrm{exact}}(-(j+1)\Delta t)|$; since the

manufactured profile carries $\sqrt{1+t}$, the ratios against level 0 have a closed form, $1 : 2.006 : 3.019$, and the spatial factor cancels when taking ratios, so the prediction is mesh-independent. Measured: $1 : 2.007 : 3.020$. `ESDIRK3` seeds nothing at step 0, so its deviations are $O(1)$, two orders above `constant`. A diagnostic reading the wrong level or the wrong instant would produce none of the three.

The remedy is a self-starting method of sufficient order, and the solver uses an **ESDIRK3** — Explicit first stage, Singly Diagonally Implicit Runge–Kutta, four stages, classical order three, L-stable and stiffly accurate. Each letter earns its place:

- *Explicit first stage* raises the *stage* order to two. A plain DIRK has stage order one and suffers order reduction on stiff problems, the observed order falling towards the stage order — which is exactly the regime a depolarisation front lives in.
- *Singly diagonally implicit* means every implicit stage shares the same γ, so one operator $\mathbf{M}/(\gamma\Delta t) - \mathbf{L}$ and one cached factorisation serve all of them, assembled exactly like $\mathbf{A}$.
- *Stiffly accurate* means b equals the last row and $c_s = 1$, so $\mathsf{V}^{n+1} = \mathsf{Y}_s$: the last stage is the answer, no separate combination step is needed, and $R(\infty) = 0$.

Runge–Kutta methods are not bound by Dahlquist's barrier, which applies to linear multistep methods only; L-stability at order three is available here and provably not in the BDF family.

With γ the root of $\gamma^3 - 3\gamma^2 + \frac{3}{2}\gamma - \frac{1}{6} = 0$ in $(\frac{1}{4}, \frac{1}{2})$ and $c = (0,\ 2\gamma,\ \frac{3}{5},\ 1)$, stage $i \geq 2$ solves

$$\left(\frac{\mathbf{M}}{\gamma\Delta t} - \mathbf{L}\right)\mathsf{Y}_i = \frac{1}{\gamma}\left[\mathbf{B}\mathsf{V}^n + \sum_{j<i} a_{ij}\mathsf{F}_j\right] + \mathsf{b}(t_i) + \mathsf{S}(\mathsf{Y}_i), \qquad t_i = t^n + c_i\Delta t, \tag{34}$$

with $\mathsf{F}_j = \mathbf{L}\mathsf{Y}_j + \mathsf{b}(t_j) + \mathsf{S}(\mathsf{Y}_j)$. Note that b enters with coefficient one while the other terms carry $1/\gamma$; this follows from dividing the stage equation by $\gamma\Delta t$ and would be invisible on any case with homogeneous Neumann data, where $\mathsf{b} \equiv 0$.

The remaining coefficients were obtained by solving the order conditions for this structure rather than transcribed from a table, and verified: all four third-order conditions hold to $\sim 10^{-41}$, while the fourth-order condition $\sum b_i c_i^3 = 1/4$ misses by 1.4×10^{-3}, confirming the method is exactly third order and not accidentally higher.

Since a one-step method of order p leaves starting values accurate to $O(\Delta t^{p+1})$, ESDIRK3 covers every BDF the solver offers, and it is the **default**. With it, every scheme reaches its nominal order on the same $N = 40$ mesh where seeding from the exact solution capped everything at one (§4.6):

Scheme	Nominal	Self-convergence, route A (`M02`)	vs analytic, route B (`M02`)
`BDF2`	2	1.979	**1.992**
`BDF3`	3	2.932	**2.971**
`BDF4`	4	3.891	**3.908**

With `exact` or `constant` instead, the observed order of BDF3 and BDF4 collapses to one, and nothing in the run reports it.

What each startup mode actually does. The distinction matters for reading a log. `exact` and `constant` take *no steps at all*: they fabricate the history before the time loop begins — `exact` by evaluating the manufactured solution at the negative times $t^0 - j\Delta t$, `constant` by repeating V^0 — so step 0 is already a BDF step. `ESDIRK3` is the only mode that computes anything, and it does so by taking the first $k-1$ steps of the run with a different scheme.

`ESDIRK3` is also the **default**, and the only mode meant to be used. It is the only one that preserves the order of BDF3 and BDF4, and the only one that needs no exact solution — which a physical solver never has. `exact` and `constant` are kept solely so that the cost of a bad startup remains measurable (suite test M20), not as options.

How much of the run the startup occupies. The guard is $n_{\text{steps}} < k - 1$, and each ESDIRK step contains four stages of which the first is explicit, so three carry an implicit solve:

`implicitScheme`	Startup steps	Fixed-point solves
`backwardEuler`, `crankNicolson`, `BDF1`	0	0
`BDF2`	1	3
`BDF3`	2	6
`BDF4`	3	9

The second column is not the first: a BDF4 run starts with three steps but nine fixed-point solves, each iterating to `nonlinearVmTolerance` exactly as a normal step does.

Two consequences follow, and both have already caused a wrong reading.

First, **the startup leaves no trace in** `nonlinearResiduals.dat`. The ESDIRK block does not call the residual writer, so on a five-step BDF4 run the file contains rows for steps 4 and 5 only. Over the thousands of steps of a production run this is immaterial, but any comparison built on that file — a parallel-invariance check, a nonlinear-method equivalence test — is comparing the BDF steps and not the startup.

Second, **the startup cost is fixed, not proportional**. Nine solves is the ceiling whether the run has fifty steps or fifty thousand. On a *convergence sweep at coarse* Δt, however, that fixed cost is a large fraction of a short run, and the measured order then mixes two schemes. This is not hypothetical: at $\Delta t = 0.025$ with $t_{\text{end}} = 0.05$, a BDF3 run has two steps and *both* are startup, so the run is pure ESDIRK and its fitted order says nothing about BDF. The coarse pairs of such a sweep must be discarded, or the run lengthened until the startup is a small fraction of it.

Nonlinear methods. BDF places no restriction on `nonlinearMethod`: it changes only the scalar a_0 multiplying $\mathbf{M}$ in $\mathbf{A}$ and the history vector H^n, both of which Picard, JFNK and `diagonalIion` consume alike. Verified across all three for BDF2, BDF3 and BDF4, with agreement at the level of the nonlinear tolerance. The one exception is the startup itself: the ESDIRK stages run their own fixed-point iteration and ignore `nonlinearMethod`, so the first $k - 1$ steps are Picard whatever was requested. With `bdfStartup exact` or `constant` there is no ESDIRK and every step uses the requested method.

Why ESDIRK is not offered as a scheme. It is implemented and verified, and it is nevertheless reachable only as a startup. Two measurements decide this. First, order reduction is real: ESDIRK4 was implemented, its tableau verified in exact rational arithmetic against all eight order conditions and the stage-order condition $C(2)$, and it measures 3.03 — capped at stage order plus one — where BDF4 reaches 4.23. A multistep method has stage order equal to its order and suffers no such reduction. Second, the cost metric that matters is not linear solves but ODE integrations: each ESDIRK stage re-integrates the states over $[t^n, t^n + c_i \Delta t]$, so a four-stage step costs $\sum_i c_i \approx 2.5$ times the ODE work of a BDF step, and the ODE integrations are 74–99% of the runtime in this solver's physiological sibling. BDF therefore wins on accuracy *and* on the dominant cost, and ESDIRK's uniquely valuable property — being self-starting — is exactly what the startup needs.

4.4 ODE states inside one PDE time step

For a voltage candidate $V_{\text{m}}^{n+1,k}$, the state ODEs are advanced from the same time-step-start states $\mathbf{u}^n$. Inside the ODE integrator, voltage is interpolated linearly between the old and candidate values:

$$V_{\text{m}}(\tau) = (1 - \alpha) V_{\text{m}}^n + \alpha V_{\text{m}}^{n+1,k}, \qquad \alpha = \frac{\tau}{\Delta t}, \quad 0 \le \tau \le \Delta t. \tag{35}$$

Here τ is the local time inside the PDE time step, that is, a temporal coordinate running from $\tau = 0$ at t^n to $\tau = \Delta t$ at t^{n+1}, so that the physical time is $t = t^n + \tau$. It is introduced because the state-ODE integrator is autonomous with respect to the PDE solver: within a single step $[t^n, t^{n+1}]$

the state integrator may take many sub-steps $h \le \Delta t$ (adaptive in the RKF45 case), and τ is the variable that tracks how much of Δt has already been covered, accumulating as $\tau \leftarrow \tau + h$ until it reaches Δt.

Its specific role is to supply the voltage that the ODE sees at each Runge–Kutta stage. This interpolation is what makes the PDE–ODE system genuinely coupled inside the step: every change of the voltage candidate changes the whole state trajectory, which is why each residual evaluation must re-integrate the ODEs from $\mathbf{u}^n$.

The accuracy of the driver caps the whole solver. The linear ramp in (35) is wrong by $O(\Delta t^2)$ pointwise, which is $O(\Delta t^3)$ once integrated across the step and therefore $O(\Delta t^2)$ globally: it is compatible with Crank–Nicolson but bounds the attainable temporal order at two, however good the PDE integrator or the state integrator is. That is not a theoretical caveat, it is measured — suite test `M21` runs BDF4 with each driver and nothing else changed, and the ramp fits **2.04** against **3.89** for the interpolant, with the error at the finest pair larger by a factor of 44.7.

The general form. The driver is therefore no longer a pair of endpoints. The solver evaluates

$$V_{\mathrm{m}}(\tau) = \sum_{j=0}^{m} L_j(\alpha)\, \mathsf{V}_j, \qquad L_j(\alpha) = \prod_{l \neq j} \frac{\alpha - s_l}{s_j - s_l}, \qquad \alpha = \frac{\tau}{\Delta t_{\mathrm{step}}}, \tag{36}$$

the Lagrange polynomial through whatever voltages the outer scheme already knows, at their normalised times s_j. Nothing extra is computed: the nodes are values the scheme has produced anyway.

- **θ methods and BDF1** supply two nodes, $s = (0, 1)$ with V_{m}^n and the candidate. That is (35) exactly. The two-node case is coded separately rather than left to the general loop, so these schemes remain byte-identical to what they produced before the generalisation.
- **BDF-k** supplies $k + 1$ nodes at $s = -(k-1), \ldots, -1, 0, 1$ from its own stored history levels and the candidate. The ODEs are integrated over $\alpha \in [0, 1]$, inside the node hull, so this interpolates rather than extrapolates.
- **The ESDIRK startup** supplies $i + 1$ nodes at $s_j = c_j/c_i$ from its already-computed stage values, normalised by the current stage's interval.

The effect on the measured order is the difference between a solver that stops at two and one that does not (`M21`, BDF4, 2-D hexa $N = 40$, $p3$ triad, `bdfStartup ESDIRK3`):

`stateODEDriver`	Driver	Measured order	Error, finest pair
`linear`	two-node ramp	2.04	4.47×10^{-9}
`history`	full interpolant	**3.89**	9.99×10^{-11}

The ramp lands on 2.04, which is the order the argument above predicts, and the interpolant recovers the scheme's own order. `history` is the default; `linear` exists so that this cost stays measurable, because a capped order is all the run reports — there is no error and no warning.

The Gauss-point path (2026-10). With `gaussPointODE` the driver was always the two-node ramp, because raising it seemed to require storing every level at every quadrature point. With `stateODEGaussPointHistory true` that path uses the same polynomial as the cell path without storing anything between steps: each history level (or each ESDIRK stage value) is reconstructed at the Gauss points at the start of the step. See Section 12.3.

4.5 The full triad grid, and the cost of each combination

`M01` sweeps the diagonal — four triads that raise all three links together — over five mesh levels. Suite test `M25` sweeps the whole space over three fine ones: 4 levels of V_m $\times$ 4 of I_ion $\times$ $\{\mathrm{CC}p1, \mathrm{CC}p2, \mathrm{CC}p3, \mathrm{GP}\}$ for the states, which is 52 distinguishable triads per mesh family (the states axis does not exist when I_ion is low order, because there are no quadrature points to reconstruct to), on hexahedra and unstructured triangles at $N = 100, 120, 140$. That is 312 runs.

It answers what `M01` cannot: what happens when the links do *not* match. The claim is that the observed order is the minimum over the three, and the consequence is sharp. Spread of the fitted order over all twelve (states, I_ion) combinations, per level of V_m:

Mesh	V_m	min	max	spread
hexahedral	`NO`	2.000	2.000	0.000
hexahedral	`p1`	2.711	2.711	0.001
hexahedral	`p2`	2.039	2.066	0.027
hexahedral	`p3`	2.005	4.281	**2.276**
unstructured	`NO`	2.984	2.988	0.005
unstructured	`p1`	2.432	2.432	0.000
unstructured	`p2`	2.004	2.011	0.007
unstructured	`p3`	2.002	4.230	**2.228**

Below $p3$ in V_m the ionic path is *irrelevant* — raising the quadrature of I_ion or the reconstruction degree of the states buys nothing, to three decimals over twelve combinations — and at $p3$ it becomes the only thing that matters, taking the order from 2.0 to 4.28.

Order alone does not let you choose. Every row of `M25` carries its wall time, split into setup and loop: the setup builds the stencils and factorises one least-squares problem per cell and per quadrature point and grows with the degree, while the loop grows with the number of steps. The cheapest configuration reaching each target order, against the diagonal:

Mesh	Cheapest at order ≥ 3	Order	Total	Diagonal $(p3, \mathrm{CC}p3, p3)$
hexahedral	$(p3, \mathrm{CC}p2, p2)$	4.280	9.1 s	10.9 s
unstructured	$(p3, \mathrm{CC}p2, p2)$	4.229	22.2 s	25.4 s

The diagonal is what symmetry suggests and it is not optimal: $(p3, \mathrm{CC}p2, p2)$ reaches the same order for less. That is a design result, and it cannot be seen in a table that carries only orders.

The tables also give the pairwise orders of `u1_L2` and `u2_L2` beside V_m's, because the states are what separate the two state modes — averaging them back from the quadrature points costs order, while V_m converges the same either way.

Locating the cap rather than observing it. `M21` shows the cap; it cannot show *where* it lives. The environment variable `CF_EXACT_STATES` overwrites the integrated states with their exact manufactured values at every stage, which removes the state path from the chain entirely, so the temporal order measured with it on is that of the PDE step alone. Suite test `M24` uses it as a bisection instrument, on the same configuration as the table above:

`stateODEDriver`	`CF_EXACT_STATES`	Measured order	Cap present?
`history`	off	3.890	—
`history`	on	3.889	—
`linear`	off	2.038	yes
`linear`	on	**3.889**	**gone**

The jump from 2.038 to 3.889 on the last two rows is the demonstration: the same scheme, the same Δt, everything else identical, and the only change is whether the states enter the chain. The two rows with the instrument on agree to sixteen digits, which they must — the driver only controls how the states are integrated, and with the instrument on they are not integrated at all. That exact agreement is the strongest check in `M24`, because an instrument that perturbed anything else would break it.

The instrument falsifies the problem on purpose: injecting the exact solution at every stage is not a valid run of anything, and the only value of each number is the contrast against its twin. The orders that the paper reports come from `M02`.

One path deliberately keeps the two-node ramp: `gaussPointODE`, where the ODE is solved at every quadrature point. Raising it would require reconstructing and storing every stage or history level *at every quadrature point*, n_{stages} times the integration-point memory. The cell-centred path, which the recommended triad uses, carries the full interpolant.

The available state solvers are:

`stateODESolver`	Method	Nominal order
`Euler`	explicit Euler	1
`RK4`	classical Runge–Kutta	4
`RKF45`	adaptive embedded Cash–Karp pair	4/5

For `RKF45`, the fifth-order candidate is accepted and the fourth-order embedded estimate supplies the local error indicator

$$e = \max_j \frac{|u_j^{(5)} - u_j^{(4)}|}{\texttt{stateODEAbsTol} + \texttt{stateODERelTol}\,\max(|u_j^{(5)}|, |u_j^{(4)}|)}. \tag{37}$$

A sub-step is accepted if $e \le 1$ or if the minimum step has been reached; the next sub-step is scaled by a safety factor proportional to $e^{-1/4}$. In convergence studies, the ODE tolerances must be tight enough that the state integration error does not mask the spatial or PDE time-integration order.

4.6 Measured temporal orders

The orders above are what the schemes are supposed to deliver. What follows is what they deliver, measured two different ways, because neither way is sufficient on its own.

4.7 Two things the convergence study measures against, and how they were wrong

A measured order is a ratio of two logarithms, $\ln(e_1/e_2)/\ln(h_1/h_2)$. Both halves turned out to be misreported, and both were fixed on 2026-09-29. Neither was a solver defect; both changed every number the sweeps had produced.

`Nn` is the stencil cells ADDED on top of the minimum, not the total. The library builds the stencil as minNn() + `Nn`, where minNn() is the number of terms in the Taylor polynomial — $(N{+}1)(N{+}2)/2$ in 2-D, $(N{+}1)(N{+}2)(N{+}3)/6$ in 3-D — i.e. the number of unknowns the local least-squares fit determines. The sweeps had been writing the intended *total* into that field: the 2-D entries were 13/16/20 for $p1/p2/p3$, which are exactly minNn() + 10, so the library added the minimum again and the real stencils were 16/22/30 — about 50 % larger than intended. Nothing failed; a larger stencil is a valid fit, just more expensive and more smoothing.

It was not harmless either. The oversized stencils reached past the halo depth, and on cells against a processor cut the stencil lost a member, which is what made suite tests `M11` and `M15` fail — a failure that had been attributed to solids4foam and classed as measured-not-fixed for months. With `Nn` = 10 in 2-D and 16/30/40 in 3-D, giving stencils of 13/16/20 and 20/40/60 and a redundancy of 2× or better everywhere, `M11` measures zero differing stencils and `M15` drops from 4.981×10^{-6} to 6.953×10^{-15}. The suite is 24/24.

Algorithm 1 Adaptive sub-step control of the embedded 5(4) pair inside one PDE time step

Input: States $\mathbf{u}^n$ at t^n, voltages V_{m}^n and $V_{\mathrm{m}}^{n+1,k}$, PDE step Δt

$\tau \leftarrow 0$; $h \leftarrow \min(\Delta t,\ \texttt{stateODEInitialStep})$; $\mathbf{u} \leftarrow \mathbf{u}^n$
$h_{\min} \leftarrow \max(10^{-12}\Delta t,\ \texttt{SMALL})$
for $s = 1, \ldots, \texttt{stateODEMaxSteps}$ **do**
  if $\tau \geq \Delta t$ **then**
    break
  end if
  $h \leftarrow \min(h,\ \Delta t - \tau)$ ▷ never overshoot t^{n+1}
  Evaluate the 6 stages with $V_{\mathrm{m}}(\tau + c_s h) = (1 - \frac{\tau + c_s h}{\Delta t})V_{\mathrm{m}}^n + \frac{\tau + c_s h}{\Delta t} V_{\mathrm{m}}^{n+1,k}$
  Form $\mathbf{u}^{(5)}$ (propagated) and $\mathbf{u}^{(4)}$ (embedded)
  $e \leftarrow \max_j \dfrac{|u_j^{(5)} - u_j^{(4)}|}{\texttt{absTol} + \texttt{relTol}\,\max(|u_j^{(5)}|, |u_j^{(4)}|)}$
  if $e \leq 1$ **or** $h \leq h_{\min}$ **then** ▷ accept; the 2nd branch is an anti-stall safeguard
    $\mathbf{u} \leftarrow \mathbf{u}^{(5)}$; $\tau \leftarrow \tau + h$
  else ▷ reject: $\mathbf{u}$ and τ are left untouched
  end if
  $\varphi \leftarrow \min\big(4,\ \max(0.1,\ 0.84\, e^{-1/4})\big)$ ▷ $e^{-1/4}$ since the error per unit step is $O(h^4)$
  $h \leftarrow \max\big(h_{\min},\ \min(\Delta t - \tau,\ h\,\varphi)\big)$
end for
if $\tau < \Delta t$ **then**
  Emit a warning: sub-step budget exhausted
end if

Output: $\mathbf{u}^{n+1,k} \leftarrow \mathbf{u}$

h is measured, not assumed to be $1/N$. On a hexahedral mesh every edge is exactly $1/N$ and the distinction is empty. On an unstructured triangulation it is not: gmsh is asked for a uniform $1/N$ and honours it *on average*, but at $N = 160$ the shortest edge is 0.00433 and the longest 0.00839, so the shortest is 31 % shorter than the longest and there is no single h.

What enters the rate is the *ratio* between refinement levels, and the four candidate definitions behave differently. Measured maximum deviation from the ideal 2: mean 0.022, rms 0.024, minimum 0.292, maximum 0.246. The mean and the rms scale as $1/N$ to about 1 %, so for an L_2 norm $1/N$ was a good proxy — recomputing `M01` with the real h moves the L_2 orders by 0.006 to 0.011. The minimum and the maximum do not: their ratios run from 1.82 to 2.29, and with them the order of $p3$ on the unstructured mesh moves between 4.31 and 4.74. For an L_∞ norm, which looks at the worst element, $1/N$ introduced a bias of that size.

So the harness now measures the edge lengths of every mesh it generates — in 2-D discarding the edges parallel to the extrusion, whose axis is detected rather than assumed — stores `h_mean`, `h_rms`, `h_min` and `h_max` alongside the errors, and fits each metric against the h that belongs to it: `h_rms` for the L_2 norms, `h_max` for the L_∞ ones. Both rate files record which h they used, because an order without that is not reproducible.

Why not simply compare against the exact solution. At a fixed mesh the error against the manufactured solution is temporal *plus* spatial, and the spatial part does not fall with Δt. On the sine solution with the $p3$ triad on a 2-D hexahedral $N = 40$ mesh it is 8.3×10^{-6} for every scheme, while BDF4's temporal error at the finest Δt of its sweep is $\sim 10^{-10}$: four orders of magnitude below the floor. Any fitted order is then the floor's, not the scheme's — an earlier version of the temporal test fitted 0.563 for Crank–Nicolson this way and reported a failure that was not the solver's. Pushing the spatial error below the temporal one would require $N \sim 1000$ in 2-D, about an hour per run.

Self-convergence, with diffusion active. The first measurement avoids any reference: at a fixed mesh, sweep Δt by halves and take $d(\Delta t) = \|\mathsf{V}(\Delta t) - \mathsf{V}(\Delta t/2)\|$. If the temporal error is $C\Delta t^p$ then $d = C\Delta t^p(1 - 2^{-p})$, so its slope is p, and the spatial error cancels because both runs share the mesh. This is suite test `M02`, on the sine solution, with the diffusion operator fully active.

Against the analytic solution, with the spatial error removed. The second measurement is the statement one actually wants, and it requires a manufactured solution the discretisation reproduces *exactly*: `manufacturedSolution uniform`, where $V_\mathrm{m} = \sqrt{1+t}$ with every spatial factor equal to one. Every reconstruction, quadrature and flux here is exact on a constant field, so the spatial error is zero to roundoff and the whole error is temporal. Route B of suite test `M02` verifies that rather than assuming it: at a fixed Δt the error on $N = 20$ and $N = 40$ agrees to 1.7×10^{-15}, and the residual spatial variation of the final field ($\sim 10^{-11}$, relative) falls by 83–246× when the solver tolerances are tightened 100×, which is what identifies it as iterative noise rather than discretisation error.

Its limitation is structural and worth stating: the Laplacian of a constant is zero, so under the uniform solution the diffusion operator contributes nothing and what is verified is the time integration of the coupled reaction system — V_m, the three states, the ESDIRK3 startup, the BDF history and the ODE driver. Diffusion is covered by `M02`. A solution that is both exactly representable and diffusive would have to be a discrete eigenfunction of the LRE operator, which has no closed form, and the `zeroGradient` boundaries rule out the linear profile that would otherwise serve.

`implicitScheme`	Nominal	Self-convergence, route A (`M02`)	vs analytic, route B (`M02`)
`backwardEuler`	1	1.000	1.000
`BDF1`	1	1.000	1.000
`crankNicolson`	2	1.998	2.000
`BDF2`	2	1.979	1.992
`BDF3`	3	2.932	2.971
`BDF4`	4	3.891	3.908

Both columns use `bdfStartup ESDIRK3`, the $p3$ triad, $\alpha = 0$, Picard, $t_\mathrm{end} = 0.512$, and the coarsest Δt of every sweep leaves at least ten steps so that the run is not mostly startup. `BDF1` reproduces `backwardEuler` bit for bit, which is checked rather than asserted.

The floor of the judged quantity, measured. Both quantities stop reflecting the scheme once they reach the noise the solver's own tolerances inject. That level is measured rather than guessed: the finest Δt is run again with tolerances 100× tighter, and the floor is how much the judged quantity itself moved. A value counts only if it exceeds $K = 100$ times that floor, K being fixed in advance by the bound it implies — such a separation perturbs a pairwise order by at most $(2/K)/\ln 2 = 0.029$, a seventh of the ±0.2 margin the tests allow. Measured floors are 2×10^{-14} to 3×10^{-13}. An earlier fixed floor of 10^{-9}, estimated and not measured, discarded a BDF4 difference that sat 292× above the real one.

5 Nonlinear coupled solve

5.1 Residual and diagnostics

For any voltage candidate x, the solver evaluates the corresponding state fields and source by calling `evaluateNonlinearFields`. This means: reconstruct V_m to I_ion quadrature points if required, reset/evolve states from t^n to t^{n+1} using x, compute I_ion, and assemble the source. The coupled voltage residual is

$$\mathsf{F}(\mathsf{x}) = \mathbf{A}\mathsf{x} - \left[\mathbf{B}\mathsf{V}^n + \theta\left(\mathsf{S}(\mathsf{x}) + \mathsf{b}^{n+1}\right) + (1-\theta)\left(\mathsf{S}^n + \mathsf{b}^n\right)\right], \tag{38}$$

where b^n and b^{n+1} are known boundary/source contributions from the manufactured boundary treatment.

The solver records several relative diagnostics:

$$R_{\mathrm{cpl}} = \frac{\|\mathsf{F}(\mathsf{x})\|_2}{\|\mathsf{r}(\mathsf{x})\|_2 + \epsilon}, \qquad R_{V_{\mathrm{m}}} = \frac{\left\|\mathsf{V}^{k+1} - \mathsf{V}^k\right\|_2}{\|\mathsf{V}^k\|_2 + \epsilon}, \tag{39}$$

$$R_{I_{\mathrm{ion}}} = \frac{\left\|\mathsf{I}_{\mathrm{ion}}^{k+1} - \mathsf{I}_{\mathrm{ion}}^{k}\right\|_2}{\left\|\mathsf{I}_{\mathrm{ion}}^{k}\right\|_2 + \epsilon}, \qquad R_{u_j} = \frac{\left\|\mathsf{u}_j^{k+1} - \mathsf{u}_j^{k}\right\|_2}{\left\|\mathsf{u}_j^{k}\right\|_2 + \epsilon}. \tag{40}$$

Here ϵ is the OpenFOAM `SMALL` safeguard. When high-order I_{ion} quadrature uses `gaussPointODE`, state residuals are measured at the quadrature points. In the default `cellCentredReconstruct` mode, and in the low-order source path, the authoritative states are cell-centred and the state residuals are measured at cell centres. The maximum state residual is

$$R_{\mathbf{u}} = \max(R_{u_1}, R_{u_2}, R_{u_3}). \tag{41}$$

The residual history is written to `postProcessing/highOrderManufacturedFDAImplicitPETSc/nonlinearResiduals.dat` when `writeNonlinearResiduals=true`.

5.2 Picard fixed-point method

Picard treats the nonlinear source as lagged during each linear voltage solve. Given V^k, the solver computes $\mathsf{S}^k = \mathsf{S}(\mathsf{V}^k, \mathbf{u}(\mathsf{V}^k))$ and then solves

$$\mathbf{A}\widehat{\mathsf{V}}^{k+1} = \mathbf{B}\mathsf{V}^n + \theta(\mathsf{S}^k + \mathsf{b}^{n+1}) + (1-\theta)(\mathsf{S}^n + \mathsf{b}^n). \tag{42}$$

The relaxed update is

$$\mathsf{V}^{k+1} = \mathsf{V}^k + \omega\left(\widehat{\mathsf{V}}^{k+1} - \mathsf{V}^k\right), \qquad \omega = \texttt{nonlinearRelaxation}. \tag{43}$$

Convergence requires at least `minNonlinearIterations`. **Since 2026-10** Picard stops with the same test as JFNK and `diagonalIion`, the shared function `nonlinearIterationExit` (Section 12.2): the V_{m} and I_{ion} increments — and the state increments if `nonlinearRequireStatesConvergence` — and the coupled residual $R_{\mathrm{cpl}} \le$ `nonlinearTolerance`, with a floor exit when the residual stops falling. Before, Picard stopped on the V_{m} and state increments alone and the coupled residual was not a stopping criterion. If Picard exhausts the iteration budget it follows the same semantics as the other two methods: by default the step is rolled back, and with `nonlinearAcceptUnconverged` the iterate is accepted (the separate branch that always accepted the last iterate was removed in September).

It is useful to have the expected convergence rate at hand, because it governs the iteration counts observed in the results. Linearising the iteration around its fixed point, the error $\mathsf{e}^k = \mathsf{V}^k - \mathsf{V}^{n+1}$ obeys

$$\mathsf{e}^{k+1} = \left[(1-\omega)\mathbf{I} + \omega\,\theta\,\mathbf{A}^{-1}\mathbf{J}_S\right]\mathsf{e}^k, \qquad \mathbf{J}_S = \frac{\partial \mathsf{S}}{\partial \mathsf{V}}. \tag{44}$$

With lumped mass, $\mathbf{A} = \mathbf{M}/\Delta t - \theta\mathbf{L}$, so that for moderate Δt one has $\mathbf{A}^{-1} \approx \Delta t\,\mathbf{I}$ and the spectral radius of the iteration operator is approximately

$$\rho_{\mathrm{Picard}} \approx \omega\,\theta\,\Delta t\,\max_i |d_i|, \qquad d_i = \frac{\partial S_i}{\partial V_i}. \tag{45}$$

Picard therefore converges linearly at a rate proportional to Δt times the Lipschitz constant of the source, and *diverges* when $\omega\theta\Delta t \max_i |d_i| \gtrsim 1$. Even though the θ scheme is A-stable, the lagged coupling imposes an explicit-like step restriction tied to the stiffness of the reaction term; that is the limitation removed in §5.3. The numerical verification of (45) over the full sweep is in §13.1.

Algorithm 2 Picard nonlinear coupling

Input: Old solution $\mathsf{V}^n, \mathbf{u}^n$, source S^n

Build $\mathbf{A}$ and $\mathbf{B}$ from $\mathbf{M}$, $\mathbf{L}$, θ, and Δt
Initialise V^0 from V^n and impose exact boundary values at t^{n+1}
for $k = 0, \ldots, N_{\mathrm{nl}} - 1$ **do**
    Evaluate states and source S^k from V^k
    Solve $\mathbf{A}\widehat{\mathsf{V}}^{k+1} = \mathsf{r}(\mathsf{S}^k)$
    Relax $\mathsf{V}^{k+1} = \mathsf{V}^k + \omega(\widehat{\mathsf{V}}^{k+1} - \mathsf{V}^k)$
    Apply exact boundary values; the source is refreshed at the next Picard corrector
    Compute $R_{V_{\mathrm{m}}}$, R_{u_1}, R_{u_2}, R_{u_3}, $R_{I_{\mathrm{ion}}}$, R_{cpl}
    if `nonlinearIterationExit` reports converged or floor **then**
        accept and exit loop
    end if
end for

Output: If not converged: roll back to t^n, or accept the iterate with `nonlinearAcceptUnconverged`

5.3 Diagonal-I_{ion} linearisation

`diagonalIion` keeps the same scalar voltage unknowns but adds an approximate row-wise source sensitivity to the matrix. The implementation perturbs all internal voltage unknowns by the same scalar `diagonalIionEpsilon`, re-imposes the exact boundary values, and uses the result as a diagonal surrogate:

$$d_i \approx \frac{S_i(\mathsf{V}^k + \varepsilon \mathbf{1}) - S_i(\mathsf{V}^k)}{\varepsilon}, \qquad \varepsilon = \texttt{diagonalIionEpsilon}. \tag{46}$$

The linearised source is

$$S_i(\mathsf{V}^{k+1}) \approx S_i(\mathsf{V}^k) + d_i \left(V_i^{k+1} - V_i^k \right). \tag{47}$$

Moving the unknown part to the left gives

$$(\mathbf{A} - \theta \operatorname{diag}(\mathbf{d}))\, \widehat{\mathsf{V}}^{k+1} = \mathsf{r}(\mathsf{S}^k) - \theta\, \mathbf{d} \odot \mathsf{V}^k, \tag{48}$$

where $\odot$ is componentwise multiplication and

$$\mathsf{r}(\mathsf{S}^k) = \mathbf{B}\mathsf{V}^n + \theta \left(\mathsf{S}^k + \mathsf{b}^{n+1} \right) + (1 - \theta) \left(\mathsf{S}^n + \mathsf{b}^n \right). \tag{49}$$

This is the same right-hand side used by the Picard voltage solve; the `diagonalIion` branch only moves the local linearised source contribution to the left-hand side. This is not a full Newton Jacobian. For a strictly cell-local source it approximates the local voltage sensitivity, including the effect of re-integrating the states for the perturbed candidate. With high-order reconstruction/quadrature active, it should be read as a diagonal surrogate for a uniform-direction row sensitivity: off-diagonal Jacobian couplings from reconstruction and state reconstruction are not assembled. It is often more implicit than Picard while remaining cheaper and simpler than JFNK.

5.4 Jacobian-free Newton–Krylov

JFNK seeks a zero of (38):

$$\mathsf{F}(\mathsf{x}) = 0. \tag{50}$$

Newton's method would solve

$$\mathbf{J}(\mathsf{x}_k)\delta_k = -\mathsf{F}(\mathsf{x}_k), \qquad \mathsf{x}_{k+1} = \mathsf{x}_k + \alpha\delta_k, \tag{51}$$

but the full Jacobian is not assembled. Instead, a Krylov method only needs products $\mathbf{J}\mathbf{v}$, approximated by the finite difference

$$\mathbf{J}(\mathsf{x})\mathbf{v} \approx \frac{\mathsf{F}(\mathsf{x} + \epsilon\mathbf{v}) - \mathsf{F}(\mathsf{x})}{\epsilon}, \qquad \epsilon = \texttt{jfnkEpsilon}\, \frac{1 + \|\mathsf{x}\|_2}{\|\mathbf{v}\|_2 + \epsilon_{\mathrm{mach}}}. \tag{52}$$

Algorithm 3 `diagonalIion` nonlinear coupling

Input: Old solution $\mathsf{V}^n, \mathbf{u}^n$, source S^n

Initialise V^0 from V^n and impose exact boundary values at t^{n+1}
for $k = 0, \ldots, N_{\mathrm{nl}} - 1$ **do**
    Evaluate S^k and I_{ion}^k from V^k
    Perturb all internal V_{m} values by `diagonalIionEpsilon`, re-apply boundary values, and re-evaluate source
    Form the diagonal surrogate d_i by the row-wise finite difference
    Solve $(\mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d}))\,\widehat{\mathsf{V}}^{k+1} = \mathsf{r}(\mathsf{S}^k) - \theta\mathbf{d} \odot \mathsf{V}^k$
    Relax the voltage update
    Re-evaluate source/states at the relaxed voltage
    Check coupled, voltage, I_{ion}, and optional state criteria
    if minimum iterations reached and diagonal tolerances hold **then**
        accept and exit loop
    end if
end for
if not converged and `nonlinearAcceptUnconverged=false` **then**
    roll back V_{m}, I_{ion}, and states to the beginning of the time step
end if

Each matrix-free Krylov product therefore performs a full nonlinear residual evaluation: ODE solve, source evaluation, source quadrature if active, and matrix-vector application. This is the main cost of JFNK.

The Krylov solve is delegated to a selectable linear-algebra backend, which decides which library materially executes the iterations without altering the mathematics of the method. With the default backend, PETSc, the Newton system is solved through a persistent `KSP` object (*Krylov SPace solver*): the `KSP` encapsulates the iterative method, its stopping criteria, and its work vectors, and is coupled to a matrix of type `MatShell`, that is, a matrix that stores no entries at all but merely registers the function computing the matrix-vector product. That function is precisely the finite difference (52), so the Jacobian is never formed. This combination is possible because Krylov methods build the solution in the subspace $\mathcal{K}_m = \mathrm{span}\{\mathsf{r}_0, \mathbf{J}\mathsf{r}_0, \ldots, \mathbf{J}^{m-1}\mathsf{r}_0\}$ and therefore only need to apply the operator to vectors, never to inspect its coefficients. The default `jfnkPetscKspType=gmres` selects GMRES (*Generalized Minimal RESidual*), the Krylov method appropriate for non-symmetric operators such as this Jacobian: at each iteration it returns the vector of the subspace that minimises the 2-norm of the linear residual, building an orthonormal basis by the Arnoldi process and solving a small least-squares problem over it. Since the cost and memory of Arnoldi grow with the iteration count — every accumulated basis vector must be stored and orthogonalised against — *restarted* GMRES is used: after `jfnkPetscRestart` iterations the basis is discarded and the process is relaunched from the current solution, bounding the memory at the price of losing global minimisation over the whole history. Other PETSc `KSP` types can be selected (for example `bcgs`, BiCGSTAB, of fixed memory) through the dictionary or the PETSc options database.

The alternative backend, `Eigen`, replaces PETSc by an equivalent in-file implementation: restarted GMRES(m) with $m =$ `jfnkMaxKrylovIterations` and `jfnkMaxRestarts` restarts, with the Arnoldi basis orthogonalised explicitly in the code and the projected least-squares problem solved with Eigen's column-pivoted Householder QR — the pivoting adds robustness when the Arnoldi basis becomes nearly rank-deficient, which can happen because the finite-difference matrix-vector product carries round-off noise. In either backend, the Krylov solve stops when the relative norm of its residual falls below `jfnkLinearTolerance` or when the iteration and restart budget is exhausted; it is worth emphasising that this tolerance is a *linear* one and only controls how accurately the Newton correction δ_k is solved for, whereas convergence of the nonlinear problem is governed by the tolerances of §8. The two backends are numerically equivalent for the same matrix and tolerances: they differ in the linear-solver implementation, not in the PDE residual.

5.5 The top-of-iteration exit, and the defect it hid

The JFNK loop evaluates its convergence test *twice* per iteration: once at the top, on the incoming iterate, so that a Krylov solve can be skipped if the iterate is already converged, and once at the bottom, after the Newton update. The two evaluations are not interchangeable, and the top one is the delicate case: the increments it needs are those accumulated by the *previous* iteration, because the current one has not moved anything yet.

An earlier revision recomputed them in place, against a copy of the fields taken a few lines above, with only one evaluation of $\mathsf{F}(\mathsf{x}_k)$ in between. That evaluation re-uses the same x_k that the bottom of the previous iteration had already evaluated, so the four "increments" measured the difference between two evaluations of one point and were identically zero. The test

$$k{+}1 \geq \texttt{minNonlinearIterations} \wedge R_{\mathrm{cpl}} \leq \texttt{nonlinearTolerance} \wedge \underbrace{R_{V_{\mathrm{m}}} \leq \texttt{nonlinearVmTolerance}}_{0 \leq \mathrm{tol}} \wedge \cdots$$

therefore reduced, unconditionally, to its first two terms: four of the five configured tolerances were dead, comparing zero against a positive number.

The consequence is not cosmetic. Because the Newton iterate is seeded by extrapolation (`jfnkInitGuessOrder`), JFNK accepted that extrapolation without taking a single Newton step whenever its coupled residual happened to fall below `nonlinearTolerance`. Measured on the 3-D configuration of M08 ($N = 10$, p3 triplet, $\Delta t = 2 \times 10^{-4}$, 50 steps): 24 of the 50 steps exited that way with zero Krylov calls. The solver's own transient summary recorded the symptom and nobody read it — `max Vm_relL2` was exactly zero over the whole run, that is, the largest voltage increment of the entire integration was reported as 0.

The fix is to carry the increments measured at the *end* of the previous iteration into the next top-of-loop test, initialised so that the first pass can never exit there; at least one Newton step is then always taken. Two independent measurements confirm the diagnosis. Forcing `minNonlinearIterations` = 2, which imposes the same thing from the outside without touching the code, reproduces the corrected number; and raising it to 3 changes nothing further, so a single Newton step is all that was missing.

quantity	before	after
`max coupled_relL2` over the run	9.493×10^{-9}	5.655×10^{-16}
`max Vm_relL2` over the run	0	9.847×10^{-9}
$\lvert$Vm_L2(JFNK) − Vm_L2(Picard)$\rvert$	6.309×10^{-8}	2.015×10^{-11}
steps with zero Krylov calls	24/50	0/50

Picard and `diagonalIion` are bit-identical before and after, which is what verifies that the change is confined to the branch that carried the defect.

The general lesson is about two-sided convergence tests rather than about JFNK. When the same predicate is evaluated at two points of a loop, the quantities it reads have different meanings at each, and a term that is structurally zero at one of them does not announce itself: the test keeps passing, the tolerance keeps being accepted from the dictionary, and the only visible trace is a diagnostic field reported as an exact zero.

5.6 Armijo line search and rollback

This subsection applies to the JFNK branch only. After the Krylov solve proposes a Newton correction δ_k, the solver optionally uses Armijo backtracking. Starting from $\alpha = \texttt{nonlinearRelaxation}$, it accepts the trial step if

$$\|\mathsf{F}(\mathsf{x}_k + \alpha\delta_k)\|_2^2 \leq (1 - 2c\alpha)\,\|\mathsf{F}(\mathsf{x}_k)\|_2^2\,, \qquad c = \texttt{jfnkArmijoC}. \tag{53}$$

If the condition fails, α is halved until either the condition is met, `jfnkLineSearchMaxIter` is reached, or `jfnkLineSearchAlphaMin` is reached. This globalises Newton by discouraging steps that increase the nonlinear residual.

The damped Newton update is:

$$\mathsf{x}_{k+1} = \mathsf{x}_k + \alpha\,\delta_k.$$

α is the fraction of the Newton step that is actually applied, and c fixes how demanding the "sufficient decrease" criterion is — asking for only a fraction c of the reduction predicted by the linear model. The factor $(1 - 2c\alpha)$ is very close to 1, so in practice the condition only rejects steps that do not reduce the residual.

For `JFNK` and `diagonalIion`, if the nonlinear solver reaches the maximum number of iterations without convergence, the default behaviour is a rollback: V_m, I_ion, and the state fields are restored to the beginning of the time step. Setting `nonlinearAcceptUnconverged=true` disables this rollback and accepts the last iterate. Picard now follows the same semantics: its own branch, which always accepted the last iterate with a warning, was removed.

When the line search exhausts `jfnkLineSearchMaxIter`, the last trial point is accepted even though it failed the Armijo test. The rollback freezes the state while time advances anyway, and therefore a single rollback invalidates the convergence rate of that run.

Algorithm 4 Jacobian-free Newton–Krylov nonlinear coupling

Input: Old solution $\mathsf{V}^n, \mathbf{u}^n$, old source S^n
Build initial guess x_0 from V^n or extrapolation
for $k = 0, \ldots, N_\mathrm{nl} - 1$ **do**
    Evaluate $\mathsf{F}(\mathsf{x}_k)$, states, source, and I_ion
    if minimum iterations reached and nonlinear tolerances hold **then**
        accept x_k and exit loop
    end if
    Define matrix-free product $\mathbf{v} \mapsto [\mathsf{F}(\mathsf{x}_k + \epsilon\mathbf{v}) - \mathsf{F}(\mathsf{x}_k)]/\epsilon$
    Use the configured matrix-free Krylov solve to approximate $\mathbf{J}\delta_k = -\mathsf{F}(\mathsf{x}_k)$
    if `jfnkLineSearch=true` **then**
        Set $\alpha \leftarrow$ `nonlinearRelaxation`
        while Armijo condition (53) fails and budget remains **do**
            $\alpha \leftarrow \max(\alpha/2,$ `jfnkLineSearchAlphaMin`$)$
        end while
    else
        $\alpha \leftarrow$ `nonlinearRelaxation`
    end if
    Update $\mathsf{x}_{k+1} = \mathsf{x}_k + \alpha\delta_k$
    Re-evaluate states/source at x_{k+1} and write residual diagnostics
    if minimum iterations reached and nonlinear tolerances hold **then**
        accept x_{k+1} and exit loop
    end if
end for
if not converged and `nonlinearAcceptUnconverged=false` **then**
    roll back V_m, I_ion, and states to their time-step-start values
end if

5.7 Method comparison

The comparison is made by looking at the iteration operator of each method.

Picard With the lagged source, the iteration is $\mathsf{V}^{k+1} = \mathsf{V}^k + \omega(\mathbf{A}^{-1}\mathsf{r}(\mathsf{S}^k) - \mathsf{V}^k)$. Linearising around the fixed point, the error $\mathsf{e}^k = \mathsf{V}^k - \mathsf{V}^{n+1}$ evolves as

$$\mathsf{e}^{k+1} = \big[(1-\omega)\mathbf{I} + \omega\,\theta\,\mathbf{A}^{-1}\mathbf{J}_S\big]\mathsf{e}^k, \qquad \mathbf{J}_S = \frac{\partial \mathsf{S}}{\partial \mathsf{V}}.$$

With lumped mass, $\mathbf{A} = \mathbf{I}/\Delta t - \theta\mathbf{L}$, so that for moderate Δt one has $\mathbf{A}^{-1} \approx \Delta t\,\mathbf{I}$ and the spectral radius is approximately

$$\rho_{\text{Picard}} \approx \omega\,\theta\,\Delta t\,\max_i |d_i|.$$

Conclusion: Picard converges linearly at a rate proportional to Δt times the Lipschitz constant of the source, and diverges if $\theta\,\Delta t\,\max_i |d_i| \gtrsim 1$. That is, even though the θ scheme is A-stable, the coupling imposes an explicit-like step restriction on the reaction term. This is precisely the Achilles heel in electrophysiology, where $\partial I_{\text{ion}}/\partial V_{\text{m}}$ is enormous during the upstroke.

diagonalIion Moving $\theta\,\text{diag}(\mathbf{d})$ to the left-hand side turns the iteration operator into

$$\mathsf{e}^{k+1} = \big[(1-\omega)\mathbf{I} + \omega\,\theta\,(\mathbf{A} - \theta\,\text{diag}(\mathbf{d}))^{-1}\big(\mathbf{J}_S - \text{diag}(\mathbf{d})\big)\big]\,\mathsf{e}^{k}.$$

What multiplies the error is no longer $\mathbf{J}_S$ but only the part that the diagonal surrogate fails to capture, $\mathbf{J}_S - \text{diag}(\mathbf{d})$. Hence the concrete theoretical advantages:

1. **It removes the step restriction due to reaction stiffness.** On the low-order path (`useHighOrder_Iion=false`) the source is strictly local per cell, $\mathbf{J}_S$ is exactly diagonal, and therefore $\mathbf{J}_S - \text{diag}(\mathbf{d}) \to \mathbf{0}$ up to the $O(\varepsilon)$ error of the finite difference. In that case diagonalIion is Newton's method for the reaction part: it converges in essentially one or two iterations and its convergence is independent of $\Delta t\,\max_i |d_i|$. Picard cannot do that at any cost.

2. **On the high-order path the rate shrinks without vanishing.** With reconstruction of V_{m} and of the states at the quadrature points, $\mathbf{J}_S$ acquires off-diagonal couplings through the LRE stencil. diagonalIion does not assemble them, so convergence is linear again — but at a rate $\rho \approx \omega\theta\Delta t\;\|\mathbf{J}_S - \text{diag}(\mathbf{d})\|$, where the norm is that of the non-local couplings, typically orders of magnitude smaller than $\|\mathbf{J}_S\|$ because the dominant terms are the on-point ones.

3. **A fine point in its favour: d_i is the row sum, not the diagonal.** The perturbation is taken along the uniform direction $\varepsilon\mathbf{1}$, so

$$d_i = \frac{S_i(\mathsf{V} + \varepsilon\mathbf{1}) - S_i(\mathsf{V})}{\varepsilon} \longrightarrow \sum_j \frac{\partial S_i}{\partial V_j},$$

   that is, the row sum of the source Jacobian. For a local source this coincides with the diagonal; for the high-order path it is a better choice than the pure diagonal, because it is exact on constant modes ($\mathbf{J}_S\mathbf{1} = \text{diag}(\mathbf{d})\mathbf{1}$ by construction). In other words, the surrogate captures exactly the response to smooth, long-wavelength perturbations, which are the ones that dominate slow convergence.

4. **Better conditioning of the linear solve.** When the reaction is stabilising ($d_i < 0$, the regime of the term $-\beta(V_{\text{m}} - u_3)/\chi C_{\text{m}}$ with $\beta < 0$), the shift $-\theta\,\text{diag}(\mathbf{d})$ adds to the diagonal, increasing the diagonal dominance of the system matrix. GMRES/ILU then converge faster. Picard always solves with $\mathbf{A}$, which gets no such help.

5. **A stronger stopping criterion.** This is a property of the branch rather than of the method, but it is real and it affects how the tables should be read: Picard only checks $R_{V_{\text{m}}}$ and the state residuals, whereas `diagonalIion` additionally requires $R_{\text{cpl}} \le$ `nonlinearTolerance` and $R_{I_{\text{ion}}} \le$ `nonlinearIionTolerance`. With Picard it is possible to leave the loop with small increments but a large coupled residual, which in an MMS study shows up as a nonlinear error floor that masks the order. The converse failure is the one JFNK exhibited, and it is worse because it is silent: leaving the loop with a small coupled residual and increments that were never measured at all (§5.5).

Cost. One extra source evaluation per iteration (the perturbed one) plus the diagonal refresh, that is, of the order of $2\times$ the per-iteration cost of Picard, against $\sim (m+1)\times$ for JFNK with m Krylov iterations. If diagonalIion reduces the number of nonlinear iterations by more than $2\times$ — which is the typical case in stiff regimes — it wins in total time.

When it does NOT win. If Δt is small enough that $\theta \Delta t \max_i |d_i| \ll 1$, Picard already converges in one or two iterations and the extra finite-difference work is pure overhead; §13.3 shows that this is exactly the regime of the present manufactured problem. And diagonalIion is not full Newton: in a strongly nonlinear regime with dominant high-order coupling, JFNK remains the only method with superlinear convergence.

6 Linear-algebra backend and implementation notes

The PDE-level operators $\mathbf{A}$, $\mathbf{B}$, $\mathbf{K}$, and the JFNK preconditioner $\mathbf{P}$ are stored as Eigen `SparseMatrix<scalar,RowMaj` while the heavy linear solves run through PETSc by default. The solver wraps the PETSc API in two small classes that exist in the anonymous namespace of the translation unit:

- `PetscKspMatrixSolver` owns a (matrix, KSP, preconditioner) triple. The first call to `reset()` translates an Eigen `SparseMatrix` into a PETSc `MatSeqAIJ`, builds the `KSP`, and triggers the symbolic ILU(T) factorisation. Subsequent updates call `updateValues()`: this zeros the PETSc `Mat` in place (`MatZeroEntries`), refreshes its numerical entries (`MatSetValue`/`MatAssemblyEnd`), and asks the PETSc preconditioner to recompute its numerical factor while reusing the existing symbolic structure (`KSPSetReusePreconditioner(PETSC_FALSE)` followed by `KSPSetUp`). This means $\mathbf{P}$ can be refreshed many times per time step without freeing and re-allocating the ILU(T) factor.

- `PetscShellKspSolver` wraps a matrix-free shell-matrix used by JFNK. A persistent `Mat` of type `MatShell`, a `KSP`, the right-hand side and solution vectors, and the two callback contexts (`matVec`, `applyPreconditioner`) are allocated once outside the time loop. Each Newton iteration only refreshes the `std::function` payloads of those contexts and calls `KSPSolve`; the PETSc `KSP` object and its work storage persist across solves. Numerically this is identical to constructing a fresh shell matrix per Newton iteration, but it avoids allocating and freeing roughly $40\, m\, N_c$ scalars per JFNK call (with $m =$ `jfnkPetscRestart` and N_c the number of cells).

For users who prefer to stay in pure Eigen, `linearSolverBackend` can be set to `Eigen`. In that mode the voltage solve uses `Eigen::SparseLU` or `Eigen::BiCGSTAB` and the JFNK inner GMRES uses the in-file implementations `solveGMRES` (no preconditioner) or `solveLeftPreconditionedGMRES` with the Eigen `IncompleteLUT`. The two backends are numerically equivalent for the same matrix and tolerances: they differ only in the linear solver implementation, not in the PDE residual.

Every option has exactly *one* accepted spelling, compared case-sensitively, and anything else stops the run at start-up with the list of valid values. This replaced a normalisation layer that lower-cased user input, mapped a collection of aliases (`KSPGMRES`, `JFNK`, `BCGS`, `LU`, `noPC`, `ILUT`, `jakobi`, ...) onto PETSc names, and passed anything else through unchecked. Two spellings for one thing is how a silent wrong answer went unnoticed: with `linearSolver SparseLU` one PETSc path forced a direct solve while the cached Picard solver ran a single ILU application in its place, because the two compared the same word differently.

The two vocabularies are now kept apart. `linearSolver` is backend-neutral, one of `GMRES`, `BiCGSTAB` and `SparseLU`. The PETSc keys take PETSc's own names from closed lists — KSP: `gmres`, `bcgs`, `cg`, `preonly`; PC: `none`, `jacobi`, `ilu`, `lu`, `gamg`, `hypre` — and `petscLinearKspType` defaults to the *translation* of `linearSolver`: `GMRES`→`gmres`, `BiCGSTAB`→`bcgs`, and `SparseLU`→`preonly`, with the PC forced to `lu`. If `petscUseOptions=true`, PETSc's options database can still override solver details through the configured options prefix, which is also how a preconditioner outside the list can be tried.

Right preconditioning (2026-10). In PETSc, GMRES and BiCGStab precondition on the left by default, and their tolerance applies to the *preconditioned* residual. With `ilu` at $\Delta t = 2 \times 10^{-3}$ the two differed by $\sim 4\times$: the linear solve "converged" to 10^{-12} leaving a true residual of 3.66×10^{-12}, and Picard — already under the common test — repeated that same solve 2000 times per step. `PetscKspMatrixSolver` now preconditions on the **right** (as the JFNK shell KSP already did), whose monitored norm is the true one; `cg` monitors the unpreconditioned norm. It is set before `KSPSetFromOptions`. Since October both classes live in `src/monodomainCommon/monodomainPetsc.H`, shared with the electro solver.

6.1 JFNK preconditioner: `diagonalIion` ILU(T)

When `jfnkPreconditioner=diagonalIion`, the inner GMRES applies a left preconditioner $\mathbf{P}^{-1}$ defined implicitly by an ILU(T) factorisation of

$$\mathbf{P} = \mathbf{A} - \theta \, \mathrm{diag}(\mathbf{d}), \qquad d_i = \frac{S_i(\mathsf{x} + \varepsilon \mathbf{1}) - S_i(\mathsf{x})}{\varepsilon}, \tag{54}$$

which is exactly the matrix that the `diagonalIion` nonlinear method would solve. The preconditioner solver type (`jfnkPreconditionerKspType`, `jfnkPreconditionerPcType`) defaults to a direct application of the ILU(T) factor (`preonly+ilu`). The two ILU(T) controls `jfnkPreconditionerDropTolerance` and `jfnkPreconditionerFillFactor` are passed verbatim to PETSc's `PCFactorSetDropTolerance` and `PCFactorSetFill`. The preconditioner is refreshed every `jfnkPreconditionerUpdateFrequency` Newton iterations inside a time step. Across time steps it stays alive through the cached `PetscKspMatrixSolver` above. Selecting `jfnkPreconditioner=diffusion` uses $\mathbf{A}$ as the preconditioner matrix without the diagonal source correction. Selecting `jfnkPreconditioner=none` disables the left preconditioner and runs the matrix-free Krylov solve without the shell preconditioner.

The persistent matrix $\mathbf{P}$ = `jfnkP` is allocated once. The first preconditioner update copies $\mathbf{A}$ into $\mathbf{P}$ (full sparse copy); all subsequent updates only overwrite the value buffer `P.valuePtr()` from `AImplicit.valuePtr()` via `std::copy` and then add the diagonal correction in place. This is correct because the sparsity pattern of $\mathbf{A}$ is invariant in time, and the `addDiagonalToMatrix` helper only touches diagonal entries that already exist in $\mathbf{A}$ (the identity/lumped mass path and the LRE consistent-mass self coefficients provide the diagonal entries used by the correction).

6.2 Chunked assembly of M and K

The high-order stiffness matrix is the dominant transient memory cost of the setup phase. On a 3D unstructured tetrahedral mesh with ∼290k cells, p_3 LRE reconstruction, ∼60-cell stencils, and stabilisation active, the monolithic assembly accumulates of order 4×10^8 Eigen triplets in a single `std::vector<Triplet>`. Because the initial reservation cannot match this exactly without an extra pre-pass, the vector would grow by capacity-doubling, briefly holding both the old and new buffer in memory.

To avoid that transient peak, the solver assembles both $\mathbf{M}$ and $\mathbf{K}$ in fixed-size chunks. For $\mathbf{K}$ the chunks are blocks of `faceChunk = 50 000` internal faces; for $\mathbf{M}$ they are blocks of `cellChunk = 50 000` cells. Each chunk fills a triplet buffer of bounded size, builds a chunk-local sparse matrix $\mathbf{K}_{\mathrm{local}}$ (resp. $\mathbf{M}_{\mathrm{local}}$) via `setFromTriplets`, and folds it into the accumulated matrix:

$$\mathbf{K} \leftarrow \mathbf{K} + \mathbf{K}_{\mathrm{local}}, \qquad \mathbf{M} \leftarrow \mathbf{M} + \mathbf{M}_{\mathrm{local}}, \tag{55}$$

where the first chunk uses assignment instead of accumulation. After the chunked sweep over internal faces, the boundary-face contributions are appended through the same flush mechanism. The triplet buffer is cleared between chunks and explicitly released (`std::vector<Triplet>().swap(triplets)`) at the end of the routine.

For the stiffness matrix, the chunk triplet reservation is controlled by `stiffnessTripletsPerFaceReserve`. With `memoryOptimization=off` this keeps the conservative historical value of 800 triplets per face. With `memoryOptimization=on`, or with `memoryOptimization=auto` on large 3D high-order cases, the

reserve is selected from `memoryOptimizationFluxTripletsPerFace` for flux-only assemblies and from `memoryOptimizationStabilisedTripletsPerFace` when stabilisation is active. This changes only the capacity of the temporary `std::vector`; the set of triplets that pass `addTripletIfNeeded(...)` is unchanged.

For the consistent high-order mass matrix, `compactMassAssembly` can accumulate all quadrature-point contributions to a cell row in a small local buffer before emitting triplets. The compact path therefore emits one triplet per final row/stencil entry instead of one triplet per quadrature contribution. Mathematically it assembles the same mass matrix; only the local floating-point summation order is changed. The first non-empty mass and stiffness chunks are moved into the global Eigen sparse matrix with `swap()`, avoiding an extra full sparse copy during setup.

Numerically the chunked assembly is identical to the monolithic one: sparse matrix addition is associative and commutative, and the per-triplet filter `addTripletIfNeeded(...)` is independent of the chunking because it inspects each triplet value before insertion.

6.3 Memory layout in the time loop

Several sparse matrices and PETSc objects live across the time loop, so the solver is careful about which copies are kept:

- **M** is moved (not copied) into **B** once the implicit matrices are assembled. After this point **M** is empty and only **A** and **B** hold the time-discrete operators.
- In the Picard/`diagonalIion` branches, the modified system matrix `ACurrent` $= \mathbf{A} - \theta\,\mathrm{diag}(\mathbf{d})$ is allocated once outside the corrector loop; subsequent corrector iterations refresh its value buffer from **A** via `std::copy` and re-apply the diagonal correction in place. The branch is also skipped entirely for pure Picard when the PETSc cached solver is used, since that branch never needs a separately stored linearised matrix.
- The Picard solve uses a single `PetscKspMatrixSolver` cached across the run via `reset()` on the first call. Because **A** is time-constant in the Picard branch, no further numerical updates are needed.
- The JFNK preconditioner solver uses the `updateValues()` fast path described above, so the ILU(T) factor is built once symbolically and only its numerical entries are recomputed per refresh.
- The seven ionic quadrature-point scalar fields (`VmOld`, `VmGuess`, source, `Iion`, and three states) are sized with `allocatedIionIntegrationPoints`. This is zero unless high-order ionic integration is actually active in $d > 1$, so pure cell-centred source runs do not pay for unused quadrature-point storage.
- The diagnostic high-order flux/laplacian reconstruction is not evaluated on every time step. The solver always forms the numerical laplacian used in the RHS diagnostic from the already assembled matrix expression $\mathbf{K}\mathsf{V} + \mathsf{b}$. The expensive `computeHighOrderLaplacian` path that fills `fluxVm_HO` is only run when an output time needs the flux field, and once more during final post-processing.
- On glibc systems the solver tightens the allocator before any large allocation: `M_ARENA_MAX` is set to 2 to prevent the OpenMP ODE thread pool from forcing dozens of arenas, `M_MMAP_THRESHOLD` and `M_TRIM_THRESHOLD` are lowered to 64 KiB so that mid-sized Eigen temporaries during the LRE QR factorisations are placed in `mmap`'d regions that are returned to the OS on free. When `memoryOptimizationTrimHeap=true` and the memory optimisation mode is effective, `malloc_trim(0)` is called after the mass-matrix assembly, after the stiffness-matrix assembly, and after **A**/**B** are built; this releases the heap padding left behind by the very large number of small Eigen allocations performed during LRE setup.

- When `profileTimings=true`, setup also prints peak-RSS checkpoints before and after mass assembly, before and after stiffness assembly, and after implicit matrix construction. These checkpoints report the monotone `getrusage(RUSAGE_SELF).ru_maxrss` peak, not the instantaneous RSS.

These choices do not change any numerical output; they only change peak resident memory and the cost of repeated linear solves.

7 Theoretical accuracy and practical error floors

For a smooth manufactured solution and exact nonlinear solves, the intended orders are:

Component	Expected order	Main caveat
standard FV diffusion	second order in space, on orthogonal *and* unstructured meshes	mesh quality enters the constant, not the or
LRE high-order reconstruction	increases with polynomial degree N	stencil quality and quadrature
`backwardEuler`	first order in time	numerical damping
`crankNicolson`	second order in time	requires converged nonlinear/source terms
`Euler` states	first order in time	rarely appropriate for high-order studies
`RK4` states	fourth order for smooth ODEs	fixed step over PDE Δt
`RKF45` states	adaptive fourth/fifth-order pair	tolerance sets error floor
Picard/JFNK/diagonal	nonlinear iteration, not spatial order by itself	residual tolerance must be small enough

The measured MMS order is the order of the entire chain. It can be limited by the largest of the following errors: spatial reconstruction, mass matrix approximation, source quadrature, PDE time stepping, state ODE integration, linear solver tolerance, nonlinear residual tolerance, and floating-point or conditioning effects. In particular, a high-order spatial method will not show its nominal order unless Δt, ODE tolerances, and nonlinear tolerances are sufficiently tight.

8 Solver controls

The numerical controls are read from `constant/spatialIntegrationProperties` and `constant/timeIntegrationProp` the tutorial `run_convergence.py` writes these dictionaries.

Control	Meaning
`useHighOrder_Vm`	Selects standard FV or LRE high-order diffusion/mass reconstruction for V_{m}.
`useHighOrder_Iion`	Selects cell-centred source evaluation or high-order ionic quadrature in $d > 1$. State placement is then controlled by `stateIntegrationMode`.
`stateIntegrationMode` together with `highOrderCoeffs/LRECoeffs_states`	How the states are treated. This is a *separate* control from `useHighOrder_Iion`, which selects only the quadrature of the ionic source. The former name of that key, `useHighOrder_states`, is no longer accepted: a dictionary that still carries it is refused with an error rather than silently ignored.
`implicitScheme`	`backwardEuler` or `crankNicolson`, which set θ, or `BDF1`–`BDF4`. `BDF1` is `backwardEuler` bit for bit.
`bdfStartup`	How the $k-1$ starting values of a BDF-k run are supplied: `ESDIRK3` (default, and the only one that preserves the order), `exact` or `constant`. Inert for the θ family and `BDF1`.

Control	Meaning
`stateODEDriver`	Which polynomial in time drives the state ODEs across a step: `history` (default, through the levels the scheme already stores) or `linear` (the two-node ramp, which caps the order at two).
`manufacturedSolution`	`sines` (default), the smooth non-polynomial solution whose spatial order is the scheme's, or `uniform`, exactly representable so that the error against the exact solution is purely temporal (§4.6).
`massMatrix`	`lumped` identity cell-average mass operator or LRE `consistent` high-order averaging mass.
`highOrderCoeffs/stabilisationAlpha`	Optional positive face-stabilisation strength for the high-order diffusion operator; defaults to zero.
`highOrderCoeffs/LRECoeffs_Vm`	LRE settings `N`, `Nn` and `k` for V_{m} diffusion and the consistent mass matrix. `Nn` is cells ADDED on top of the minimum the degree needs, not the total (§3.3).
`highOrderCoeffs/LRECoeffs_Iion`	Required LRE settings `N`, `Nn` and `k` for ionic quadrature.
`highOrderCoeffs/LRECoeffs_states`	Required only for `stateIntegrationMode=cellCentredReconstruct` with high-order ionic quadrature; controls reconstruction of cell-centred states to ionic quadrature points.
`stateIntegrationMode`	`cellCentredReconstruct` (default, one ODE solve per cell plus state reconstruction) or `gaussPointODE` (legacy, one ODE solve per ionic quadrature point).
`linearSolverBackend`	`PETSc` (default) routes the voltage solve through PETSc; `Eigen` keeps the original in-process Eigen path.
`linearSolver`	Eigen sparse solver name when the Eigen backend is selected: `BiCGSTAB` or `SparseLU`.
`tolerance`, `maxIterations`	Linear PDE solver stopping tolerance and iteration cap.
`petscLinearKspType`, `petscLinearPcType`, `petscLinearRestart`	PETSc KSP type (`gmres`, `bcgs`, `preonly`, ...), preconditioner type (`ilu`, `lu`, `none`, ...) and GMRES restart length for the Picard voltage solve.
`petscLinearOptionsPrefix`, `petscUseOptions`	Options-database prefix and switch that lets the user override KSP/PC settings from the PETSc options database.
`nonlinearMethod`	`Picard`, `diagonalIion`, or `JFNK`.
`nonlinearIterations`	Maximum nonlinear iterations per PDE time step.
`minNonlinearIterations`	Minimum nonlinear iterations before convergence can be accepted.
`nonlinearTolerance`	Coupled residual tolerance for JFNK/diagonal branches.
`nonlinearVmTolerance`	Relative L_2 tolerance for the voltage update.
`nonlinearStatesTolerance`	Relative L_2 tolerance for state updates.
`nonlinearRequireStatesConvergence`	If true, state residuals are part of the JFNK/diagonal stopping test.
`nonlinearAcceptUnconverged`	If false, JFNK/diagonal roll back on nonlinear failure.
`nonlinearIionTolerance`	Relative L_2 tolerance for I_{ion} changes.
`nonlinearRelaxation`	Picard relaxation or initial Newton step length.
`jfnkLinearSolverBackend`	`PETSc` (default) for the persistent shell-Krylov solver or `Eigen` for the in-file GMRES implementations.
`jfnkMaxKrylovIterations`	GMRES restart length m (also the Krylov budget for the matrix-free solve).
`jfnkMaxRestarts`	Number of restarted GMRES cycles.
`jfnkLinearTolerance`	GMRES relative residual tolerance.

Control	Meaning
`jfnkPetscKspType`, `jfnkPetscPcType`, `jfnkPetscRestart`, `jfnkPetscOptionsPrefix`	PETSc KSP/PC settings and options-database prefix used by the persistent shell-Krylov solver.
`jfnkEpsilon`	Scaling of the finite-difference Jacobian-vector perturbation.
`jfnkInitGuessOrder`	JFNK-only initial guess: constant old voltage for order 0, or linear extrapolation from the previous two committed voltages for order ≥ 1.
`jfnkInitGuessVmMin/Max`	Clamp for extrapolated guess and optionally for ODE input.
`jfnkClampODEInput`	If true, clamps voltage passed to state ODE evaluations.
`jfnkLineSearch`, `jfnkLineSearchMaxIter`, `jfnkLineSearchAlphaMin`, `jfnkArmijoC`	Armijo backtracking controls.
`jfnkPreconditioner`	Left preconditioner for the JFNK Krylov solve: `none`, `diagonalIion` or `AImplicit`.
`jfnkPreconditionerKspType`, `jfnkPreconditionerPcType`	KSP and PC types for the preconditioner solve, typically `preonly`+`ilu` to apply the cached ILU(T) factor directly.
`jfnkPreconditionerDropTolerance`, `jfnkPreconditionerFillFactor`	PETSc ILU(T) drop tolerance and fill factor for the JFNK preconditioner factorisation.
`jfnkPreconditionerTolerance`, `jfnkPreconditionerMaxIterations`	Inner KSP tolerance and iteration cap for the preconditioner solve (used by iterative inner solvers; the default `preonly` ignores them).
`jfnkPreconditionerUpdateFrequency`, `jfnkPreconditionerOptionsPrefix`	How often the preconditioner matrix and factor are refreshed (every n Newton iterations) and the PETSc options-database prefix.
`diagonalIionEpsilon`	Finite-difference perturbation for diagonal source derivative.
`stateODESolver`	`RKF45`, `RK4` or `Euler`.
`stateODEInitialStep`, `stateODEAbsTol`, `stateODERelTol`, `stateODEMaxSteps`	Adaptive ODE step and tolerance controls, read only under these names. `timeIntegrationProperties` is not read by this solver. The former names `stateODETolerance` and `stateODEMaxSubSteps` are simply not read: OpenFOAM silently ignores a key nobody looks up, so a dictionary using them runs with the defaults and says nothing.
`stateODEUseOpenMP`, `stateODENumThreads`, `stateODEOpenMPThreshold`	Enables OpenMP parallelisation of the per-cell/per-quadrature-point state ODE integration, sets the thread count and the cell threshold above which the parallel region is entered.
`memoryOptimization`	`off`, `on`, or `auto`. In `auto`, memory optimisations are effective only for 3D high-order V_m runs with at least `memoryOptimizationCellThreshold` cells.
`memoryOptimizationCellThreshold`	Cell-count threshold used by `memoryOptimization=auto`; default 8000.
`memoryOptimizationAdaptiveTripletReserve`	If effective, selects a smaller or stabilised stiffness triplet reserve instead of the conservative 800 triplets per face.
`memoryOptimizationFluxTripletsPerFace`, `memoryOptimizationStabilisedTripletsPerFace`	Triplets-per-face reserve used for high-order stiffness assembly in flux-only and stabilised cases. Values are clamped to at least 16.

Control	Meaning
`memoryOptimizationTrimHeap`	Enables gated `malloc_trim(0)` calls after large setup phases when memory optimisation is effective.
`memoryOptimizationCompactMassAssembly`	Enables compact row-wise triplet emission for the consistent high-order mass matrix when memory optimisation is effective.
`profileTimings`	Records per-phase wall-clock timings (assembly, GMRES, preconditioner setup/apply, sparse solves) for the final summary.
`writeNonlinearResiduals`	Writes per-iteration nonlinear residual diagnostics.

9 Output diagnostics

At the final time, the solver writes cell-centred fields, manufactured exact fields, error fields, reconstructed/quadrature diagnostics when available, and a text summary. The summary includes mesh size, dimension, Δt, time scheme, mass matrix, high-order settings for V_{m} and I_{ion}, selected nonlinear method, relevant tolerances, ODE controls, nonlinear convergence statistics, effective memory-optimisation settings, allocated ionic quadrature-point storage, wall-clock timings, and peak resident memory. These diagnostics are intended to make each convergence point auditable: the numerical method used to generate a data point is recorded alongside the error data.

10 Error Measures and Convergence Rates

The solver writes cell-centred volume-weighted L_1, L_2, and L_∞ errors for V_m, u_1, and u_2. For a generic field $q \in \{V_m, u_1, u_2\}$, let q_i be the numerical value stored at the centre of cell K_i, $q^\star(\mathbf{x}, t)$ the manufactured exact solution, and

$$e_i(q) = q_i - q^\star(\mathbf{C}_i, t), \qquad |\Omega_h| = \sum_i |K_i|. \tag{56}$$

The cell-centred errors written by the solver are

$$E_1^C(q) = \frac{1}{|\Omega_h|} \sum_i |K_i|\, |e_i(q)|, \tag{57}$$

$$E_2^C(q) = \left[\frac{1}{|\Omega_h|} \sum_i |K_i|\, e_i(q)^2 \right]^{1/2}, \tag{58}$$

$$E_\infty^C(q) = \max_i |e_i(q)|. \tag{59}$$

These are volume-weighted mean/RMS errors, not unnormalised integral norms.

The solver also writes reconstructed quadrature errors, denoted by the superscript G. Let $\mathbf{x}_{iq}$ be a quadrature point in cell K_i and let ω_{iq} be its normalised quadrature weight, so that $m_{iq} = |K_i|\omega_{iq}$ is the physical measure associated with that point. For voltage, q_{iq}^G is the LRE/Taylor reconstructed voltage $V_{m,h}(\mathbf{x}_{iq})$; for the states, when high-order ionic integration is active, q_{iq}^G is the state value used at the ionic quadrature point. In the default `cellCentredReconstruct` mode this value is reconstructed from the advanced cell-centred state; in the legacy `gaussPointODE` mode it is the state advanced directly at that quadrature point. The quadrature-point error is

$$e_{iq}^G(q) = q_{iq}^G - q^\star(\mathbf{x}_{iq}, t), \qquad |\Omega_h|_G = \sum_i \sum_q m_{iq}. \tag{60}$$

The corresponding quadrature errors are

$$E_1^G(q) = \frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq} \, |e_{iq}^G(q)|, \tag{61}$$

$$E_2^G(q) = \left[\frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq} \, \left(e_{iq}^G(q)\right)^2 \right]^{1/2}, \tag{62}$$

$$E_\infty^G(q) = \max_{i,q} |e_{iq}^G(q)|. \tag{63}$$

If the corresponding high-order diagnostic is inactive, the report maps the cell-centred errors into the same output columns.

For the quadrature diagnostics, the exact-solution RMS norm and the relative percentage error are

$$\|q^\star\|_{2,G} = \left[\frac{1}{|\Omega_h|_G} \sum_i \sum_q m_{iq} \, (q^\star(\mathbf{x}_{iq}, t))^2 \right]^{1/2}, \qquad E_{\mathrm{rel},2}^G(q) = 100 \, \frac{E_2^G(q)}{\max(\|q^\star\|_{2,G}, \texttt{SMALL})}. \tag{64}$$

Pairwise rates use consecutive mesh sizes,

$$p_{N_1,N_2} = \frac{\log(E_{N_1}/E_{N_2})}{\log(h_{N_1}/h_{N_2})}, \qquad h_N = 1/N. \tag{65}$$

The convergence tables below recompute log–log fits directly from the `errors_vs_N` files. The 2D tables use $N = 10$–100 and $N = 70$–100; the 3D tables use $N = 10$–30 and $N = 20$–30.

11 Distributed-memory parallelisation

This variant leaves the formulation of the previous sections untouched and makes the linear algebra genuinely distributed. Three solvers exist in this family, and only the last one is parallel in the algebraic sense:

Variant	Linear algebra
`...ImplicitPETSc`	Serial mesh, serial LRE, Eigen assembly, PETSc used only as a solver on `PETSC_COMM_SELF`.
`...ImplicitPETScParallel`	Parallel high-order library: the mesh is decomposed and the stencils cross ranks, but the assembly still produces one Eigen matrix per rank and hands the finished CSR to PETSc on `PETSC_COMM_SELF`. Serial by construction.
`...ImplicitPETScDistributed`	Operators become `MPIAIJ` matrices on `PETSC_COMM_WORLD` and every matrix–vector product goes through `MatMult`.

Because this solver's purpose is verification, the requirement is sharper than "it runs": at a fixed manufactured problem the discrete operator must not depend on how the mesh was partitioned. Sections 11.2 and 11.3 are about establishing exactly that, and Section 11.4 about the one place where it still fails.

11.1 Matrix layout: local rows, global columns

Each rank owns the rows of its own cells and nothing else, while columns are *global* cell indices taken from the `globalIndex` that the high-order library builds alongside its stencils. With n_{loc} the local cell count and n_{glob} the total, every assembled operator has shape $n_{\mathrm{loc}} \times n_{\mathrm{glob}}$ and this rank's row offset is

$$\texttt{gRowStart} \;=\; \texttt{globalIndex::localStart()}.$$

That is PETSc's `MPIAIJ` layout, so `buildDistributedMat` only has to offset the row index and split the preallocation into the diagonal block (columns this rank owns) and the off-diagonal one. In serial `gRowStart` $= 0$ and $n_{\mathrm{glob}} = n_{\mathrm{loc}}$, so the same code reproduces the sequential matrix exactly.

The trap that follows from it. *A local index used where a global one is expected is invisible in serial.* In this solver the whole orthogonal assembly (`assembleStandardOrthogonalStiffnessMatrix`, the `useHighOrder_Vm = false` path) wrote local cell indices as columns for its two-point fluxes while `addCellGradientDotCoeffs` wrote global ones for its stabilisation terms, and produced a square $n_{\text{loc}} \times n_{\text{loc}}$ matrix. Both are correct at one rank and neither can be assembled correctly at more than one. Four further instances of the same family were found in the physiological sibling.

11.2 Stabilisation across processor faces

The face-jump stabilisation of Section 3.3 is the only term that needs the *neighbouring cell's entire reconstruction row* — up to ~ 60 (global column, coefficient) pairs for a $p3$ stencil in 3-D. No field swap can carry that, and for a while the term was simply omitted on coupled faces: a processor patch passed the empty/zeroGradient filter, contributed its flux, and fell through the `fixedValue` test that gated the stabilisation.

Mechanism. The rank that *owns* the far cell builds the row itself and ships the pairs (`exchangeCoupledFaceRows`, one `PstreamBuffers` round per assembly):

- high-order operator: `buildCellTaylorExtrapolationRow`, evaluated at the face centre, a point both ranks have;
- orthogonal operator: `buildCellGradientDotRow`, evaluated at *minus* the sending rank's `patch.delta()`, so the row arrives already expressed in the receiving rank's orientation.

Nothing else is exchanged: no cell centres, no stencils, no coefficient tables. In particular `mesh.C()` — a `slicedVolVectorField` whose coupled values are valid only after an evaluation the assembly does not perform — is never touched; $\mathbf{d}_{PN}$ comes from `patch.delta()` and the far conductivity from `syncTools::swapBoundaryCellList`, since a processor patch field holds the interpolated *face* value rather than the neighbouring *cell* value.

Only the local row is written on each side. Writing the jump from the point of view of the local cell,

$$\frac{a}{V_c}\Big(T_{\text{far}}(\mathbf{x}_f) - T_c(\mathbf{x}_f)\Big),$$

with T_c the Taylor reconstruction of cell c, removes the owner/neighbour convention entirely: the $+$ the internal-face loop writes into the owner row and the $-$ it writes into the neighbour row are the same expression seen from either cell. The face coefficient a enters only through $|\mathbf{d}_{PN}\cdot\mathbf{n}|$ and $\mathbf{n}\cdot\mathbf{D}\cdot\mathbf{n}$, both orientation-independent, so the two ranks compute the same a without agreeing on anything. The Taylor coefficient of a single stencil entry is factored into `cellTaylorEntryCoeff` so that the shipped row and the locally assembled one cannot drift apart.

Acceptance. Deviation of $\|V_{\text{m}}\|_{L_2}$ between one and two ranks at $\alpha = 0.1$, against the $\alpha = 0$ control (`diagonalIion`; the other two nonlinear methods agree to the digits shown, which is what identified the defect as spatial rather than algebraic):

dim	mesh	before	after	$\alpha = 0$
2D	hexahedral	3.34e-05	4.55e-06	2.40e-09
2D	triangular	1.99e-04	1.30e-07	2.04e-07
3D	hexahedral	6.80e-03	7.38e-10	1.81e-10
3D	triangular	7.40e-04	9.55e-04	9.70e-04

Serial results are unchanged to every digit. Three of the four rows collapse onto their $\alpha = 0$ control, which is the acceptance criterion. The two that do not are explained below and neither is the stabilisation: the 2-D hexahedral residual is preconditioner path dependence (Section 11.3) and the 3-D triangular control is itself wrong (Section 11.4).

11.3 Verifying a parallel operator

Two checks are cheap and neither is sufficient alone:

- **Row sums.** With no-flux boundaries every row of **K** must sum to zero, since a constant field has zero gradient. This caught a `CompactListList::operator[]` that returns a view by value, which compiled but yielded an empty stencil and hence an empty operator. It cannot see an *omitted* face: omitting one contributes nothing to either of its rows, so the sums still vanish — which is precisely why the missing stabilisation of Section 11.2 survived.
- **Operator fingerprint** (global nnz, $\sum a_{ij}$, $\sum |a_{ij}|$, $\max_i |\sum_j a_{ij}|$). The stencils are geometric, so the global nnz must match the serial value exactly — but a value written into the *wrong column* changes none of these.

The check that decides is an entry-by-entry comparison. With `CF_DUMP_K` set in the environment the solver writes **K** as (global row, global column, value), one file per rank; mapping the parallel indices back through `processor`N`/constant/polyMesh/cellProcAddressing` gives a direct diff against the serial assembly. For 2-D hexahedra at $\alpha = 0.1$ the fixed high-order operator matches serial **exactly**: 47012 nonzeros on both sides, identical sparsity, no entry differing by more than 10^{-12} relative. `CF_DUMP_STENCILS` does the same one level lower, dumping stencil *membership* and cell centres, which separates a differing coefficient from a differing stencil.

Comparing iteration by iteration. Set `petscLinearPcType` to `jacobi`. PETSc wraps `ilu` and `lu` in block-Jacobi in parallel, so those preconditioners are partition-dependent *by construction*. That is the whole of the 4.55×10^{-6} left in the 2-D hexahedral row above: with `jacobi` the same case agrees between one and two ranks to all thirteen printed digits, deviation exactly zero, on a run whose error norm is 2.7×10^{-5} to begin with.

11.4 Partition dependence of the MLS stencils

Note (2026-10). Much of what follows came from the case and not from the library: the `Nn` key was written as a total when it is the number of cells added on top of the minimum, and the stencils came out 50 % larger than the halo (Section 4.7). With the correct size, in 2-D, M11 measures 0 differing stencils and M15 drops to 6.953×10^{-15}. The 3-D tetrahedral measurements in this subsection predate that fix and were not re-run.

One genuine inconsistency remains. It is independent of the stabilisation — it is present at $\alpha = 0$ — and it lives in the stencil selection of the high-order library rather than in this solver.

Mechanism. A stencil is meant to be *the N cells whose centres are nearest the face centre*. The library reaches that in two steps: it grows complete layers of `cellCells` until the candidate pool holds $\lceil 1.5N \rceil$ cells, then keeps the nearest N of the pool by Euclidean distance. The second step is geometric; **the first is topological**. On a tetrahedral mesh "within k hops" and "within radius r" are different sets, and with only $0.5N$ of slack the last positions of the stencil are decided by the connectivity rather than by the geometry.

Under decomposition the layer growth stops at the cut and the far side is supplied instead by a query returning every cell of the neighbouring rank inside a bounding box — a *geometric* set, and a generous one. Near a partition the selection therefore mixes a topological candidate set on one side with a geometric one on the other. This is why the deviation does not respond to `haloDepthScale`: the box was never the binding constraint.

Measurements (3-D unstructured tetrahedra, 4971 cells, one rank against two, $\alpha = 0$):

Observable	Value
Face stencils that differ	536 of 10676
Their size	identical (80): members are *substituted*, not lost
Position of the swapped members	98 % in the last 10 % of the distance-ordered stencil
Rows of **K** that differ	552 of 4971, up to 4.17 against $\max\|K\| = 256$
Cut faces whose two ranks disagree with *each other*	9
Cell centres, serial vs decomposed	up to 3.3×10^{-16} (the centroid sum runs over a renumbered face list)

The nine cut faces are a defect on their own terms: the two rows of one face are then built from different flux stencils, so the flux leaving a cell is not the flux entering its neighbour and the scheme is not conservative there.

Effect on the verification. Comparing the V_m *fields* cell by cell rather than the deviation of a norm, at $\alpha = 0.1$ with `jacobi`:

$$\frac{\left\| V_\mathrm{m}^{np=1} - V_\mathrm{m}^{np=2} \right\|}{\| V_\mathrm{m} - V_\mathrm{m}^{\mathrm{exact}} \|} = \frac{1.51 \times 10^{-5}}{6.84 \times 10^{-4}} = 2.2\,\%, \qquad \frac{\|\Delta V_\mathrm{m}\|}{\text{amplitude}} = 7.5 \times 10^{-6}.$$

Over a refinement sweep the deviation stays at the 10^{-4}–10^{-3} level while the error itself falls by a factor of five, i.e. the perturbation shrinks at the same rate as the error:

N	$\|V_\mathrm{m}\|_{L_2}$, $np = 1$	$np = 2$	rel. deviation
8	1.187049e-03	1.186528e-03	4.39e-04
10	6.429373e-04	6.423235e-04	9.55e-04
12	3.975660e-04	3.975437e-04	5.60e-05
14	2.275209e-04	2.273524e-04	7.41e-04

The fitted convergence order is therefore untouched:

$$p_{V_\mathrm{m},L_2}^{np=1} = 2.90722821, \qquad p_{V_\mathrm{m},L_2}^{np=2} = 2.90722635,$$

a difference of 1.9×10^{-6} against an acceptance tolerance of 0.05; every other metric agrees to 10^{-5} or better. That is what two *consistent* schemes of the same formal order predict — their difference is a same-order perturbation, not a systematic error — and it is the statement that matters for a verification study: the convergence results of Section 13 are not affected by the decomposition.

How it would be fixed.

1. Make the candidate pool geometric on both sides: query cell centres by radius (`indexedOctree`) instead of growing `cellCells` layers, with the radius taken from the layer estimate as an upper bound. The stencil then becomes a function of (face centre, cell centres, N) alone, invariant under partitioning.

2. Stop the tie clipping: the parallel branch filters candidates by radius *before* the equal-distance expansion runs, so a tie group straddling that radius is cut in half (17 members in serial against 16 at two ranks, on a structured 2-D case at $p1$). Inflating the radius by $(1 + \texttt{relTol})$ and routing the serial early return through the same selection code closes it.

3. Assert completeness instead of failing silently: check that the N-th distance lies strictly inside the queried radius.

4. One face, one stencil: let the lower-ranked side of a cut face own the stencil and send it, restoring conservation across the cut.

`relTol` (10^{-8}), `overSampleFactor` (1.5) and `searchExpansionFactor` (1.3) are hard-coded default arguments rather than dictionary entries, so none of this can be explored from a case: it is a library change.

A caveat before touching it. The serial run is the truncated one — its candidate pool is topological everywhere, not only near a cut — so fixing the selection changes serial results too, and with them every convergence table in Section 13. That is a decision to take deliberately, not a side effect to absorb.

11.5 Verification cases: what was run and where it lives

Every case below sits under `tutorials_electro/mmsTestParallelVerification/`. The driver itself is `tutorials_electro/mmsDistValidation/`; the directories here are copies of it with one setting changed each, so that a sweep never overwrites another.

Directory	Configuration	What it establishes
`matrix_mms_2D_np1`, `..._np2`	$N = 30$, $\alpha = 0.1$	the matrix at $\alpha > 0$
`matrix_mms_3D_np1`, `..._np2`	$N = 10$, $\alpha = 0.1$	idem, 3-D
`alpha0_mms_2D_np1`, `..._np2`	$N = 30$, $\alpha = 0$	the control
`alpha0_mms_3D_np1`, `..._np2`	$N = 10$, $\alpha = 0$	idem, 3-D
`halo_3D_np1`, `halo_3D_np2`	`haloDepthScale` $7.5 \to 16$	falsifies the halo hypothesis
`mmsJFNK_np1`, `_np2`, `_np4`	early JFNK attempt	superseded by the matrix

Each `matrix_*` and `alpha0_*` directory holds three results trees, `res_Picard`, `res_JFNK` and `res_diagonalIion`, because the driver takes a single nonlinear method per invocation. Both meshes (`hexa` and `triangular_Unstr`) are swept inside each run. The triad is $(p3, \mathrm{CCp3}, p3)$ throughout, with $\Delta t = 10^{-3}$.

Result. All 24 runs at $\alpha = 0.1$ and all 24 at $\alpha = 0$ completed successfully, which is what establishes that JFNK and diagonalIion work under MPI in 2-D and 3-D on both mesh types — until this sweep every parallel run had used Picard.

The deviation of $\|Vm\|_{L2}$ between $np = 1$ and $np = 2$, before the coupled-face stabilisation was implemented:

Dim	Mesh	$\alpha = 0.1$	$\alpha = 0$	Ratio
2-D	hexa	3.34×10^{-5}	2.40×10^{-9}	$13\,900\times$
2-D	triangular	1.99×10^{-4}	2.03×10^{-7}	$980\times$
3-D	hexa	6.80×10^{-3}	1.81×10^{-10}	3.8×10^{7}
3-D	triangular	7.40×10^{-4}	9.70×10^{-4}	$1\times$

The three nonlinear methods gave almost identical deviations for a given mesh — 6.80×10^{-3} for all three on 3-D hexa — which is what identified this as a spatial-discretisation defect rather than a solver one: a bug in the nonlinear iteration would differ from method to method.

The case that does not collapse. 3-D unstructured stays at 9.70×10^{-4} with the stabilisation switched off, so the missing term does not explain it. Two hypotheses were tested and both failed: the meshes are byte-identical between the two runs (same `.msh` and `polyMesh` checksums, 4971 cells), and raising `haloDepthScale` from 7.5 to 16 (`halo_3D_np1`, `halo_3D_np2`) reproduces 9.70×10^{-4} exactly. The cause is the stencil selection in solids4foam's `movingLeastSquaresStencil`, not this solver; see the discussion in `CHANGES-cardiacFoam.md`.

12 October 2026 changes and new results

This section collects what changed in the solver between 2026-09-29 and 2026-10-18 and what measures it in the suite (`run/tutorials_electro/mmsTestParallelVerification_2D/`). The earlier sections have already been corrected where they had become false. The results of Section 13 **predate** all of this.

12.1 Code shared with the electro solver

What this solver shares with `highOrderElectroActivationFoamImplicitPETScDistributed` moved to headers in `src/monodomainCommon/`:

Header	Contents
`monodomainBase.H`	global row offset, option validation, nonlinear stopping test
`monodomainAssembly.H`	assembly of $\mathbf{M}$ and $\mathbf{K}$ (orthogonal and high order)
`monodomainPetsc.H`	distributed matrices, `PetscKspMatrixSolver`, `PetscShellKspSolver`, KSP/PC selection
`monodomainKrylov.H`	in-house GMRES and JFNK
`monodomainTimeSchemes.H`	θ/BDF coefficients, ESDIRK3 tableau, V_{m} driver
`monodomainReconstruction.H`	reconstructions to the Gauss points
`openMPControls.H`, `jfnkPreconditionerControls.H`	OpenMP keys and JFNK preconditioner keys

They are not a library: they are included *inside* the anonymous namespace of each `.C`, so each executable compiles its own copy and a change there changes both solvers. What is specific to the MMS stays in the `.C`: the manufactured solution and its source, the three-state model and its ODEs, and the time loop. The code walkthrough (`codigo_*.pdf`, generated by `mkcodereport.py`) looks up each function in the `.C` and in those headers.

The shared reconstruction does the K fields in a single pass over the stencil. The first shared version did one field per call and left this solver's state reconstruction 30 % slower (3-D M18: 3.54 s $\rightarrow$ 4.61 s); with the single pass it is back to the previous cost, bit-identical to reconstructing one field at a time. OpenMP now also splits those reconstructions, and `Eigen::setNbThreads` ties Eigen's products to the same `stateODEUseOpenMP` key.

12.2 One stopping test, with a floor exit

Picard used to stop on the V_{m} and state increments alone, while JFNK and `diagonalIion` also required the coupled residual: a method comparison (M08) partly compared stopping rules. The three methods now call `nonlinearIterationExit`:

1. unless $k{+}1 \geq$ `minNonlinearIterations`, $R_{V_{\mathrm{m}}}$, $R_{I_{\mathrm{ion}}}$ and (with `nonlinearRequireStatesConvergence`) $R_{\mathbf{u}}$ pass: continue;
2. if $R_{\mathrm{cpl}} \leq$ `nonlinearTolerance`: converged;
3. if R_{cpl} did not fall since the previous iteration: **floor**, stop, count it and report it in the log;
4. otherwise continue.

The floor exists because the coupled residual can stall above the tolerance — the true residual a preconditioned Krylov solve leaves, or round-off on a large stiff system — and then every iteration repeats the same solve up to the cap. It was found here: Picard, `ilu`, $\Delta t = 2 \times 10^{-3}$, 2000 identical iterations per step with $R_{\mathrm{cpl}} = 3.66 \times 10^{-12}$ against 10^{-12}. The cause of that case was the preconditioning side (Section 6), but the floor stays as a safety net. The increment tests are never loosened, so the worst case is the old Picard criterion.

12.3 The V_{m} history at the Gauss points

With `gaussPointODE` the ODE is integrated at every quadrature point and the driver was always the two-node ramp, which caps BDF3/BDF4. The `stateODEGaussPointHistory` key (default `false` in the solver, `True` in the suite harness) gives it the same polynomial as the cell path without storing anything between steps: at the start of every step the BDF history levels — or, during startup, the accepted ESDIRK stage values — are reconstructed at the Gauss points with the same operator that gives V_{m}^n there (`reconstructVmLevelAtIionIntegrationPoints`). Within the step $k-1$ temporary arrays of Gauss-point size live. It only acts with `stateODEDriver history` and high-order I_{ion} in 2-D or 3-D. Its cost, measured in the electro solver (E36, BDF4), is +8.9 % of run time on GP with no measurable memory change.

A stage-step error in the ESDIRK stages, fixed along the way. On the same path, `updateIntegrationPointStatesODE` received `dt` instead of `stageDt`: inside an ESDIRK stage the states were integrated over the whole step and not over $c_i\,\Delta t$ as on the cell path. It only affected the BDF startup with `gaussPointODE`.

12.4 New results: M02 with GP

M02 (2-D hexa $N = 40$, triad $p3$, $\alpha = 0$, Picard, $t_{\mathrm{end}} = 0.512$, ESDIRK3 startup) now has the GP path next to CC. Judged order of the finest valid pair, route A (Cauchy, sines MMS) / route B (error against the analytical solution, uniform MMS):

Scheme	nominal	CC	GP with history	GP without history (B)
`backwardEuler`, `BDF1`	1	1.000 / 1.000	1.000 / 1.000	—
`crankNicolson`	2	1.998 / 2.000	2.000 / 2.000	—
`BDF2`	2	1.979 / 1.992	1.979 / 1.992	1.99
`BDF3`	3	2.932 / 2.971	2.932 / 2.971	2.90
`BDF4`	4	3.891 / 3.908	3.891 / 3.908	floor $\sim 2 \times 10^{-10}$

The last column is the contrast without the key (linear driver at the Gauss points), with no verdict: the BDF4 pairs give 3.89, 7.51, −1.59 and 1.63, i.e. the error stops falling at $\sim 2 \times 10^{-10}$. With the key, GP reaches the same order as CC by both routes. M02 is PASS. The CC figures differ slightly from the September ones (BDF4 route B 3.941 → 3.908, CN route A 1.994 → 1.998) because of the changes in this section; the tables in the time-integration sections have already been updated.

13 Results

Note (2026-10). The tables in this section were generated with the solver as it was before the changes of Section 12 (Picard's own stopping rule, left-preconditioned KSP, no history at the Gauss points). They are kept as a record and were not re-run.

This section collects the current convergence tables generated from `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/`. All result blocks in this section use `crankNicolson` time integration, the `consistent` mass matrix, and `RKF45` for the state ODEs. The tables are split by dimension, nonlinear method, mesh family, stabilisation parameter, and time step. Their first column records the actual `Vm-states-Iion` triplet used in each run.

The complete sweep comprises 139 776 runs: two dimensions, three nonlinear methods, two time steps, two mesh families, two values of α, 52 `Vm-states-Iion` triplets, and 91 mesh sizes in 2D ($N = 10$–100) or 21 in 3D ($N = 10$–30).

13.1 Rollback audit and nonlinear robustness

As discussed in §5.6, a time step that exhausts the nonlinear iteration budget is neither retried nor resolved with a smaller Δt: for `diagonalIion` and `JFNK` with `nonlinearAcceptUnconverged=false` the fields are reverted to their start-of-step values while the clock advances anyway, and for `Picard` the last iterate is accepted with a warning. In both cases the affected step introduces an $O(1)$ consistency error that is *not* recovered by later steps, so a single such event invalidates the convergence rate of that run. The rates reported in this section are therefore only interpretable once it has been verified that no step was left unconverged.

That verification was carried out by walking the `postProcessing/.../nonlinearResiduals_N*_dt_*.dat` files of all 139 776 runs and grouping their rows by time step: a step counts as converged if any of its iterations has the `converged` flag set to 1, which is exactly the condition under which the solver breaks out of the corrector loop. The outcome is summarised in Table 2 and broken down, row by row, in the `RB` column of every table in the following sections.

Table 2: Global nonlinear-convergence audit over the complete `example0` sweep. `RB` counts the time steps that exhausted the `nonlinearIterations`= 2000 budget (and that would therefore have been rolled back for `diagonalIion`/`JFNK`, or accepted with a warning for `Picard`). $\bar{n}_{\mathrm{nl}}$ and $n_{\mathrm{nl}}^{\max}$ are the mean and maximum number of nonlinear iterations per step; Σ_{LS} is the total number of Armijo line-search backtracks.

Dim.	Method	Δt	Runs	Steps	RB	$\bar{n}_{\mathrm{nl}}$	$n_{\mathrm{nl}}^{\max}$	Σ_{LS}
3D	`Picard`	10^{-2}	4368	21840	**0**	4.012	5	0
3D	`Picard`	10^{-3}	4368	218400	**0**	3.000	3	0
3D	`diagonalIion`	10^{-2}	4368	21840	**0**	3.002	4	0
3D	`diagonalIion`	10^{-3}	4368	218400	**0**	2.153	3	0
3D	`JFNK`	10^{-2}	4368	21840	**0**	2.317	3	0
3D	`JFNK`	10^{-3}	4368	218400	**0**	2.009	3	0
Total			26208	720720	**0**			

The result is categorical: **zero rollbacks in** 3 843 840 **time steps**. None of the convergence rates reported below is contaminated by a frozen step or an unconverged iterate. Moreover the margin is not marginal but three orders of magnitude: the configured budget is `nonlinearIterations`= 2000 and the maximum observed over the whole sweep is 5 iterations.

The iteration statistics of Table 2 admit a direct quantitative reading in terms of the analysis of §5.2 and §5.7. For this manufactured solution, with $u_3 = 0$, $u_1 + u_3 - V_{\mathrm{m}} = (1+t)G$ and $u_2^2 = [(1+t)^2 G]^{-1}$, the local source sensitivity is

$$d = \frac{\partial S}{\partial V_{\mathrm{m}}} = \frac{1}{2}\left[\frac{1}{1+t} - \frac{F}{(1+t)^{3/2}G}\right] - \frac{\beta}{\chi C_{\mathrm{m}}}, \tag{66}$$

which with $\mathbf{D} = \mathrm{diag}(0.1115, 0.1216, 0.0304)$ and $\chi C_{\mathrm{m}} = 6$ gives $\max_i |d_i| \approx 2.0$ in 2D and ≈ 2.4 in 3D. The Picard contraction factor predicted by (45) is then

$$\rho_{\mathrm{Picard}} \approx \omega\,\theta\,\Delta t \max_i |d_i| \approx \begin{cases} 1.0 \times 10^{-2}, & \Delta t = 10^{-2}, \\ 1.0 \times 10^{-3}, & \Delta t = 10^{-3}, \end{cases} \tag{67}$$

that is, two to three orders of magnitude below the divergence threshold $\rho = 1$. Each Picard correction gains between two and three decimal digits, and with an initial residual of $\sim 5 \times 10^{-4}$ and tolerances of 10^{-8} three or four iterations are needed. That is exactly what the data show: Picard averages 4.00 iterations per step with $\Delta t = 10^{-2}$ and 3.00 with $\Delta t = 10^{-3}$, and `diagonalIion` drops from 3.00 to 2.00 over the same change of step. The floor of 2 is structural, not numerical: the convergence criterion is evaluated *after* the update, so even a step that converges in a single correction needs a second iteration to confirm it.

Three observations follow that are worth keeping in mind when reading the tables.

- **The nonlinear iteration count is independent of the spatial discretisation.** Within a given (dimension, method, Δt) combination, the 52 `Vm-states-Iion` triplets give exactly the same n_{nl}: for instance 4.00/4 for all 52 triplets of 2D `Picard` with $\Delta t = 10^{-2}$. This is the empirical confirmation that the convergence rate of the coupling is governed by $\theta \Delta t \max_i |d_i|$, that is by the reaction term, and not by the diffusion operator or the reconstruction order.
- **The Armijo line search never engaged.** The Σ_{LS} column of Table 2 is identically zero in the six `JFNK` rows: across the 46 592 JFNK runs of the sweep, the full Newton step always satisfied condition (53) on the first try, so $\alpha = \texttt{nonlinearRelaxation} = 1$ in every case. The globalisation is present as a safeguard but does not intervene in this regime.
- **The linear-solve cost *decreases* as the spatial order rises.** The maximum number of GMRES iterations per correction, with $\Delta t = 10^{-2}$ and $\alpha = 0$, goes from 10 (`NO`) to 5, 5, and 4 for `p1`, `p2`, and `p3` on the 2D hexahedral mesh, and from 24 to 11, 9, and 8 on the unstructured triangular one. The high-order systems have a wider stencil but are not harder for PETSc's ILU(T) to solve; the extra cost of high order lies in the assembly and in the source evaluation, not in the linear algebra.

13.2 Two-dimensional manufactured problem

The tables in this subsection are generated directly from the current `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0` result tree. The method-specific roots are `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/2D_CN_cons_Picard_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/2D_CN_cons_diagonalIion_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/2D_CN_cons_JFNK_RKF45`. All entries use `crankNicolson`, the `consistent` mass matrix, and `RKF45` for the state ODEs. The first column reports `Vm-states-Iion`: the V_m reconstruction degree, the state treatment (`CCp*`, `GP`, or `na`), and the I_{ion} quadrature/reconstruction setting. Fits use $N = 10$–100 for the broad mesh range and $N = 70$–100 for the fine-mesh range. The last two columns audit the nonlinear solver: `RB` counts rolled-back steps and n_{nl} summarises the nonlinear iterations per step (median/maximum).

13.2.1 `Picard`

$\Delta t = 10^{-2}$

Table 3: 2D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 0 mesh sizes of the row (0 time steps per row in total). `Picard` accepts its last iterate with a warning instead of rolling back, so for this method the column counts steps accepted without converging. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–100						Fit $N = 70$–100								
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
No aggregated error file found for this case.																		

$\Delta t = 10^{-3}$

Table 4: 2D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 0 mesh sizes of the row (0 time steps per row in total). `Picard` accepts its last iterate with a warning instead of rolling back, so for this method the column counts steps accepted without converging. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–100						Fit $N = 70$–100								
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
No aggregated error file found for this case.																		

13.2.2 `diagonalIion`

$\Delta t = 10^{-2}$

Table 5: 2D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 0 mesh sizes of the row (0 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–100						Fit $N = 70$–100								
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
No aggregated error file found for this case.																		

$\Delta t = 10^{-3}$

Table 6: 2D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 0 mesh sizes of the row (0 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–100						Fit $N = 70$–100								
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
No aggregated error file found for this case.																		

13.2.3 `JFNK`

$\Delta t = 10^{-2}$

Table 7: 2D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 0 mesh sizes of the row (0 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–100						Fit $N = 70$–100								
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
No aggregated error file found for this case.																		

$\Delta t = 10^{-3}$

Table 8: 2D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–100; the second over $N = 70$–100. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 0 mesh sizes of the row (0 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–100						Fit $N = 70$–100								
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
No aggregated error file found for this case.																		

13.3 Discussion of the 2D results

The three nonlinear methods are numerically indistinguishable. Comparing the 4 992 (configuration, metric) pairs for which all three methods exist, the maximum spread of the fitted order between `Picard`, `diagonalIion`, and `JFNK` is 0.0030, the median is 8×10^{-7}, and *no* case exceeds 0.01. This is what one expects and is itself a verification result: the three methods solve the same discrete nonlinear equation down to tolerances of 10^{-8}, so they can only differ below that tolerance. The choice between them is a decision about cost and robustness, never about accuracy.

In this regime Picard is the cheapest, despite needing more iterations. The median wall-clock ratio over the 208 identical configurations is $t_{\texttt{diagonalIion}}/t_{\texttt{Picard}} = 1.16$ and $t_{\texttt{JFNK}}/t_{\texttt{Picard}} = 1.12$ with $\Delta t = 10^{-2}$, and 1.39 and 1.27 with $\Delta t = 10^{-3}$. The reason is the one noted in §6.3: in the Picard branch $\mathbf{A}$ is constant in time, so the cached `PetscKspMatrixSolver` factorises once for the whole run, whereas `diagonalIion` must refresh the diagonal and redo the numeric ILU(T) factorisation at every correction and additionally pay one extra source evaluation for the finite difference. With $\rho_{\mathrm{Picard}} \approx 10^{-2}$, Picard's additional iteration costs less than that per-iteration overhead. The theoretical advantage of `diagonalIion` described in §5.7 is real but manifests itself in the stiff regime $\theta \Delta t \max_i |d_i| \gtrsim 1$, which this manufactured problem never reaches; with a physiological ionic model, where $\partial I_{\mathrm{ion}}/\partial V_{\mathrm{m}}$ is several orders larger during the depolarisation front, the balance reverses.

The cheapest configuration for a given order is not the diagonal one. Plotting every run of the sweep as a point in (cost, error) makes the question "which triad should one actually use" answerable rather than a matter of symmetry. Figure 1 does that for the whole triad space, on both mesh families and both stabilisation settings. The natural choice $(p3, \mathrm{CC}p3, p3)$ — the same level everywhere — never appears on the Pareto front: $(p3, \mathrm{CC}p2, p2)$ reaches the same order 3.72 in 234 s

against 375 s, i.e. 60 % cheaper for an identical result, because the ionic quadrature and the state reconstruction stop paying once V_m is the binding link. The `gaussPointODE` family is dominated on both axes at once.

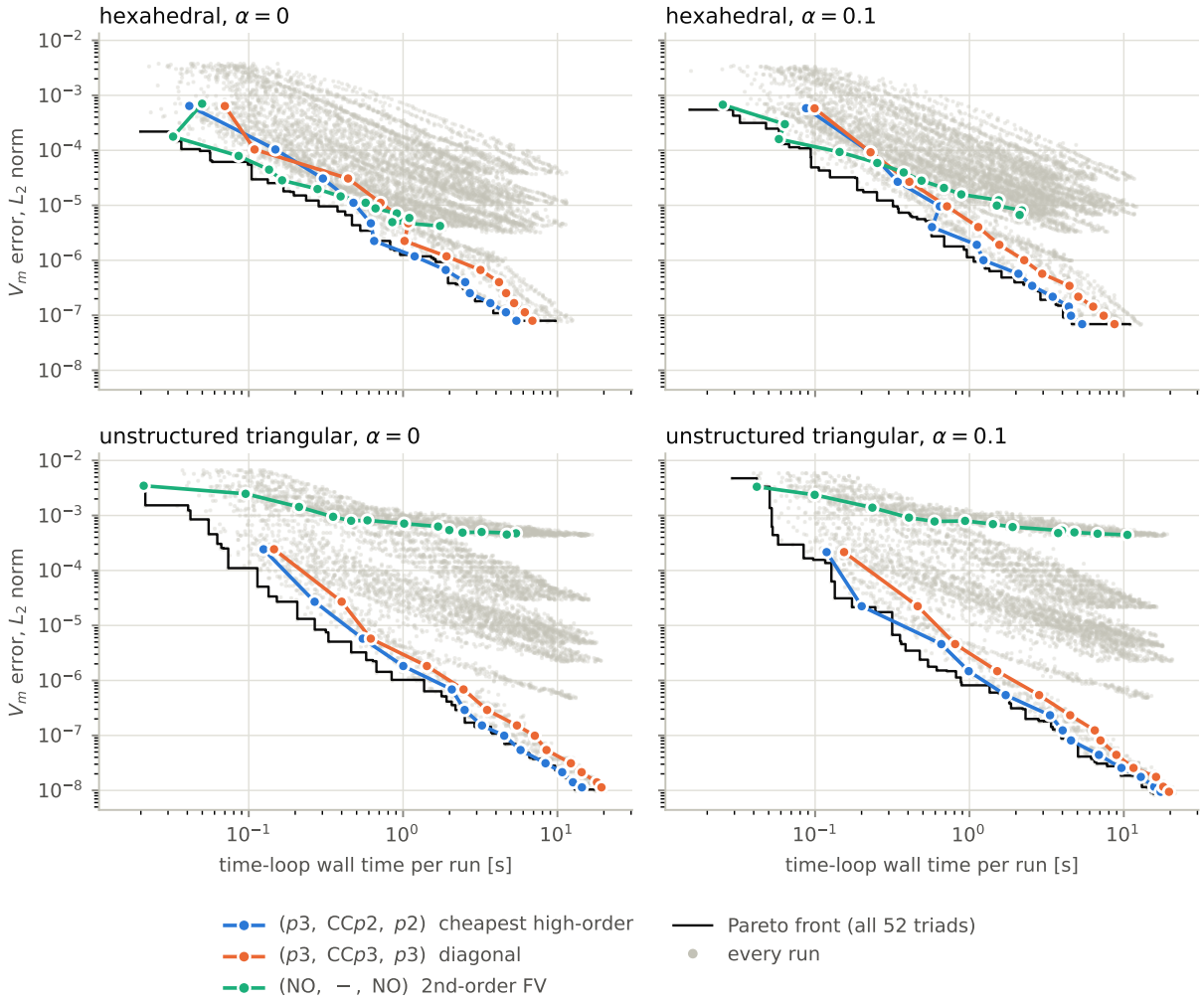


Figure 1: Accuracy–cost Pareto front over the $(V_\mathrm{m}, \text{states}, I_\mathrm{ion})$ triad space, from the `np` = 4 sweep with Picard and $\Delta t = 10^{-3}$, $N = 10$–130. Each faint dot is one run; the black step curve is the non-dominated front; the three coloured curves are mesh-refinement paths for the cheapest high-order triad $(p3, \mathrm{CC}p2, p2)$, the diagonal $(p3, \mathrm{CC}p3, p3)$, and second-order FV. The diagonal triad lies above the front in all four panels. Generated by `figures/fig_pareto_triads.py` from `figures/pareto_2d_picard_dt1e-3.csv`.

The 3-D sweep, and what it replicates. The 3-D study is a sweep of its own: seven triads on hexahedral and unstructured tetrahedral meshes, $N = 10$–30, BDF4 with the ESDIRK3 startup and $\Delta t = 10^{-3}$. Its orders, and the 2-D ones, are Figure 2. Two results carry over from 2-D unchanged. The loss of consistency of the *uncorrected* two-point flux on simplices is *worse* in 3-D — `NO` fits 0.65 on tetrahedra against 0.77 on triangles and 1.96–1.98 on hexahedra; this sweep predates the non-orthogonal correction, so the 3-D numbers need re-running before they are quoted as the operator's behaviour — and the cheap triad still matches the diagonal, 3.383 for $(p3, \mathrm{CC}p2, p2)$ against 3.381 for $(p3, \mathrm{CC}p3, p3)$.

One number must be read with care: $p3$ fits 3.38 in 3-D against 3.72–4.32 in 2-D, and that is *not* temporal contamination. Halving Δt on the finest 3-D mesh moves the error in the seventh digit ($3.9486606 \times 10^{-5} \to 3.9486604 \times 10^{-5}$ on hexa), so the temporal error is $\sim 10^{-11}$ against a spatial

error of $\sim 3 \times 10^{-5}$. The pairwise orders say the two families are in different regimes: on hexahedra they oscillate without rising $(3.67, 3.22, 3.20, 3.35)$ while on tetrahedra they climb monotonically $(2.91, 3.15, 3.40, 3.51)$, i.e. the asymptotic range is not reached by $N = 30$ — and $N > 30$ is out of reach here, since $N = 50$ on hexahedra is killed by the OOM killer during stiffness assembly with four ranks.

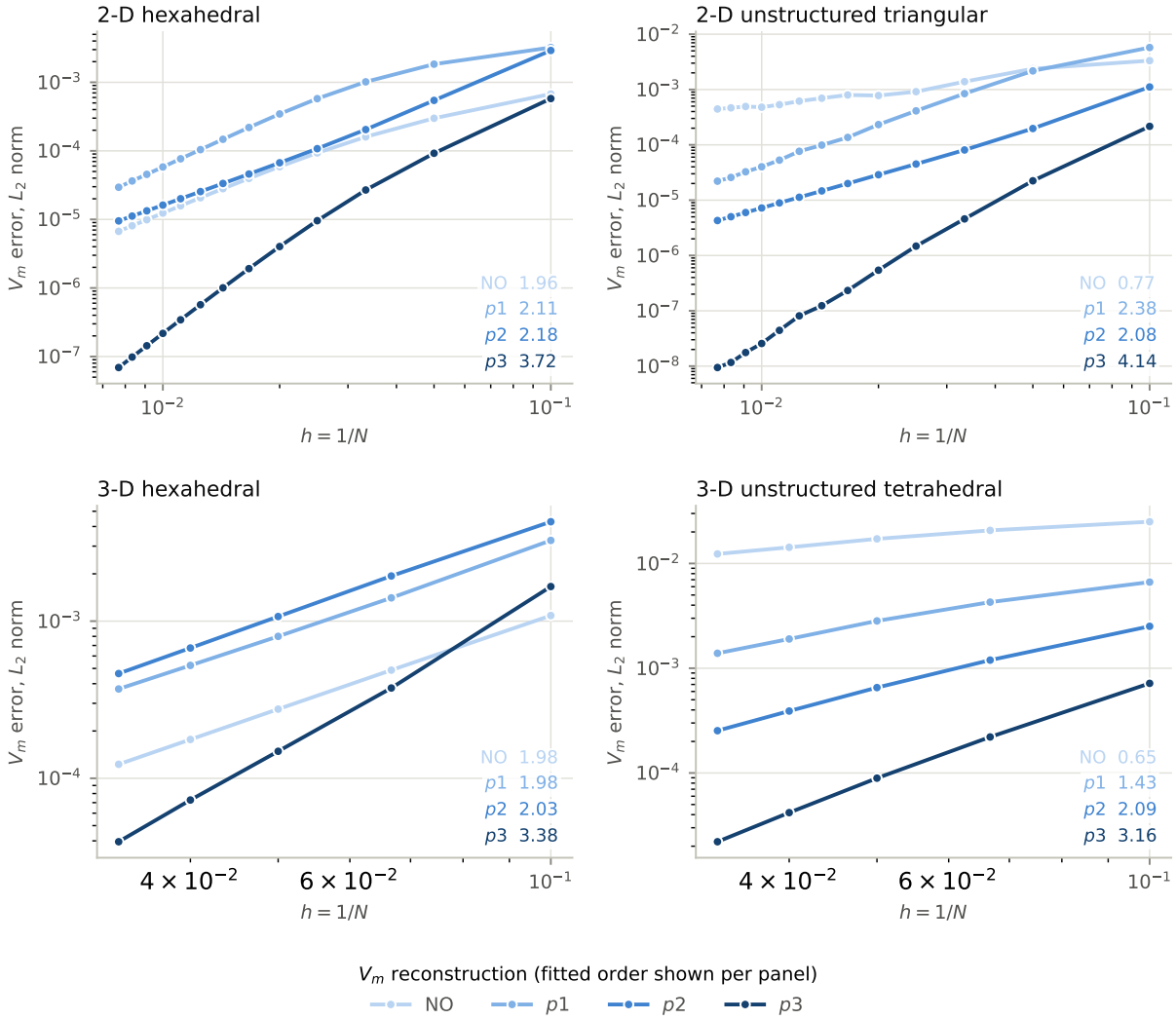


Figure 2: Spatial order of V_m for the four reconstruction levels on the diagonal, in 2-D ($N = 10$–130) and 3-D ($N = 10$–30), on structured and unstructured meshes. The number in each panel is the order fitted over that panel's whole range. The four levels are an ordered scale, so they are drawn as one hue light to dark rather than as four categorical colours. Generated by `figures/fig_orders.py` from the raw `errors_vs_N` files.

States at cell centres against states at Gauss points. The claim that the cell-centred mode is both cheaper and strictly more accurate is Figure 3, and it now holds in 3-D as well. Integrating the states at the quadrature points pins their cell-centred error at order two exactly — 2.00 and 1.99 in 2-D, 1.99 and 1.92 in 3-D — because averaging them back to the centres destroys the order, while the cell-centred mode follows the scheme (3.48, 3.87, 3.25, 2.97). V_m itself converges identically either way (3.384 against 3.381 on 3-D hexa), so nothing is lost by the cheaper path, which solves one ODE per cell instead of N_q.

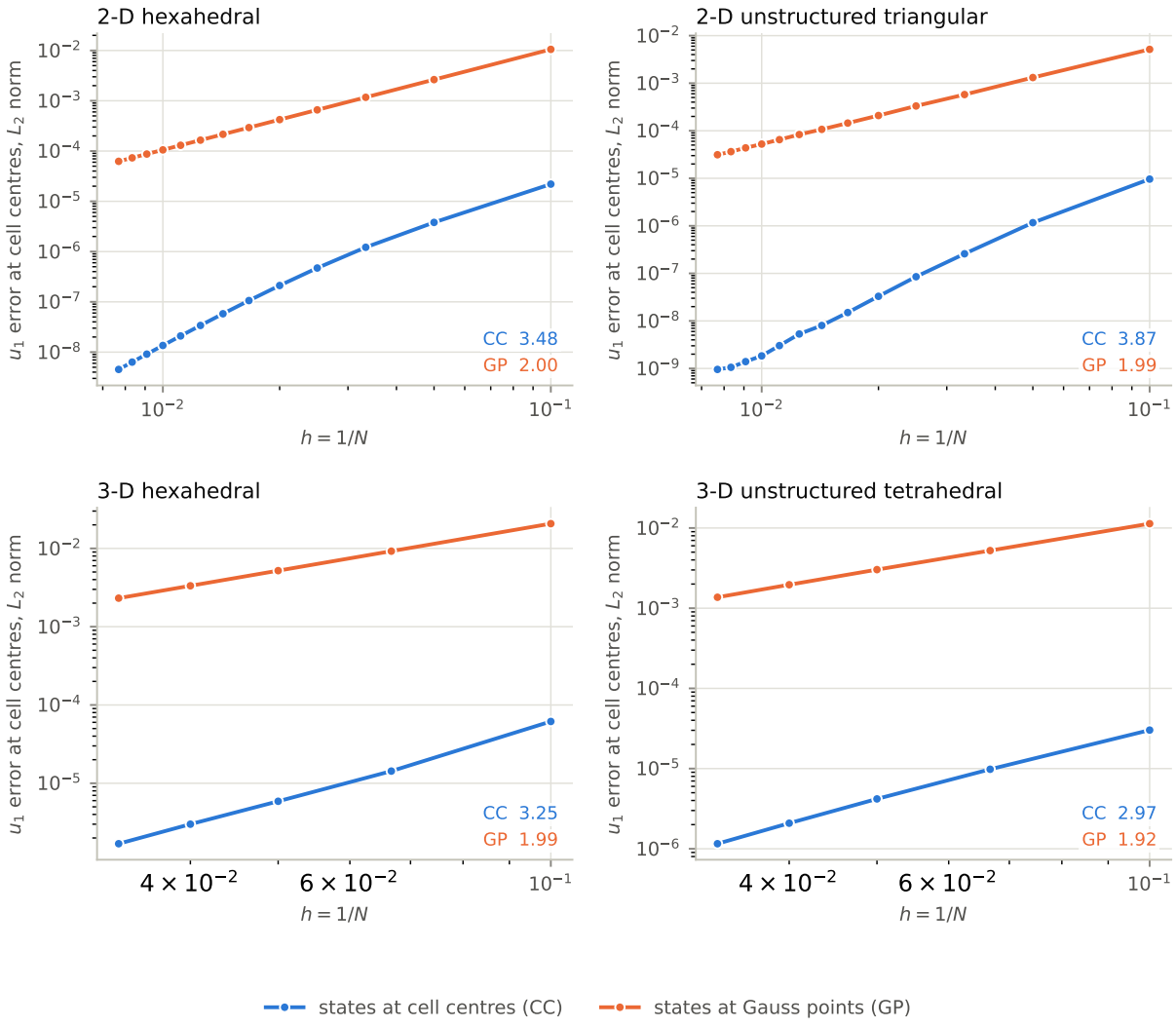


Figure 3: Error of the ionic state u_1 at cell centres, `cellCentredReconstruct` against `gaussPointODE`, with the $(p3, \cdot, p3)$ triad. The Gauss-point mode is pinned at order two in every panel; the cell-centred mode follows the scheme. Generated by `figures/fig_orders.py`.

Only odd reconstruction degrees buy a new order. Fixing the hexahedral mesh, $\alpha = 0$, and $\Delta t = 10^{-3}$, the observed order of V_m in the fine fit ($N = 70$–100) is

V_m reconstruction	NO	p1	p2	p3
Expected order (§3.3)	2	2	3	4
Observed order (hexa)	2.00	2.61	2.03	4.32
Observed order (triangular unstr)	0.96	2.60	1.99	4.48

The pattern is the classical odd/even one for cell-centred diffusion operators: the reconstructed face gradient has $O(h^N)$ truncation for degree N, but on approximately symmetric stencils the even-order error term cancels, so odd degrees gain one extra order and even degrees do not. This reproduces 2, 2, and 4 for p1, p2, and p3, and explains why p2 does *not* reach the nominal third order anticipated in §3.3: in the fine fit p2 delivers 2.03, barely above p1 but with a wider stencil and a higher cost. The practical conclusion is that, for this operator, the useful options are p1 and p3; p2 pays for a wider stencil without buying order.

The NO rows of this sweep predate the non-orthogonal correction and need re-running. The NO column of the table above was measured with a build that assembled only the orthogonal part of the face flux (§3.2), so its unstructured entry, 0.96 in the asymptotic fit and 0.80 over $N = 10$–100, describes an operator the solver no longer has. Stabilisation did not compensate for it either (0.95 with $\alpha = 0.1$), which is consistent with the omitted term being a leading-order contribution rather than a high-order perturbation.

With the correction the low-order operator holds its nominal order 2 on that mesh as well: M01 fits 2.49 and 2.74 at $\alpha = 0$ and 0.1. The comparison this report draws against the high-order path is therefore a gain of order — 2 against 4.61–4.65 for p3 on the same unstructured mesh — and not a difference in consistency. The hexahedral NO entries are unaffected: there the correction is identically zero.

The bottleneck moves to the ionic path once V_m is high order. With NO or p1 for V_m, the complete 5×4 grid of (states $\times$ I_{ion}) combinations gives the same order to three digits: 2.000 and 2.611 respectively. The ionic path is then irrelevant. With p3 the picture changes completely (hexa mesh, $\alpha = 0$, $\Delta t = 10^{-3}$, order of V_m in the fine fit):

states \ I_{ion}	NO	p1	p2	p3
na	2.01	–	–	–
CCp1	–	2.02	2.12	2.12
CCp2	–	2.02	**4.32**	**4.32**
CCp3	–	2.02	**4.32**	**4.32**
GP	–	2.02	**4.32**	**4.32**

Fourth order is unlocked if and only if the I_{ion} quadrature is at least p2 *and* the state interpolator is at least CCp2. Failing either condition degrades the result to second order, even though the V_m reconstruction is cubic and the diffusion operator is assembled to fourth order. This is the quantitative verification of the principle stated in §3.3: the observed order is the minimum of the orders of all the links. Translated into a usage rule, the triplet must be raised jointly: p3--CCp2--p2 is the cheapest configuration that attains fourth order, and raising only one of the three levels is wasted effort.

cellCentredReconstruct dominates gaussPointODE. The legacy GP mode reaches 4.19 on the quadrature-point metrics u_1^G and u_2^G, but its cell-centred state errors are stuck at 2.000: averaging the states from the quadrature points back to the cell centres destroys the order, exactly as anticipated in §3.5. The default mode CCp3 delivers 4.08 at cell centres and 4.28 at the quadrature

points simultaneously, and does so integrating one ODE per cell instead of one per quadrature point. On the quadrature-point metrics, the degree of the state interpolator sets the order directly: `CCp1`→ 2.15, `CCp2`→ 3.00, `CCp3`→ 4.28. There is no metric on which `GP` beats `CCp3`.

With $\Delta t = 10^{-2}$ the temporal floor contaminates the state rates. For the `p3--CCp2--p2` triplet on the hexa mesh, the cell-centred L_2 error of u_1 goes from 9.68×10^{-8} ($N = 70$) to 7.47×10^{-8} ($N = 100$) with $\Delta t = 10^{-2}$: it has stalled. With $\Delta t = 10^{-3}$ the same two points give 6.71×10^{-8} and 1.57×10^{-8}, i.e. it is still converging at fourth order. The Crank–Nicolson error plus that of the linear V_m ramp in τ, (35), produces a floor of $\approx 7\times10^{-8}$ that the fine meshes already reach. Quantified over the whole 2D sweep, the fine fit changes by more than 0.5 when going from $\Delta t = 10^{-2}$ to 10^{-3} in 120 out of 156 `p3` configurations for u_1 and u_2, and in 36 out of 156 for V_m. **The u_1 and u_2 rates of the 2D tables with $\Delta t = 10^{-2}$ must not be read as spatial orders**; only the $\Delta t = 10^{-3}$ tables are. For V_m, whose error is larger, the contamination is marginal.

Stabilisation is unnecessary in 2D. Except for the `NO` case on the hexahedral mesh, where $\alpha = 0.1$ artificially raises the fine fit from 2.00 to 2.29 (a pre-asymptotic effect, not an order improvement), the median effect of going from $\alpha = 0$ to $\alpha = 0.1$ on the order of V_m is smaller than 0.05 in every other level/mesh combination. Since $\alpha > 0$ additionally adds terms to the assembly and raises the run time by between 40% and 80%, $\alpha = 0$ is the preferable choice in 2D.

13.4 Three-dimensional manufactured problem

The tables in this subsection are generated directly from the current `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0` result tree. The method-specific roots are `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/3D_CN_cons_Picard_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/3D_CN_cons_diagonalIion_RKF45`, `/home/pablo/OpenFOAM/pablo-v2312/run/tutorials_electro/highOrderManufacturedFDAImplicitPETSc/results/example0/3D_CN_cons_JFNK_RKF45`. All entries use `crankNicolson`, the `consistent` mass matrix, and `RKF45` for the state ODEs. The first column reports `Vm-states-Iion`: the V_m reconstruction degree, the state treatment (`CCp*`, `GP`, or `na`), and the I_ion quadrature/reconstruction setting. Fits use $N = 10$–30 for the broad mesh range and $N = 20$–30 for the fine-mesh range. The last two columns audit the nonlinear solver: `RB` counts rolled-back steps and n_nl summarises the nonlinear iterations per step (median/maximum).

13.4.1 `Picard`

$\Delta t = 10^{-2}$

Table 9: 3D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 21 mesh sizes of the row (105 time steps per row in total). `Picard` accepts its last iterate with a warning instead of rolling back, so for this method the column counts steps accepted without converging. n_nl reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–30						Fit $N = 20$–30									
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS$_\Sigma$ [MB]	RB	n_nl
`NO--na--NO`	hexa	0.0	1.985	1.985	1.994	1.994	1.994	1.994	1.993	1.993	2.009	2.009	2.009	2.009	13.701	4034.4	0	4.00/4
`NO--na--NO`	hexa	0.1	1.436	1.436	1.297	1.297	1.306	1.306	1.411	1.411	1.214	1.214	1.223	1.223	22.097	6312.2	0	4.00/4
`NO--na--NO`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	21.144	4988.3	0	4.00/5
`NO--na--NO`	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	105.72	12055.9	0	4.00/5
`NO--CCp1--p1`	hexa	0.0	1.985	1.985	1.998	2.166	1.995	2.246	1.992	1.992	2.010	2.208	2.009	2.189	28.255	4710.5	0	4.00/4
`NO--CCp1--p1`	hexa	0.1	1.436	1.436	1.287	2.167	1.306	2.230	1.411	1.411	1.205	2.208	1.224	2.159	46.542	6565.3	0	4.00/4
`NO--CCp1--p1`	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	116.61	6199.0	0	4.00/5

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--CCp1--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	186.46	12250.5	0	4.00/5
NO--CCp1--p2	hexa	0.0	1.985	1.985	1.999	2.098	1.994	2.090	1.992	1.992	2.011	2.130	2.009	2.062	31.539	5888.2	0	4.00/4
NO--CCp1--p2	hexa	0.1	1.436	1.436	1.283	2.098	1.307	2.087	1.411	1.411	1.201	2.130	1.224	2.056	44.590	7233.9	0	4.00/4
NO--CCp1--p2	tetra unstr	0.0	0.664	0.664	0.375	1.924	0.373	0.457	0.826	0.826	0.617	1.896	0.616	0.669	128.54	6649.6	0	4.00/5
NO--CCp1--p2	tetra unstr	0.1	0.677	0.677	0.398	1.928	0.396	0.483	0.823	0.823	0.629	1.903	0.628	0.683	199.14	12498.6	0	4.00/5
NO--CCp1--p3	hexa	0.0	1.985	1.985	1.999	2.094	1.994	2.088	1.992	1.992	2.011	2.127	2.009	2.061	35.704	8248.0	0	4.00/4
NO--CCp1--p3	hexa	0.1	1.436	1.436	1.283	2.095	1.307	2.085	1.411	1.411	1.201	2.127	1.224	2.054	57.374	9568.3	0	4.00/4
NO--CCp1--p3	tetra unstr	0.0	0.664	0.664	0.375	1.923	0.373	0.460	0.826	0.826	0.617	1.906	0.616	0.670	123.28	7555.6	0	4.00/5
NO--CCp1--p3	tetra unstr	0.1	0.677	0.677	0.398	1.926	0.396	0.485	0.823	0.823	0.629	1.912	0.628	0.684	217.11	12994.2	0	4.00/5
NO--CCp2--p1	hexa	0.0	1.977	1.977	2.009	3.171	1.987	2.618	1.989	1.989	2.014	3.091	2.005	2.390	37.567	5489.6	0	4.00/4
NO--CCp2--p1	hexa	0.1	1.447	1.447	1.257	3.172	1.318	2.180	1.426	1.426	1.176	3.092	1.239	1.626	56.054	6775.8	0	4.00/4
NO--CCp2--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	190.82	9543.3	0	4.00/5
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	244.84	13294.2	0	4.00/5
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.010	3.247	1.979	2.753	1.985	1.985	2.013	3.146	2.001	2.494	55.354	6662.9	0	4.00/4
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.248	3.248	1.333	2.320	1.441	1.441	1.170	3.147	1.257	1.718	58.989	7953.8	0	4.00/4
NO--CCp2--p2	tetra unstr	0.0	0.664	0.664	0.375	2.080	0.373	0.465	0.826	0.826	0.617	1.273	0.616	0.673	183.95	9999.5	0	4.00/5
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.398	2.128	0.396	0.486	0.823	0.823	0.629	1.333	0.628	0.683	257.01	13767.7	0	4.00/5
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.010	3.252	1.979	2.758	1.985	1.985	2.013	3.149	2.001	2.501	68.040	9017.1	0	4.00/4
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.248	3.253	1.333	2.328	1.441	1.441	1.170	3.150	1.257	1.726	66.357	10307.2	0	4.00/4
NO--CCp2--p3	tetra unstr	0.0	0.664	0.664	0.375	1.992	0.373	0.466	0.826	0.826	0.617	1.241	0.616	0.675	180.62	10904.7	0	4.00/5
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.398	2.039	0.396	0.487	0.823	0.823	0.629	1.295	0.628	0.686	257.54	14649.4	0	4.00/5
NO--CCp3--p1	hexa	0.0	1.980	1.980	2.001	4.033	1.989	2.062	1.991	1.991	2.011	4.072	2.007	2.017	90.356	7390.4	0	4.00/4
NO--CCp3--p1	hexa	0.1	1.452	1.452	1.244	4.015	1.323	1.276	1.429	1.429	1.170	4.028	1.241	1.186	89.988	8677.1	0	4.00/4
NO--CCp3--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	358.12	18313.3	0	4.00/5
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	385.12	22079.8	0	4.00/5
NO--CCp3--p2	hexa	0.0	1.976	1.976	2.000	3.980	1.985	2.107	1.990	1.990	2.011	4.053	2.005	2.022	94.372	8563.2	0	4.00/4
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.233	3.966	1.343	1.305	1.447	1.447	1.163	4.019	1.261	1.187	88.556	9854.5	0	4.00/4
NO--CCp3--p2	tetra unstr	0.0	0.664	0.664	0.375	0.890	0.373	0.418	0.826	0.826	0.617	0.674	0.616	0.657	346.61	18765.0	0	4.00/5
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.398	0.930	0.396	0.441	0.823	0.823	0.629	0.689	0.628	0.670	399.32	22514.1	0	4.00/5
NO--CCp3--p3	hexa	0.0	1.976	1.976	2.000	3.973	1.985	2.108	1.990	1.990	2.011	4.054	2.005	2.022	94.607	10920.3	0	4.00/4
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.233	3.959	1.343	1.305	1.447	1.447	1.163	4.021	1.261	1.186	106.47	12209.7	0	4.00/4
NO--CCp3--p3	tetra unstr	0.0	0.664	0.664	0.375	0.888	0.373	0.432	0.826	0.826	0.617	0.684	0.616	0.665	348.72	19672.6	0	4.00/5
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.398	0.926	0.396	0.454	0.823	0.823	0.629	0.697	0.628	0.677	418.61	23417.4	0	4.00/5
NO--GP--p1	hexa	0.0	1.984	1.984	1.992	2.072	1.992	2.077	1.992	1.992	1.996	2.038	1.997	2.041	28.095	4465.1	0	4.00/4
NO--GP--p1	hexa	0.1	1.459	1.459	1.992	2.073	1.993	2.079	1.429	1.429	1.996	2.038	1.996	2.041	37.380	6564.9	0	4.00/4
NO--GP--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	29.082	5234.8	0	4.00/5
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	97.186	12251.2	0	4.00/5
NO--GP--p2	hexa	0.0	1.983	1.983	1.984	2.014	1.985	2.018	1.992	1.992	1.992	2.007	1.993	2.009	72.453	5644.5	0	4.00/4
NO--GP--p2	hexa	0.1	1.482	1.482	1.984	2.015	1.985	2.018	1.447	1.447	1.992	2.007	1.993	2.010	84.100	7214.1	0	4.00/4
NO--GP--p2	tetra unstr	0.0	0.664	0.664	1.832	0.861	0.516	0.859	0.826	0.826	1.792	0.867	0.652	0.866	43.013	5699.3	0	4.00/5
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.844	0.883	0.546	0.881	0.823	0.823	1.812	0.884	0.668	0.883	115.92	12501.1	0	4.00/5
NO--GP--p3	hexa	0.0	1.983	1.983	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.993	2.008	159.70	7997.1	0	4.00/4
NO--GP--p3	hexa	0.1	1.482	1.482	1.984	2.011	1.985	2.015	1.447	1.447	1.992	2.006	1.992	2.008	156.73	9198.0	0	4.00/4
NO--GP--p3	tetra unstr	0.0	0.664	0.664	1.828	0.863	0.513	0.862	0.826	0.826	1.790	0.868	0.651	0.868	71.823	6602.5	0	4.00/5
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.542	0.884	0.823	0.823	1.810	0.885	0.667	0.885	146.28	12996.8	0	4.00/5
p1--na--NO	hexa	0.0	1.967	2.095	1.927	1.927	1.930	1.930	1.919	2.125	1.838	1.838	1.845	1.845	77.496	7145.3	0	4.00/4
p1--na--NO	hexa	0.1	2.067	2.093	2.047	2.047	2.048	2.048	2.066	2.124	2.033	2.033	2.035	2.035	67.389	8698.8	0	4.00/4
p1--na--NO	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	207.91	9888.9	0	4.00/4
p1--na--NO	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	209.36	11566.5	0	4.00/4
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.926	2.166	1.930	2.193	1.919	2.125	1.836	2.207	1.845	2.115	73.944	7263.0	0	4.00/4
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.046	2.167	2.048	2.225	2.066	2.124	2.033	2.208	2.035	2.168	93.736	8854.4	0	4.00/4
p1--CCp1--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	315.13	10862.6	0	4.00/4
p1--CCp1--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	289.28	12081.8	0	4.00/4
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.925	2.098	1.930	2.083	1.919	2.125	1.835	2.130	1.845	2.049	83.398	8101.2	0	4.00/4
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.046	2.098	2.047	2.090	2.066	2.124	2.033	2.130	2.035	2.061	87.516	9506.3	0	4.00/4
p1--CCp1--p2	tetra unstr	0.0	1.474	1.866	1.139	1.973	1.142	1.493	1.771	1.928	1.419	1.984	1.428	1.590	330.99	11305.1	0	4.00/4
p1--CCp1--p2	tetra unstr	0.1	1.613	1.909	1.262	1.973	1.264	1.621	1.889	1.963	1.556	1.983	1.558	1.721	309.50	12482.1	0	4.00/4
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.925	2.094	1.930	2.081	1.919	2.125	1.835	2.127	1.845	2.048	105.81	10456.5	0	4.00/4
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.046	2.094	2.047	2.088	2.066	2.124	2.033	2.127	2.035	2.060	106.65	11155.3	0	4.00/4
p1--CCp1--p3	tetra unstr	0.0	1.474	1.866	1.139	1.965	1.142	1.505	1.771	1.928	1.419	1.982	1.428	1.597	307.94	12243.6	0	4.00/4
p1--CCp1--p3	tetra unstr	0.1	1.613	1.909	1.262	1.965	1.264	1.632	1.889	1.963	1.556	1.982	1.558	1.727	309.45	13293.2	0	4.00/4
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.926	3.169	1.929	2.043	1.919	2.125	1.831	3.089	1.845	1.844	115.71	7558.1	0	4.00/4
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.050	3.170	2.046	2.200	2.066	2.124	2.033	3.090	2.034	2.090	96.811	8857.6	0	4.00/4
p1--CCp2--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	356.84	14195.6	0	4.00/4
p1--CCp2--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	342.43	14883.3	0	4.00/4
p1--CCp2--p2	hexa	0.0	1.965	2.095	1.925	3.244	1.929	2.095	1.919	2.125	1.828	3.143	1.845	1.846	115.21	8770.9	0	4.00/4
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.051	3.245	2.045	2.265	2.065	2.124	2.033	3.145	2.034	2.115	109.97	9580.9	0	4.00/4
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.139	2.917	1.142	1.199	1.771	1.928	1.418	2.913	1.428	1.465	379.66	14646.2	0	4.00/4
p1--CCp2--p2	tetra unstr	0.1	1.613	1.909	1.262	2.928	1.264	1.336	1.889	1.963	1.555	2.938	1.558	1.614	374.84	15337.7	0	4.00/4
p1--CCp2--p3	hexa	0.0	1.965	2.095	1.925	3.250	1.929	2.098	1.919	2.125	1.828	3.146	1.845	1.848	108.78	11158.3	0	4.00/4
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.051	3.251	2.045	2.269	2.065	2.124	2.033	3.148	2.034	2.117	114.59	11791.9	0	4.00/4
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.139	2.922	1.142	1.211	1.771	1.928	1.418	2.906	1.428	1.476	352.44	15569.3	0	4.00/4
p1--CCp2--p3	tetra unstr	0.1	1.613	1.909	1.262	2.938	1.264	1.348	1.889	1.963	1.555	2.938	1.558	1.624	367.63	16260.6	0	4.00/4
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.922	4.067	1.929	1.877	1.919	2.125	1.830	4.138	1.845	1.729	114.34	9452.8	0	4.00/4
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.046	4.057	2.047	2.042	2.066	2.124	2.032	4.119	2.034	2.007	131.72	9844.0	0	4.00/4
p1--CCp3--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	499.68	22974.6	0	4.00/4
p1--CCp3--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	473.73	23656.3	0	4.00/4
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.920	4.009	1.930	1.853	1.920	2.125	1.827	4.112	1.846	1.682	127.48	10692.9	0	4.00/4
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.045	4.000	2.047	2.039	2.066	2.124	2.031	4.092	2.034	1.993	142.00	10999.5	0	4.00/4
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.139	2.936	1.142	1.202	1.771	1.928	1.418	2.110	1.428	1.472	498.40	23422.0	0	4.00/4
p1--CCp3--p2	tetra unstr	0.1	1.613	1.909	1.262	3.163	1.264	1.332	1.889	1.963	1.555	2.437	1.558	1.614	470.69	24109.9	0	4.00/4
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.920	4.003	1.930	1.852	1.920	2.125	1.827	4.114	1.846	1.681	162.36	13075.5	0	4.00/4

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.045	3.992	2.047	2.039	2.066	2.124	2.031	4.094	2.034	1.993	143.37	13660.3	0	4.00/4
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.139	2.837	1.142	1.240	1.771	1.928	1.418	2.066	1.428	1.496	507.99	24331.7	0	4.00/4
p1--CCp3--p3	tetra unstr	0.1	1.613	1.909	1.262	3.084	1.264	1.375	1.889	1.963	1.555	2.395	1.558	1.644	490.47	25022.1	0	4.00/4
p1--GP--p1	hexa	0.0	1.967	2.095	1.992	2.157	1.999	2.164	1.919	2.125	1.997	2.201	2.001	2.201	87.931	7353.8	0	4.00/4
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.002	2.162	2.066	2.124	1.997	2.199	2.005	2.199	78.820	8903.7	0	4.00/4
p1--GP--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	222.68	9942.5	0	4.00/4
p1--GP--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	219.88	11623.1	0	4.00/4
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.920	2.125	1.993	2.129	1.997	2.128	127.68	8029.7	0	4.00/4
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.093	1.991	2.096	2.066	2.124	1.993	2.128	1.998	2.126	120.62	9552.9	0	4.00/4
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.771	1.928	1.953	1.877	1.777	1.880	215.29	10400.8	0	4.00/4
p1--GP--p2	tetra unstr	0.1	1.613	1.909	1.920	1.897	1.790	1.900	1.889	1.963	1.955	1.921	1.848	1.922	230.12	11946.2	0	4.00/4
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.920	2.125	1.993	2.126	1.996	2.125	195.10	10159.3	0	4.00/4
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.066	2.124	1.993	2.124	1.998	2.123	195.10	10948.2	0	4.00/4
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.874	1.771	1.928	1.951	1.889	1.775	1.892	264.06	11323.2	0	4.00/4
p1--GP--p3	tetra unstr	0.1	1.613	1.909	1.916	1.900	1.786	1.903	1.889	1.963	1.953	1.927	1.846	1.929	256.05	12695.5	0	4.00/4
p2--na--NO	hexa	0.0	1.947	3.025	1.944	1.944	1.948	1.948	2.001	2.766	2.003	2.003	2.005	2.005	126.94	25863.3	0	4.00/4
p2--na--NO	hexa	0.1	1.946	3.053	1.943	1.943	1.947	1.947	2.000	2.807	2.002	2.002	2.005	2.005	127.33	23781.8	0	4.00/4
p2--na--NO	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	329.21	17527.9	0	4.00/4
p2--na--NO	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	299.85	21384.0	0	4.00/4
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.995	2.168	2.003	2.183	2.035	2.884	2.035	2.209	2.040	2.147	162.66	13057.9	0	4.00/4
p2--CCp1--p1	hexa	0.1	2.008	3.133	2.000	2.167	2.008	2.193	2.037	2.925	2.037	2.209	2.042	2.153	154.56	15809.6	0	4.00/4
p2--CCp1--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	401.07	18278.8	0	4.00/4
p2--CCp1--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	409.42	21360.5	0	4.00/4
p2--CCp1--p2	hexa	0.0	2.040	3.142	2.028	2.098	2.039	2.087	2.059	2.936	2.057	2.131	2.064	2.063	153.06	13705.3	0	4.00/4
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.037	2.098	2.049	2.088	2.063	2.975	2.062	2.131	2.069	2.063	144.90	16447.5	0	4.00/4
p2--CCp1--p2	tetra unstr	0.0	2.103	2.703	1.863	1.971	1.864	1.960	2.383	2.718	2.113	1.982	2.121	1.974	369.74	18791.5	0	4.00/4
p2--CCp1--p2	tetra unstr	0.1	2.347	2.823	2.013	1.971	2.016	1.970	2.727	2.903	2.362	1.982	2.372	1.985	404.73	21713.1	0	4.00/4
p2--CCp1--p3	hexa	0.0	2.040	3.142	2.028	2.095	2.039	2.085	2.059	2.936	2.057	2.127	2.064	2.061	146.72	15161.4	0	4.00/4
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.037	2.095	2.049	2.086	2.063	2.975	2.062	2.127	2.069	2.061	168.64	17752.1	0	4.00/4
p2--CCp1--p3	tetra unstr	0.0	2.103	2.703	1.863	1.963	1.864	1.958	2.383	2.718	2.112	1.981	2.121	1.972	408.34	19711.2	0	4.00/4
p2--CCp1--p3	tetra unstr	0.1	2.347	2.823	2.013	1.963	2.016	1.968	2.727	2.903	2.362	1.981	2.372	1.983	451.98	22462.7	0	4.00/4
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.999	3.170	2.004	2.077	2.035	2.884	2.038	3.091	2.040	2.074	151.33	13056.4	0	4.00/4
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.005	3.171	2.009	2.096	2.037	2.925	2.040	3.091	2.043	2.083	144.64	15812.2	0	4.00/4
p2--CCp2--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	414.32	21601.2	0	4.00/4
p2--CCp2--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	459.74	23596.6	0	4.00/4
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.036	3.246	2.040	2.173	2.059	2.936	2.063	3.146	2.065	2.129	183.26	13759.2	0	4.00/4
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.046	3.246	2.050	2.209	2.064	2.975	2.068	3.146	2.070	2.147	160.11	16458.6	0	4.00/4
p2--CCp2--p2	tetra unstr	0.0	2.103	2.704	1.863	2.944	1.864	1.932	2.384	2.719	2.114	2.972	2.121	2.120	456.55	22081.1	0	4.00/4
p2--CCp2--p2	tetra unstr	0.1	2.349	2.824	2.014	2.945	2.016	2.103	2.732	2.904	2.367	2.972	2.373	2.405	446.99	24005.8	0	4.00/4
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.036	3.251	2.040	2.176	2.059	2.936	2.063	3.149	2.065	2.130	193.86	15535.7	0	4.00/4
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.046	3.251	2.050	2.212	2.064	2.975	2.068	3.149	2.070	2.149	164.90	17811.0	0	4.00/4
p2--CCp2--p3	tetra unstr	0.0	2.103	2.704	1.863	2.958	1.864	1.943	2.384	2.719	2.114	2.980	2.121	2.133	428.34	22991.9	0	4.00/4
p2--CCp2--p3	tetra unstr	0.1	2.349	2.824	2.014	2.959	2.016	2.110	2.732	2.904	2.367	2.981	2.373	2.411	463.22	24901.5	0	4.00/4
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.999	4.087	2.004	2.022	2.035	2.884	2.038	4.142	2.040	2.059	188.83	13728.6	0	4.00/4
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.004	4.082	2.009	2.028	2.037	2.925	2.040	4.137	2.043	2.062	176.63	15813.4	0	4.00/4
p2--CCp3--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	572.02	30371.1	0	4.00/4
p2--CCp3--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	607.78	31489.2	0	4.00/4
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.035	4.022	2.040	2.074	2.059	2.936	2.063	4.109	2.065	2.104	182.15	14822.2	0	4.00/4
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.045	4.017	2.050	2.087	2.064	2.975	2.068	4.103	2.070	2.111	177.15	16475.9	0	4.00/4
p2--CCp3--p2	tetra unstr	0.0	2.103	2.704	1.863	3.893	1.864	1.896	2.384	2.719	2.114	3.841	2.121	2.127	586.59	30834.3	0	4.00/4
p2--CCp3--p2	tetra unstr	0.1	2.349	2.824	2.014	3.911	2.016	2.046	2.732	2.904	2.367	3.890	2.373	2.389	634.18	31950.0	0	4.00/4
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.035	4.014	2.040	2.075	2.059	2.936	2.063	4.110	2.065	2.105	244.85	17100.4	0	4.00/4
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.045	4.010	2.050	2.087	2.064	2.975	2.068	4.104	2.070	2.112	199.29	18144.8	0	4.00/4
p2--CCp3--p3	tetra unstr	0.0	2.103	2.704	1.863	3.888	1.864	1.929	2.384	2.719	2.114	3.831	2.121	2.146	611.73	31741.8	0	4.00/4
p2--CCp3--p3	tetra unstr	0.1	2.349	2.824	2.014	3.907	2.016	2.072	2.732	2.904	2.367	3.883	2.373	2.403	632.63	32867.4	0	4.00/4
p2--GP--p1	hexa	0.0	2.003	3.107	2.011	3.133	2.845	3.122	2.035	2.884	2.004	3.005	2.449	2.999	164.63	26074.9	0	4.00/4
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.888	3.131	2.037	2.925	2.004	3.021	2.476	3.015	138.88	23991.3	0	4.00/4
p2--GP--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	342.41	17573.5	0	4.00/4
p2--GP--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	330.06	21400.4	0	4.00/4
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.898	3.215	2.059	2.936	2.001	3.092	2.432	3.083	236.16	26715.1	0	4.00/4
p2--GP--p2	hexa	0.1	2.050	3.166	2.004	3.236	2.920	3.222	2.064	2.975	2.001	3.105	2.440	3.095	179.04	24639.4	0	4.00/4
p2--GP--p2	tetra unstr	0.0	2.103	2.704	1.928	2.794	2.100	2.796	2.384	2.719	1.960	2.779	2.102	2.784	342.91	17961.8	0	4.00/4
p2--GP--p2	tetra unstr	0.1	2.349	2.824	1.928	2.831	2.146	2.833	2.732	2.904	1.960	2.861	2.166	2.866	336.90	21691.1	0	4.00/4
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.897	3.220	2.059	2.936	2.000	3.095	2.432	3.085	355.85	28015.7	0	4.00/4
p2--GP--p3	hexa	0.1	2.050	3.166	2.003	3.242	2.920	3.227	2.064	2.975	2.000	3.108	2.439	3.097	288.86	25931.7	0	4.00/4
p2--GP--p3	tetra unstr	0.0	2.103	2.704	1.924	2.802	2.082	2.802	2.384	2.719	1.958	2.769	2.088	2.773	414.66	18813.0	0	4.00/4
p2--GP--p3	tetra unstr	0.1	2.349	2.824	1.924	2.847	2.128	2.847	2.732	2.904	1.958	2.864	2.152	2.868	497.14	22280.7	0	4.00/4
p3--na--NO	hexa	0.0	2.313	3.204	2.308	2.308	2.312	2.312	2.193	2.830	2.197	2.197	2.200	2.200	897.53	77071.6	0	4.00/4
p3--na--NO	hexa	0.1	2.278	3.207	2.274	2.274	2.278	2.278	2.169	2.826	2.174	2.174	2.176	2.176	755.12	60518.2	0	4.00/4
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1241.6	45021.5	0	4.00/4
p3--na--NO	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1292.1	53563.8	0	4.00/4
p3--CCp1--p1	hexa	0.0	2.761	3.638	2.698	2.166	2.745	2.264	2.550	3.526	2.508	2.207	2.553	2.199	878.49	44825.6	0	4.00/4
p3--CCp1--p1	hexa	0.1	2.717	3.647	2.659	2.166	2.706	2.261	2.501	3.534	2.465	2.207	2.509	2.198	790.19	45947.9	0	4.00/4
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1339.5	45670.1	0	4.00/4
p3--CCp1--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1387.9	53118.8	0	4.00/4
p3--CCp1--p2	hexa	0.0	3.129	3.794	3.046	2.097	3.078	2.094	2.914	3.838	2.857	2.130	2.869	2.063	1000.0	45474.5	0	4.00/4
p3--CCp1--p2	hexa	0.1	3.103	3.804	3.033	2.097	3.064	2.093	2.862	3.851	2.828	2.130	2.835	2.063	818.24	46594.6	0	4.00/4
p3--CCp1--p2	tetra unstr	0.0	3.009	3.711	2.897	1.970	2.893	1.968	3.092	3.661	3.022	1.982	3.021	1.961	1488.2	45933.5	0	4.00/4
p3--CCp1--p2	tetra unstr	0.1	3.120	3.786	3.019	1.970	3.015	1.968	3.099	3.741	3.149	1.982	3.147	1.960	1324.3	53368.7	0	4.00/4
p3--CCp1--p3	hexa	0.0	3.129	3.794	3.046	2.094	3.078	2.092	2.914	3.838	2.857	2.127	2.869	2.062	821.92	46769.5	0	4.00/4
p3--CCp1--p3	hexa	0.1	3.103	3.804	3.033	2.094	3.064	2.091	2.862	3.851	2.828	2.127	2.835	2.062	829.13	47889.4	0	4.00/4
p3--CCp1--p3	tetra unstr	0.0	3.009	3.711	2.897	1.963	2.892	1.966	3.091	3.661	3.021	1.981	3.021	1.960	1619.5	46505.3	0	4.00/4

			Fit $N = 10$–30						Fit $N = 20$–30									
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p3--CCp1--p3	tetra unstr	0.1	3.120	3.786	3.018	1.963	3.015	1.966	3.099	3.741	3.149	1.981	3.147	1.960	1344.9	53870.2	0	4.00/4
p3--CCp2--p1	hexa	0.0	2.781	3.643	2.756	3.172	2.763	2.955	2.572	3.538	2.566	3.092	2.573	2.804	663.90	44829.9	0	4.00/4
p3--CCp2--p1	hexa	0.1	2.738	3.653	2.719	3.172	2.726	2.951	2.523	3.546	2.524	3.092	2.530	2.794	837.36	45948.4	0	4.00/4
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1489.8	46456.2	0	4.00/4
p3--CCp2--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1414.7	53257.0	0	4.00/4
p3--CCp2--p2	hexa	0.0	3.332	3.812	3.211	3.247	3.230	3.235	3.260	3.882	3.083	3.147	3.113	3.176	668.73	45478.7	0	4.00/4
p3--CCp2--p2	hexa	0.1	3.350	3.823	3.234	3.247	3.253	3.247	3.275	3.896	3.103	3.147	3.135	3.190	833.85	46594.5	0	4.00/4
p3--CCp2--p2	tetra unstr	0.0	3.196	3.775	2.986	2.942	2.983	2.997	3.461	3.809	3.184	2.970	3.189	3.072	1497.0	46813.3	0	4.00/4
p3--CCp2--p2	tetra unstr	0.1	3.450	3.862	3.163	2.942	3.162	3.045	3.777	3.920	3.434	2.970	3.440	3.120	1290.8	53564.8	0	4.00/4
p3--CCp2--p3	hexa	0.0	3.332	3.812	3.211	3.253	3.230	3.234	3.260	3.882	3.083	3.150	3.113	3.175	678.52	46771.1	0	4.00/4
p3--CCp2--p3	hexa	0.1	3.350	3.823	3.234	3.253	3.253	3.246	3.275	3.896	3.103	3.150	3.135	3.189	812.26	47892.2	0	4.00/4
p3--CCp2--p3	tetra unstr	0.0	3.195	3.776	2.985	2.957	2.983	3.007	3.461	3.809	3.184	2.980	3.188	3.076	1498.5	47514.6	0	4.00/4
p3--CCp2--p3	tetra unstr	0.1	3.450	3.862	3.163	2.957	3.161	3.054	3.777	3.921	3.434	2.980	3.440	3.122	1273.4	54155.3	0	4.00/4
p3--CCp3--p1	hexa	0.0	2.780	3.644	2.753	4.033	2.762	2.775	2.572	3.538	2.566	4.067	2.573	2.597	1165.1	44825.3	0	4.00/4
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.716	4.033	2.725	2.746	2.522	3.545	2.523	4.068	2.530	2.555	804.75	45947.1	0	4.00/4
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1615.4	52498.7	0	4.00/4
p3--CCp3--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1395.3	57649.7	0	4.00/4
p3--CCp3--p2	hexa	0.0	3.330	3.812	3.205	3.978	3.228	3.112	3.256	3.881	3.086	4.042	3.110	3.026	962.21	45471.4	0	4.00/4
p3--CCp3--p2	hexa	0.1	3.347	3.823	3.226	3.978	3.250	3.143	3.271	3.895	3.106	4.042	3.131	3.046	818.88	46596.1	0	4.00/4
p3--CCp3--p2	tetra unstr	0.0	3.196	3.775	2.987	3.943	2.983	3.011	3.461	3.808	3.184	3.959	3.188	3.209	1542.9	52918.9	0	4.00/4
p3--CCp3--p2	tetra unstr	0.1	3.450	3.862	3.164	3.942	3.162	3.181	3.777	3.920	3.434	3.959	3.440	3.451	1398.8	58052.9	0	4.00/4
p3--CCp3--p3	hexa	0.0	3.330	3.812	3.205	3.971	3.228	3.112	3.256	3.881	3.086	4.044	3.110	3.027	1042.9	46765.3	0	4.00/4
p3--CCp3--p3	hexa	0.1	3.347	3.823	3.226	3.971	3.250	3.143	3.271	3.895	3.106	4.044	3.131	3.046	864.30	47889.9	0	4.00/4
p3--CCp3--p3	tetra unstr	0.0	3.195	3.776	2.986	3.944	2.982	3.056	3.460	3.809	3.184	3.961	3.188	3.240	1623.8	53767.0	0	4.00/4
p3--CCp3--p3	tetra unstr	0.1	3.449	3.862	3.163	3.944	3.161	3.224	3.777	3.920	3.434	3.961	3.439	3.481	1407.7	58877.6	0	4.00/4
p3--GP--p1	hexa	0.0	2.780	3.644	2.003	3.792	2.717	3.810	2.572	3.538	2.000	3.687	2.355	3.691	930.65	77273.8	0	4.00/4
p3--GP--p1	hexa	0.1	2.737	3.653	2.003	3.799	2.707	3.817	2.522	3.546	2.000	3.695	2.347	3.700	905.37	77345.2	0	4.00/4
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1124.9	45255.4	0	4.00/4
p3--GP--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1323.8	53639.5	0	4.00/4
p3--GP--p2	hexa	0.0	3.331	3.812	1.994	3.893	2.454	3.912	3.257	3.881	1.996	3.958	2.171	3.969	979.58	77917.7	0	4.00/4
p3--GP--p2	hexa	0.1	3.348	3.823	1.994	3.897	2.447	3.916	3.272	3.896	1.996	3.964	2.167	3.975	853.75	77986.4	0	4.00/4
p3--GP--p2	tetra unstr	0.0	3.196	3.775	1.927	3.844	2.032	3.845	3.461	3.808	1.961	3.840	1.977	3.846	1028.3	45508.8	0	4.00/4
p3--GP--p2	tetra unstr	0.1	3.450	3.862	1.927	3.883	2.028	3.884	3.777	3.920	1.961	3.904	1.974	3.909	1038.4	53893.4	0	4.00/4
p3--GP--p3	hexa	0.0	3.331	3.812	1.994	3.886	2.452	3.905	3.257	3.881	1.995	3.960	2.170	3.970	1098.9	79214.1	0	4.00/4
p3--GP--p3	hexa	0.1	3.348	3.823	1.993	3.890	2.446	3.909	3.272	3.896	1.995	3.966	2.167	3.976	769.90	79286.6	0	4.00/4
p3--GP--p3	tetra unstr	0.0	3.195	3.776	1.923	3.842	2.021	3.844	3.460	3.809	1.958	3.837	1.974	3.844	1068.5	46006.7	0	4.00/4
p3--GP--p3	tetra unstr	0.1	3.450	3.862	1.923	3.884	2.017	3.885	3.777	3.921	1.958	3.904	1.971	3.911	1068.8	54392.0	0	4.00/4

$\Delta t = 10^{-3}$

Table 10: 3D manufactured results for `Picard`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 21 mesh sizes of the row (1050 time steps per row in total). `Picard` accepts its last iterate with a warning instead of rolling back, so for this method the column counts steps accepted without converging. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–30						Fit $N = 20$–30									
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--na--NO	hexa	0.0	1.985	1.985	1.985	1.985	1.985	1.985	1.993	1.993	1.993	1.993	1.993	1.993	86.614	4035.7	0	3.00/3
NO--na--NO	hexa	0.1	1.438	1.438	1.293	1.293	1.302	1.302	1.413	1.413	1.211	1.211	1.220	1.220	67.106	6315.1	0	3.00/3
NO--na--NO	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	135.14	4989.6	0	3.00/3
NO--na--NO	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	252.56	12057.8	0	3.00/3
NO--CCp1--p1	hexa	0.0	1.986	1.986	1.989	2.167	1.986	2.246	1.993	1.993	1.995	2.208	1.993	2.188	121.46	4714.8	0	3.00/3
NO--CCp1--p1	hexa	0.1	1.438	1.438	1.298	2.167	1.302	2.230	1.413	1.413	1.212	2.208	1.220	2.158	144.23	6566.5	0	3.00/3
NO--CCp1--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	358.87	6202.6	0	3.00/3
NO--CCp1--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	401.30	12253.2	0	3.00/3
NO--CCp1--p2	hexa	0.0	1.986	1.986	1.991	2.098	1.986	2.090	1.993	1.993	1.995	2.130	1.993	2.062	126.95	5894.5	0	3.00/3
NO--CCp1--p2	hexa	0.1	1.438	1.438	1.300	2.098	1.303	2.086	1.413	1.413	1.212	2.130	1.220	2.055	174.32	7235.8	0	3.00/3
NO--CCp1--p2	tetra unstr	0.0	0.663	0.663	0.377	1.924	0.375	0.459	0.829	0.829	0.617	1.896	0.617	0.670	375.15	6653.6	0	3.00/3
NO--CCp1--p2	tetra unstr	0.1	0.677	0.677	0.400	1.928	0.398	0.484	0.828	0.828	0.629	1.903	0.629	0.684	461.73	12500.1	0	3.00/3
NO--CCp1--p3	hexa	0.0	1.986	1.986	1.991	2.094	1.986	2.088	1.993	1.993	1.995	2.127	1.993	2.060	189.07	8249.2	0	3.00/3
NO--CCp1--p3	hexa	0.1	1.438	1.438	1.300	2.095	1.303	2.085	1.413	1.413	1.212	2.127	1.220	2.054	232.83	9568.0	0	3.00/3
NO--CCp1--p3	tetra unstr	0.0	0.663	0.663	0.377	1.922	0.375	0.462	0.829	0.829	0.617	1.906	0.617	0.671	435.28	7561.8	0	3.00/3
NO--CCp1--p3	tetra unstr	0.1	0.677	0.677	0.400	1.926	0.398	0.487	0.828	0.828	0.629	1.912	0.629	0.685	491.97	12998.0	0	3.00/3
NO--CCp2--p1	hexa	0.0	1.977	1.977	2.002	3.171	1.978	2.610	1.989	1.989	1.999	3.091	1.989	2.376	198.39	5489.1	0	3.00/3
NO--CCp2--p1	hexa	0.1	1.447	1.447	1.258	3.172	1.313	2.174	1.428	1.428	1.175	3.092	1.236	1.620	211.28	6772.3	0	3.00/3
NO--CCp2--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	646.90	9551.6	0	3.00/3
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	642.48	13261.1	0	3.00/3
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.004	3.247	1.970	2.745	1.985	1.985	1.998	3.146	1.985	2.481	282.85	6666.9	0	3.00/3
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.251	3.248	1.326	2.314	1.443	1.443	1.170	3.147	1.253	1.712	229.83	7952.0	0	3.00/3
NO--CCp2--p2	tetra unstr	0.0	0.663	0.663	0.377	2.079	0.375	0.467	0.829	0.829	0.617	1.272	0.617	0.673	654.87	10004.3	0	3.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.400	2.127	0.398	0.488	0.828	0.828	0.629	1.333	0.629	0.684	716.55	13729.4	0	3.00/3
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.004	3.252	1.970	2.751	1.985	1.985	1.998	3.149	1.985	2.488	339.01	9025.0	0	3.00/3
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.251	3.253	1.326	2.323	1.443	1.443	1.170	3.150	1.253	1.720	294.83	10309.6	0	3.00/3
NO--CCp2--p3	tetra unstr	0.0	0.663	0.663	0.378	1.991	0.375	0.469	0.829	0.829	0.617	1.240	0.617	0.676	675.80	10910.1	0	3.00/3
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.400	2.038	0.398	0.489	0.828	0.828	0.629	1.295	0.629	0.686	703.70	14607.9	0	3.00/3
NO--CCp3--p1	hexa	0.0	1.980	1.980	1.993	4.033	1.980	2.052	1.992	1.992	1.995	4.072	1.992	2.002	346.12	7391.3	0	3.00/3
NO--CCp3--p1	hexa	0.1	1.453	1.453	1.241	4.015	1.318	1.271	1.431	1.431	1.167	4.028	1.238	1.184	328.43	8669.8	0	3.00/3
NO--CCp3--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	1243.0	18315.6	0	3.00/3
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	1293.8	22044.6	0	3.00/3
NO--CCp3--p2	hexa	0.0	1.975	1.975	1.992	3.981	1.975	2.098	1.990	1.990	1.995	4.053	1.990	2.007	374.09	8567.8	0	3.00/3
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.230	3.966	1.336	1.300	1.449	1.449	1.160	4.019	1.258	1.185	407.91	9852.1	0	3.00/3
NO--CCp3--p2	tetra unstr	0.0	0.663	0.663	0.378	0.891	0.375	0.420	0.829	0.829	0.617	0.675	0.617	0.658	1250.2	18767.3	0	3.00/3
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.400	0.930	0.398	0.443	0.828	0.828	0.629	0.689	0.629	0.671	1248.8	22474.0	0	3.00/3
NO--CCp3--p3	hexa	0.0	1.975	1.975	1.992	3.973	1.975	2.099	1.991	1.991	1.995	4.055	1.990	2.007	428.80	10913.1	0	3.00/3
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.230	3.959	1.336	1.300	1.449	1.449	1.160	4.021	1.258	1.184	503.35	12204.4	0	3.00/3
NO--CCp3--p3	tetra unstr	0.0	0.663	0.663	0.378	0.888	0.375	0.434	0.829	0.829	0.617	0.684	0.617	0.666	1289.0	19673.6	0	3.00/3
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.400	0.927	0.398	0.456	0.828	0.828	0.629	0.698	0.629	0.678	1303.2	23376.9	0	3.00/3
NO--GP--p1	hexa	0.0	1.985	1.985	1.992	2.071	1.992	2.077	1.993	1.993	1.996	2.038	1.996	2.041	168.28	4465.5	0	3.00/3
NO--GP--p1	hexa	0.1	1.461	1.461	1.992	2.073	1.992	2.079	1.431	1.431	1.996	2.038	1.995	2.041	162.93	6564.0	0	3.00/3
NO--GP--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	140.66	5234.5	0	3.00/3
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	230.04	12251.4	0	3.00/3
NO--GP--p2	hexa	0.0	1.984	1.984	1.984	2.014	1.984	2.018	1.992	1.992	1.992	2.007	1.993	2.009	438.08	5647.7	0	3.00/3
NO--GP--p2	hexa	0.1	1.483	1.483	1.984	2.014	1.985	2.018	1.450	1.450	1.992	2.007	1.992	2.009	441.07	7215.9	0	3.00/3
NO--GP--p2	tetra unstr	0.0	0.663	0.663	1.832	0.862	0.518	0.860	0.829	0.829	1.792	0.867	0.653	0.866	253.95	5704.6	0	3.00/3
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.843	0.883	0.547	0.882	0.828	0.828	1.812	0.884	0.668	0.883	327.91	12501.2	0	3.00/3
NO--GP--p3	hexa	0.0	1.984	1.984	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.992	2.008	1011.8	7998.2	0	3.00/3
NO--GP--p3	hexa	0.1	1.483	1.483	1.984	2.011	1.984	2.015	1.450	1.450	1.992	2.005	1.992	2.008	957.54	9195.7	0	3.00/3
NO--GP--p3	tetra unstr	0.0	0.663	0.663	1.828	0.864	0.514	0.862	0.829	0.829	1.790	0.869	0.652	0.868	457.00	6601.5	0	3.00/3
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.543	0.884	0.828	0.828	1.810	0.886	0.667	0.885	564.41	13000.6	0	3.00/3
p1--na--NO	hexa	0.0	1.967	2.095	1.924	1.924	1.927	1.927	1.920	2.125	1.833	1.833	1.840	1.840	155.87	7144.0	0	3.00/3
p1--na--NO	hexa	0.1	2.067	2.093	2.044	2.044	2.044	2.044	2.067	2.124	2.028	2.028	2.029	2.029	136.81	8698.4	0	3.00/3
p1--na--NO	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	398.80	9891.6	0	3.00/3
p1--na--NO	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	397.09	11569.4	0	3.00/3
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.924	2.166	1.926	2.192	1.920	2.125	1.833	2.207	1.840	2.114	166.93	7261.7	0	3.00/3
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.043	2.167	2.044	2.224	2.067	2.124	2.028	2.208	2.029	2.167	166.18	8857.0	0	3.00/3
p1--CCp1--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	610.94	10859.4	0	3.00/3
p1--CCp1--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	539.25	12082.9	0	3.00/3
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.924	2.098	1.926	2.083	1.920	2.125	1.833	2.130	1.840	2.049	201.25	8095.6	0	3.00/3
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.043	2.098	2.044	2.090	2.067	2.124	2.028	2.130	2.029	2.061	249.86	9499.3	0	3.00/3
p1--CCp1--p2	tetra unstr	0.0	1.475	1.866	1.142	1.973	1.144	1.493	1.768	1.927	1.422	1.984	1.432	1.592	596.95	11306.5	0	3.00/3
p1--CCp1--p2	tetra unstr	0.1	1.612	1.909	1.265	1.973	1.267	1.622	1.884	1.962	1.558	1.983	1.561	1.723	658.33	12480.0	0	3.00/3
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.924	2.094	1.926	2.081	1.920	2.125	1.833	2.127	1.840	2.048	301.30	10445.0	0	3.00/3
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.043	2.094	2.044	2.088	2.067	2.124	2.028	2.127	2.029	2.059	274.68	11154.7	0	3.00/3
p1--CCp1--p3	tetra unstr	0.0	1.475	1.866	1.142	1.965	1.144	1.506	1.768	1.927	1.422	1.982	1.432	1.599	640.80	12241.1	0	3.00/3
p1--CCp1--p3	tetra unstr	0.1	1.612	1.909	1.265	1.965	1.267	1.633	1.884	1.962	1.558	1.982	1.561	1.729	597.66	13293.2	0	3.00/3
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.924	3.169	1.926	2.040	1.920	2.125	1.827	3.089	1.840	1.840	269.04	7542.1	0	3.00/3
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.048	3.170	2.043	2.196	2.066	2.124	2.028	3.090	2.028	2.085	273.06	8855.9	0	3.00/3
p1--CCp2--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	812.43	14195.8	0	3.00/3
p1--CCp2--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	770.23	14886.6	0	3.00/3
p1--CCp2--p2	hexa	0.0	1.966	2.095	1.924	3.244	1.925	2.092	1.920	2.125	1.824	3.143	1.840	1.841	289.16	8760.2	0	3.00/3
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.050	3.245	2.042	2.261	2.065	2.124	2.028	3.145	2.028	2.109	272.25	9575.8	0	3.00/3
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.141	2.917	1.145	1.201	1.768	1.927	1.422	2.912	1.432	1.469	834.07	14648.6	0	3.00/3
p1--CCp2--p2	tetra unstr	0.1	1.612	1.909	1.264	2.928	1.267	1.339	1.884	1.962	1.558	2.938	1.561	1.617	805.97	15334.0	0	3.00/3
p1--CCp2--p3	hexa	0.0	1.966	2.095	1.924	3.250	1.925	2.095	1.920	2.125	1.824	3.146	1.840	1.843	334.46	11158.4	0	3.00/3
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.050	3.251	2.042	2.265	2.065	2.124	2.028	3.148	2.028	2.111	380.06	11779.2	0	3.00/3
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.141	2.922	1.145	1.213	1.768	1.927	1.422	2.906	1.432	1.479	862.07	15565.5	0	3.00/3
p1--CCp2--p3	tetra unstr	0.1	1.612	1.909	1.264	2.938	1.267	1.350	1.884	1.962	1.558	2.938	1.561	1.627	820.54	16257.5	0	3.00/3
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.919	4.067	1.926	1.874	1.920	2.125	1.826	4.139	1.840	1.725	319.04	9440.7	0	3.00/3
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.043	4.057	2.043	2.038	2.067	2.124	2.026	4.119	2.029	2.002	353.23	9829.8	0	3.00/3
p1--CCp3--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	1325.7	22972.5	0	3.00/3
p1--CCp3--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1347.4	23653.0	0	3.00/3
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.917	4.010	1.926	1.850	1.921	2.125	1.823	4.112	1.841	1.677	423.99	10685.6	0	3.00/3
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.042	4.000	2.043	2.035	2.067	2.124	2.025	4.092	2.029	1.988	428.66	10985.7	0	3.00/3
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.141	2.935	1.145	1.205	1.768	1.927	1.422	2.111	1.432	1.475	1319.9	23418.0	0	3.00/3
p1--CCp3--p2	tetra unstr	0.1	1.612	1.909	1.264	3.163	1.267	1.335	1.884	1.962	1.558	2.438	1.561	1.616	1339.2	24104.2	0	3.00/3
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.917	4.003	1.926	1.849	1.921	2.125	1.823	4.114	1.841	1.676	577.60	13061.6	0	3.00/3
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.042	3.992	2.043	2.035	2.067	2.124	2.025	4.094	2.029	1.988	535.66	13643.6	0	3.00/3
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.141	2.836	1.145	1.243	1.768	1.927	1.422	2.067	1.432	1.500	1304.3	24333.6	0	3.00/3
p1--CCp3--p3	tetra unstr	0.1	1.612	1.909	1.264	3.084	1.267	1.378	1.884	1.962	1.558	2.395	1.561	1.647	1434.8	25014.7	0	3.00/3
p1--GP--p1	hexa	0.0	1.968	2.095	1.992	2.157	1.999	2.164	1.920	2.125	1.997	2.201	2.001	2.200	254.63	7353.9	0	3.00/3
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.001	2.162	2.067	2.124	1.997	2.199	2.004	2.199	222.49	8905.5	0	3.00/3
p1--GP--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	435.69	9944.3	0	3.00/3
p1--GP--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	427.55	11621.8	0	3.00/3
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.921	2.125	1.993	2.129	1.996	2.128	481.37	8024.5	0	3.00/3
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.092	1.991	2.096	2.067	2.124	1.993	2.128	1.998	2.126	472.63	9555.5	0	3.00/3
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.768	1.927	1.953	1.877	1.777	1.880	493.81	10403.2	0	3.00/3
p1--GP--p2	tetra unstr	0.1	1.612	1.909	1.920	1.897	1.790	1.900	1.884	1.962	1.955	1.921	1.849	1.923	531.81	11946.4	0	3.00/3
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.921	2.125	1.993	2.126	1.996	2.124	900.71	10144.0	0	3.00/3
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.067	2.124	1.993	2.124	1.998	2.123	963.73	10940.8	0	3.00/3
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.873	1.768	1.927	1.951	1.889	1.775	1.892	780.16	11337.6	0	3.00/3
p1--GP--p3	tetra unstr	0.1	1.612	1.909	1.916	1.900	1.786	1.903	1.884	1.962	1.953	1.927	1.847	1.929	679.56	12707.2	0	3.00/3
p2--na--NO	hexa	0.0	1.947	3.025	1.942	1.942	1.946	1.946	2.001	2.766	2.000	2.000	2.003	2.003	210.78	25865.1	0	3.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p2--na--NO	hexa	0.1	1.946	3.053	1.941	1.941	1.946	1.946	2.000	2.807	1.999	1.999	2.002	2.002	207.61	23781.5	0	3.00/3
p2--na--NO	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	654.46	17527.1	0	3.00/3
p2--na--NO	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	551.02	21382.3	0	3.00/3
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.993	2.168	2.001	2.183	2.035	2.884	2.031	2.209	2.036	2.145	276.91	13059.6	0	3.00/3
p2--CCp1--p1	hexa	0.1	2.008	3.133	1.997	2.167	2.006	2.192	2.037	2.925	2.033	2.209	2.038	2.152	269.12	15810.2	0	3.00/3
p2--CCp1--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	791.52	18277.7	0	3.00/3
p2--CCp1--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	805.96	21365.2	0	3.00/3
p2--CCp1--p2	hexa	0.0	2.040	3.141	2.024	2.098	2.036	2.087	2.059	2.936	2.052	2.131	2.060	2.062	289.55	13706.2	0	3.00/3
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.033	2.098	2.046	2.088	2.064	2.974	2.056	2.131	2.065	2.062	333.13	16453.6	0	3.00/3
p2--CCp1--p2	tetra unstr	0.0	2.105	2.705	1.863	1.971	1.864	1.960	2.382	2.719	2.113	1.982	2.122	1.974	740.31	18792.7	0	3.00/3
p2--CCp1--p2	tetra unstr	0.1	2.348	2.824	2.014	1.971	2.018	1.970	2.721	2.901	2.364	1.982	2.374	1.986	866.90	21712.1	0	3.00/3
p2--CCp1--p3	hexa	0.0	2.040	3.141	2.024	2.095	2.036	2.085	2.059	2.936	2.052	2.127	2.060	2.061	344.81	15168.3	0	3.00/3
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.033	2.095	2.046	2.086	2.064	2.974	2.056	2.127	2.065	2.061	368.42	17748.2	0	3.00/3
p2--CCp1--p3	tetra unstr	0.0	2.105	2.705	1.863	1.963	1.864	1.958	2.382	2.719	2.113	1.981	2.122	1.973	819.47	19712.8	0	3.00/3
p2--CCp1--p3	tetra unstr	0.1	2.347	2.824	2.014	1.963	2.018	1.968	2.721	2.901	2.364	1.981	2.374	1.984	901.98	22464.9	0	3.00/3
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.998	3.170	2.002	2.075	2.035	2.884	2.035	3.091	2.037	2.071	370.35	13057.8	0	3.00/3
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.003	3.171	2.007	2.094	2.037	2.925	2.037	3.091	2.039	2.079	306.27	15810.1	0	3.00/3
p2--CCp2--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	961.03	21600.9	0	3.00/3
p2--CCp2--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1023.0	23594.7	0	3.00/3
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.034	3.246	2.037	2.170	2.059	2.936	2.059	3.146	2.061	2.125	420.00	13761.2	0	3.00/3
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.044	3.246	2.047	2.206	2.064	2.975	2.064	3.146	2.066	2.143	349.30	16459.6	0	3.00/3
p2--CCp2--p2	tetra unstr	0.0	2.105	2.705	1.863	2.944	1.864	1.932	2.384	2.720	2.116	2.972	2.122	2.121	954.64	22082.1	0	3.00/3
p2--CCp2--p2	tetra unstr	0.1	2.350	2.825	2.016	2.945	2.018	2.104	2.727	2.903	2.369	2.972	2.376	2.408	1040.0	23997.5	0	3.00/3
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.034	3.251	2.037	2.173	2.059	2.936	2.059	3.149	2.061	2.126	457.56	15535.8	0	3.00/3
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.044	3.251	2.047	2.209	2.064	2.975	2.064	3.149	2.066	2.144	401.16	17813.2	0	3.00/3
p2--CCp2--p3	tetra unstr	0.0	2.105	2.705	1.863	2.958	1.864	1.943	2.384	2.720	2.116	2.980	2.122	2.134	1051.8	22990.9	0	3.00/3
p2--CCp2--p3	tetra unstr	0.1	2.350	2.825	2.016	2.959	2.018	2.111	2.727	2.903	2.369	2.981	2.376	2.414	1088.7	24901.2	0	3.00/3
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.997	4.087	2.002	2.020	2.035	2.884	2.035	4.142	2.037	2.056	411.75	13728.9	0	3.00/3
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.002	4.082	2.007	2.026	2.037	2.925	2.037	4.137	2.039	2.058	401.83	15810.7	0	3.00/3
p2--CCp3--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1500.5	30375.9	0	3.00/3
p2--CCp3--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1552.3	31492.0	0	3.00/3
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.033	4.022	2.037	2.072	2.060	2.936	2.059	4.109	2.061	2.100	520.74	14822.2	0	3.00/3
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.043	4.017	2.047	2.084	2.064	2.975	2.064	4.103	2.066	2.107	460.62	16473.4	0	3.00/3
p2--CCp3--p2	tetra unstr	0.0	2.105	2.705	1.863	3.893	1.864	1.897	2.384	2.720	2.116	3.841	2.123	2.129	1501.0	30835.0	0	3.00/3
p2--CCp3--p2	tetra unstr	0.1	2.350	2.825	2.016	3.911	2.018	2.048	2.727	2.903	2.369	3.890	2.376	2.392	1579.6	31952.0	0	3.00/3
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.033	4.014	2.037	2.073	2.060	2.936	2.059	4.110	2.061	2.101	681.86	17100.2	0	3.00/3
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.043	4.010	2.047	2.085	2.064	2.975	2.064	4.104	2.066	2.107	571.45	18143.3	0	3.00/3
p2--CCp3--p3	tetra unstr	0.0	2.105	2.705	1.863	3.888	1.864	1.929	2.384	2.720	2.116	3.831	2.123	2.148	1532.3	31743.1	0	3.00/3
p2--CCp3--p3	tetra unstr	0.1	2.350	2.825	2.016	3.907	2.018	2.074	2.727	2.903	2.369	3.884	2.376	2.406	1612.3	32870.0	0	3.00/3
p2--GP--p1	hexa	0.0	2.004	3.107	2.011	3.133	2.844	3.122	2.035	2.884	2.004	3.004	2.447	2.998	435.91	26067.6	0	3.00/3
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.886	3.131	2.037	2.925	2.004	3.021	2.473	3.014	313.10	23991.7	0	3.00/3
p2--GP--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	657.39	17584.1	0	3.00/3
p2--GP--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	692.09	21405.3	0	3.00/3
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.897	3.215	2.060	2.936	2.001	3.092	2.431	3.082	759.22	26717.6	0	3.00/3
p2--GP--p2	hexa	0.1	2.051	3.166	2.004	3.236	2.920	3.221	2.064	2.975	2.001	3.105	2.439	3.094	623.79	24635.8	0	3.00/3
p2--GP--p2	tetra unstr	0.0	2.105	2.705	1.928	2.794	2.100	2.795	2.384	2.720	1.960	2.779	2.102	2.783	887.93	17964.1	0	3.00/3
p2--GP--p2	tetra unstr	0.1	2.350	2.825	1.928	2.831	2.146	2.833	2.727	2.903	1.960	2.862	2.167	2.866	776.48	21693.1	0	3.00/3
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.896	3.220	2.060	2.936	2.000	3.095	2.430	3.085	1431.6	28015.9	0	3.00/3
p2--GP--p3	hexa	0.1	2.051	3.166	2.003	3.241	2.919	3.226	2.064	2.975	2.000	3.108	2.438	3.097	1111.8	25928.3	0	3.00/3
p2--GP--p3	tetra unstr	0.0	2.105	2.705	1.924	2.801	2.082	2.802	2.384	2.720	1.958	2.768	2.089	2.773	1108.6	18824.2	0	3.00/3
p2--GP--p3	tetra unstr	0.1	2.350	2.825	1.924	2.846	2.128	2.847	2.727	2.903	1.958	2.864	2.153	2.868	1250.4	22275.1	0	3.00/3
p3--na--NO	hexa	0.0	2.313	3.204	2.303	2.303	2.307	2.307	2.193	2.830	2.189	2.189	2.192	2.192	1163.8	77061.2	0	3.00/3
p3--na--NO	hexa	0.1	2.278	3.207	2.269	2.269	2.273	2.273	2.169	2.827	2.166	2.166	2.168	2.168	1137.9	77131.2	0	3.00/3
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	1409.3	45038.0	0	3.00/3
p3--na--NO	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	1785.5	53562.0	0	3.00/3
p3--CCp1--p1	hexa	0.0	2.758	3.637	2.675	2.166	2.727	2.263	2.547	3.524	2.476	2.207	2.525	2.199	1051.8	44827.2	0	3.00/3
p3--CCp1--p1	hexa	0.1	2.713	3.646	2.635	2.166	2.687	2.261	2.497	3.532	2.433	2.207	2.481	2.197	965.91	45949.0	0	3.00/3
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2254.3	45663.0	0	3.00/3
p3--CCp1--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	1997.6	53125.5	0	3.00/3
p3--CCp1--p2	hexa	0.0	3.096	3.790	2.987	2.097	3.017	2.094	2.864	3.829	2.773	2.130	2.782	2.063	952.29	45473.8	0	3.00/3
p3--CCp1--p2	hexa	0.1	3.064	3.800	2.965	2.097	2.994	2.093	2.807	3.842	2.734	2.130	2.739	2.063	981.62	46587.4	0	3.00/3
p3--CCp1--p2	tetra unstr	0.0	2.979	3.699	2.857	1.970	2.857	1.968	3.035	3.633	2.949	1.982	2.960	1.961	2125.4	45933.6	0	3.00/3
p3--CCp1--p2	tetra unstr	0.1	3.071	3.771	2.960	1.971	2.963	1.968	3.012	3.707	3.038	1.982	3.053	1.960	2058.2	53371.6	0	3.00/3
p3--CCp1--p3	hexa	0.0	3.096	3.790	2.987	2.094	3.017	2.092	2.864	3.829	2.773	2.127	2.782	2.062	1048.1	46766.9	0	3.00/3
p3--CCp1--p3	hexa	0.1	3.064	3.800	2.965	2.094	2.994	2.091	2.807	3.842	2.734	2.127	2.739	2.062	1049.4	47890.8	0	3.00/3
p3--CCp1--p3	tetra unstr	0.0	2.978	3.699	2.856	1.963	2.857	1.966	3.034	3.633	2.949	1.981	2.960	1.960	2280.9	46502.7	0	3.00/3
p3--CCp1--p3	tetra unstr	0.1	3.071	3.772	2.959	1.963	2.963	1.966	3.011	3.707	3.037	1.981	3.053	1.960	2033.4	53871.1	0	3.00/3
p3--CCp2--p1	hexa	0.0	2.782	3.644	2.745	3.172	2.752	2.948	2.574	3.538	2.546	3.092	2.553	2.790	823.62	44830.9	0	3.00/3
p3--CCp2--p1	hexa	0.1	2.739	3.653	2.708	3.172	2.715	2.943	2.524	3.546	2.503	3.092	2.509	2.778	982.87	45950.0	0	3.00/3
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2270.3	46460.9	0	3.00/3
p3--CCp2--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2137.6	53103.6	0	3.00/3
p3--CCp2--p2	hexa	0.0	3.335	3.812	3.202	3.247	3.220	3.232	3.264	3.882	3.065	3.147	3.095	3.171	867.38	45475.8	0	3.00/3
p3--CCp2--p2	hexa	0.1	3.352	3.823	3.223	3.247	3.241	3.244	3.280	3.896	3.083	3.147	3.114	3.186	1040.2	46594.9	0	3.00/3
p3--CCp2--p2	tetra unstr	0.0	3.198	3.777	2.984	2.942	2.981	2.996	3.463	3.810	3.182	2.970	3.187	3.073	2331.7	46813.0	0	3.00/3
p3--CCp2--p2	tetra unstr	0.1	3.452	3.863	3.162	2.942	3.161	3.045	3.775	3.921	3.433	2.970	3.439	3.121	2028.2	53563.8	0	3.00/3
p3--CCp2--p3	hexa	0.0	3.335	3.812	3.202	3.253	3.220	3.231	3.264	3.882	3.065	3.150	3.095	3.171	1096.9	46770.9	0	3.00/3
p3--CCp2--p3	hexa	0.1	3.352	3.823	3.223	3.253	3.241	3.243	3.279	3.896	3.083	3.150	3.114	3.185	1155.1	47889.2	0	3.00/3
p3--CCp2--p3	tetra unstr	0.0	3.198	3.777	2.984	2.957	2.981	3.007	3.463	3.810	3.182	2.980	3.187	3.076	2405.5	47513.2	0	3.00/3
p3--CCp2--p3	tetra unstr	0.1	3.451	3.863	3.162	2.957	3.160	3.055	3.776	3.921	3.433	2.980	3.439	3.123	2108.3	54156.0	0	3.00/3
p3--CCp3--p1	hexa	0.0	2.781	3.644	2.742	4.034	2.751	2.764	2.573	3.538	2.546	4.068	2.552	2.578	1355.6	44827.4	0	3.00/3
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.705	4.034	2.713	2.735	2.523	3.546	2.502	4.068	2.508	2.535	1142.3	45947.2	0	3.00/3
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2810.5	52501.8	0	3.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p3--CCp3--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2501.8	57652.3	0	3.00/3
p3--CCp3--p2	hexa	0.0	3.332	3.813	3.194	3.978	3.216	3.106	3.260	3.882	3.069	4.042	3.091	3.018	1421.8	45470.9	0	3.00/3
p3--CCp3--p2	hexa	0.1	3.349	3.823	3.214	3.978	3.236	3.135	3.275	3.896	3.087	4.042	3.109	3.036	1175.8	46593.2	0	3.00/3
p3--CCp3--p2	tetra unstr	0.0	3.198	3.777	2.985	3.943	2.981	3.010	3.463	3.810	3.182	3.960	3.187	3.210	2790.9	52920.7	0	3.00/3
p3--CCp3--p2	tetra unstr	0.1	3.452	3.863	3.163	3.943	3.161	3.182	3.775	3.921	3.433	3.960	3.438	3.453	2518.8	58052.7	0	3.00/3
p3--CCp3--p3	hexa	0.0	3.332	3.813	3.194	3.971	3.216	3.106	3.260	3.882	3.069	4.044	3.091	3.018	1510.6	46764.8	0	3.00/3
p3--CCp3--p3	hexa	0.1	3.349	3.823	3.214	3.971	3.236	3.136	3.275	3.896	3.087	4.044	3.109	3.037	1445.5	47890.3	0	3.00/3
p3--CCp3--p3	tetra unstr	0.0	3.197	3.777	2.984	3.944	2.980	3.056	3.462	3.810	3.182	3.962	3.186	3.240	2937.0	53766.4	0	3.00/3
p3--CCp3--p3	tetra unstr	0.1	3.451	3.863	3.162	3.944	3.160	3.224	3.775	3.921	3.433	3.962	3.438	3.483	2602.0	58874.8	0	3.00/3
p3--GP--p1	hexa	0.0	2.781	3.644	2.003	3.789	2.716	3.807	2.573	3.538	2.000	3.679	2.354	3.683	1183.8	77275.7	0	3.00/3
p3--GP--p1	hexa	0.1	2.738	3.653	2.003	3.796	2.706	3.814	2.523	3.546	2.000	3.687	2.345	3.692	1122.7	77341.2	0	3.00/3
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	1767.9	45258.5	0	3.00/3
p3--GP--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	1954.2	53638.4	0	3.00/3
p3--GP--p2	hexa	0.0	3.332	3.813	1.994	3.892	2.454	3.910	3.260	3.882	1.996	3.955	2.171	3.965	1548.8	77916.9	0	3.00/3
p3--GP--p2	hexa	0.1	3.350	3.823	1.994	3.896	2.447	3.914	3.275	3.896	1.996	3.961	2.167	3.971	1229.3	77991.0	0	3.00/3
p3--GP--p2	tetra unstr	0.0	3.198	3.777	1.927	3.839	2.032	3.841	3.463	3.810	1.961	3.829	1.977	3.835	1666.6	45510.1	0	3.00/3
p3--GP--p2	tetra unstr	0.1	3.452	3.863	1.927	3.878	2.028	3.879	3.775	3.921	1.961	3.893	1.974	3.898	1704.4	53890.6	0	3.00/3
p3--GP--p3	hexa	0.0	3.332	3.813	1.994	3.884	2.452	3.904	3.260	3.882	1.995	3.956	2.170	3.967	2130.9	79212.1	0	3.00/3
p3--GP--p3	hexa	0.1	3.349	3.823	1.993	3.889	2.446	3.908	3.275	3.896	1.995	3.962	2.167	3.973	1565.6	79288.0	0	3.00/3
p3--GP--p3	tetra unstr	0.0	3.197	3.777	1.923	3.838	2.021	3.840	3.462	3.810	1.958	3.826	1.974	3.833	1853.5	46006.8	0	3.00/3
p3--GP--p3	tetra unstr	0.1	3.451	3.863	1.923	3.879	2.017	3.880	3.775	3.921	1.958	3.893	1.971	3.900	1872.5	54389.9	0	3.00/3

13.4.2 `diagonalIion`

$\Delta t = 10^{-2}$

Table 11: 3D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 21 mesh sizes of the row (105 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--na--NO	hexa	0.0	1.985	1.985	1.994	1.994	1.994	1.994	1.993	1.993	2.009	2.009	2.009	2.009	17.839	4049.9	0	3.00/3
NO--na--NO	hexa	0.1	1.436	1.436	1.297	1.297	1.306	1.306	1.411	1.411	1.214	1.214	1.223	1.223	31.231	6313.3	0	3.00/3
NO--na--NO	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	38.256	5022.6	0	3.00/3
NO--na--NO	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	152.32	12058.9	0	3.00/3
NO--CCp1--p1	hexa	0.0	1.985	1.985	1.998	2.166	1.995	2.246	1.992	1.992	2.010	2.208	2.009	2.189	33.377	4757.2	0	3.00/3
NO--CCp1--p1	hexa	0.1	1.436	1.436	1.287	2.167	1.306	2.230	1.411	1.411	1.205	2.208	1.224	2.159	60.609	6569.4	0	3.00/3
NO--CCp1--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	131.10	6192.3	0	3.00/3
NO--CCp1--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	195.78	12251.9	0	3.00/3
NO--CCp1--p2	hexa	0.0	1.985	1.985	1.999	2.098	1.994	2.090	1.992	1.992	2.011	2.130	2.009	2.062	40.973	6194.9	0	3.00/3
NO--CCp1--p2	hexa	0.1	1.436	1.436	1.283	2.098	1.307	2.087	1.411	1.411	1.201	2.130	1.224	2.056	76.370	7373.5	0	3.00/3
NO--CCp1--p2	tetra unstr	0.0	0.664	0.664	0.375	1.924	0.373	0.457	0.826	0.826	0.617	1.896	0.616	0.669	117.69	6648.8	0	3.00/3
NO--CCp1--p2	tetra unstr	0.1	0.677	0.677	0.398	1.928	0.396	0.483	0.823	0.823	0.629	1.903	0.628	0.683	197.93	12502.6	0	3.00/3
NO--CCp1--p3	hexa	0.0	1.985	1.985	1.999	2.094	1.994	2.088	1.992	1.992	2.011	2.127	2.009	2.061	54.810	9025.9	0	3.00/3
NO--CCp1--p3	hexa	0.1	1.436	1.436	1.283	2.095	1.307	2.085	1.411	1.411	1.201	2.127	1.224	2.054	103.38	10166.2	0	3.00/3
NO--CCp1--p3	tetra unstr	0.0	0.664	0.664	0.375	1.923	0.373	0.460	0.826	0.826	0.617	1.906	0.616	0.670	119.98	7667.8	0	3.00/3
NO--CCp1--p3	tetra unstr	0.1	0.677	0.677	0.398	1.926	0.396	0.485	0.823	0.823	0.629	1.912	0.628	0.684	203.25	12998.5	0	3.00/3
NO--CCp2--p1	hexa	0.0	1.977	1.977	2.009	3.171	1.987	2.618	1.989	1.989	2.014	3.091	2.005	2.390	52.204	5530.9	0	3.00/3
NO--CCp2--p1	hexa	0.1	1.447	1.447	1.257	3.172	1.318	2.180	1.426	1.426	1.176	3.092	1.239	1.626	76.123	6923.8	0	3.00/3
NO--CCp2--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	202.22	9409.5	0	3.00/3
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	256.48	13682.7	0	3.00/3
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.010	3.247	1.979	2.753	1.985	1.985	2.013	3.146	2.001	2.494	59.701	6947.9	0	3.00/3
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.248	3.248	1.333	2.320	1.441	1.441	1.170	3.147	1.257	1.718	100.29	8180.6	0	3.00/3
NO--CCp2--p2	tetra unstr	0.0	0.664	0.664	0.375	2.080	0.373	0.465	0.826	0.826	0.617	1.273	0.616	0.673	213.68	9939.8	0	3.00/3
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.398	2.128	0.396	0.486	0.823	0.823	0.629	1.333	0.628	0.683	257.37	14142.1	0	3.00/3
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.010	3.252	1.979	2.758	1.985	1.985	2.013	3.149	2.001	2.501	76.927	9777.2	0	3.00/3
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.248	3.253	1.333	2.328	1.441	1.441	1.170	3.150	1.257	1.726	109.29	10937.0	0	3.00/3
NO--CCp2--p3	tetra unstr	0.0	0.664	0.664	0.375	1.992	0.373	0.466	0.826	0.826	0.617	1.241	0.616	0.675	191.52	11027.8	0	3.00/3
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.398	2.039	0.396	0.487	0.823	0.823	0.629	1.295	0.628	0.686	264.47	15030.7	0	3.00/3
NO--CCp3--p1	hexa	0.0	1.980	1.980	2.001	4.033	1.989	2.062	1.991	1.991	2.011	4.072	2.007	2.017	96.054	7428.4	0	3.00/3
NO--CCp3--p1	hexa	0.1	1.452	1.452	1.244	4.015	1.323	1.276	1.429	1.429	1.170	4.028	1.241	1.186	122.78	8775.2	0	3.00/3
NO--CCp3--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	407.78	18177.9	0	3.00/3
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	422.09	22222.9	0	3.00/3
NO--CCp3--p2	hexa	0.0	1.976	1.976	2.000	3.980	1.985	2.107	1.990	1.990	2.011	4.053	2.005	2.022	104.58	8844.7	0	3.00/3
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.233	3.966	1.343	1.305	1.447	1.447	1.163	4.019	1.261	1.187	157.18	9986.5	0	3.00/3
NO--CCp3--p2	tetra unstr	0.0	0.664	0.664	0.375	0.890	0.373	0.418	0.826	0.826	0.617	0.674	0.616	0.657	397.25	18709.8	0	3.00/3
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.398	0.930	0.396	0.441	0.823	0.823	0.629	0.689	0.628	0.670	427.69	22681.6	0	3.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--CCp3--p3	hexa	0.0	1.976	1.976	2.000	3.973	1.985	2.108	1.990	1.990	2.011	4.054	2.005	2.022	125.21	11668.9	0	3.00/3
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.233	3.959	1.343	1.305	1.447	1.447	1.163	4.021	1.261	1.186	159.54	12808.5	0	3.00/3
NO--CCp3--p3	tetra unstr	0.0	0.664	0.664	0.375	0.888	0.373	0.432	0.826	0.826	0.617	0.684	0.616	0.665	428.12	19793.8	0	3.00/3
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.398	0.926	0.396	0.454	0.823	0.823	0.629	0.697	0.628	0.677	432.58	23574.8	0	3.00/3
NO--GP--p1	hexa	0.0	1.984	1.984	1.992	2.072	1.992	2.077	1.992	1.992	1.996	2.038	1.997	2.041	59.598	4506.2	0	3.00/3
NO--GP--p1	hexa	0.1	1.459	1.459	1.992	2.073	1.993	2.079	1.429	1.429	1.996	2.038	1.996	2.041	71.966	6563.7	0	3.00/3
NO--GP--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	51.433	5256.8	0	3.00/3
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	147.95	12251.1	0	3.00/3
NO--GP--p2	hexa	0.0	1.983	1.983	1.984	2.014	1.985	2.018	1.992	1.992	1.992	2.007	1.993	2.009	158.85	5951.4	0	3.00/3
NO--GP--p2	hexa	0.1	1.482	1.482	1.984	2.015	1.985	2.018	1.447	1.447	1.992	2.007	1.993	2.010	155.23	7217.8	0	3.00/3
NO--GP--p2	tetra unstr	0.0	0.664	0.664	1.832	0.861	0.516	0.859	0.826	0.826	1.792	0.867	0.652	0.866	77.643	5716.7	0	3.00/3
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.844	0.883	0.546	0.881	0.823	0.823	1.812	0.884	0.668	0.883	175.86	12501.7	0	3.00/3
NO--GP--p3	hexa	0.0	1.983	1.983	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.993	2.008	399.48	8785.1	0	3.00/3
NO--GP--p3	hexa	0.1	1.482	1.482	1.984	2.011	1.985	2.015	1.447	1.447	1.992	2.006	1.992	2.008	325.33	9791.0	0	3.00/3
NO--GP--p3	tetra unstr	0.0	0.664	0.664	1.828	0.863	0.513	0.862	0.826	0.826	1.790	0.868	0.651	0.868	138.65	6716.1	0	3.00/3
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.542	0.884	0.823	0.823	1.810	0.885	0.667	0.885	256.75	12995.9	0	3.00/3
p1--na--NO	hexa	0.0	1.967	2.095	1.927	1.927	1.930	1.930	1.919	2.125	1.838	1.838	1.845	1.845	76.120	7133.9	0	3.00/3
p1--na--NO	hexa	0.1	2.067	2.093	2.047	2.047	2.048	2.048	2.066	2.124	2.033	2.033	2.035	2.035	77.548	8693.9	0	3.00/3
p1--na--NO	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	261.81	10336.8	0	3.00/3
p1--na--NO	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	251.13	11875.1	0	3.00/3
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.926	2.166	1.930	2.193	1.919	2.125	1.836	2.207	1.845	2.115	102.19	7284.7	0	3.00/3
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.046	2.167	2.048	2.225	2.066	2.124	2.033	2.208	2.035	2.168	108.52	8861.8	0	3.00/3
p1--CCp1--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	326.80	11245.2	0	3.00/3
p1--CCp1--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	280.71	12509.6	0	3.00/3
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.925	2.098	1.930	2.083	1.919	2.125	1.835	2.130	1.845	2.049	112.64	8199.6	0	3.00/3
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.046	2.098	2.047	2.090	2.066	2.124	2.033	2.130	2.035	2.061	102.35	9508.0	0	3.00/3
p1--CCp1--p2	tetra unstr	0.0	1.474	1.866	1.139	1.973	1.142	1.493	1.771	1.928	1.419	1.984	1.428	1.590	360.40	11702.0	0	3.00/3
p1--CCp1--p2	tetra unstr	0.1	1.613	1.909	1.262	1.973	1.264	1.621	1.889	1.963	1.556	1.983	1.558	1.721	283.85	12931.0	0	3.00/3
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.925	2.094	1.930	2.081	1.919	2.125	1.835	2.127	1.845	2.048	126.80	11006.7	0	3.00/3
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.046	2.094	2.047	2.088	2.066	2.124	2.033	2.127	2.035	2.060	121.17	11450.9	0	3.00/3
p1--CCp1--p3	tetra unstr	0.0	1.474	1.866	1.139	1.965	1.142	1.505	1.771	1.928	1.419	1.982	1.428	1.597	366.81	12637.4	0	3.00/3
p1--CCp1--p3	tetra unstr	0.1	1.613	1.909	1.262	1.965	1.264	1.632	1.889	1.963	1.556	1.982	1.558	1.727	288.00	13736.6	0	3.00/3
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.926	3.169	1.929	2.043	1.919	2.125	1.831	3.089	1.845	1.844	108.14	7668.9	0	3.00/3
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.050	3.170	2.046	2.200	2.066	2.124	2.033	3.090	2.034	2.090	133.55	8866.9	0	3.00/3
p1--CCp2--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	383.95	14523.2	0	3.00/3
p1--CCp2--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	340.65	15430.1	0	3.00/3
p1--CCp2--p2	hexa	0.0	1.965	2.095	1.925	3.244	1.929	2.095	1.919	2.125	1.828	3.143	1.845	1.846	119.40	8935.3	0	3.00/3
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.051	3.245	2.045	2.265	2.065	2.124	2.033	3.145	2.034	2.115	124.52	9632.8	0	3.00/3
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.139	2.917	1.142	1.199	1.771	1.928	1.418	2.913	1.428	1.465	406.79	14966.3	0	3.00/3
p1--CCp2--p2	tetra unstr	0.1	1.613	1.909	1.262	2.928	1.264	1.336	1.889	1.963	1.555	2.938	1.558	1.614	343.96	15843.4	0	3.00/3
p1--CCp2--p3	hexa	0.0	1.965	2.095	1.925	3.250	1.929	2.098	1.919	2.125	1.828	3.146	1.845	1.848	138.47	11754.3	0	3.00/3
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.051	3.251	2.045	2.269	2.065	2.124	2.033	3.148	2.034	2.117	146.88	12096.8	0	3.00/3
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.139	2.922	1.142	1.211	1.771	1.928	1.418	2.906	1.428	1.476	419.30	15835.9	0	3.00/3
p1--CCp2--p3	tetra unstr	0.1	1.613	1.909	1.262	2.938	1.264	1.348	1.889	1.963	1.555	2.938	1.558	1.624	349.25	16789.7	0	3.00/3
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.922	4.067	1.929	1.877	1.919	2.125	1.830	4.138	1.845	1.729	154.68	9571.5	0	3.00/3
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.046	4.057	2.047	2.042	2.066	2.124	2.032	4.119	2.034	2.007	181.97	9945.3	0	3.00/3
p1--CCp3--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	587.62	23052.2	0	3.00/3
p1--CCp3--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	503.46	24024.5	0	3.00/3
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.920	4.009	1.930	1.853	1.920	2.125	1.827	4.112	1.846	1.682	204.40	10809.0	0	3.00/3
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.045	4.000	2.047	2.039	2.066	2.124	2.031	4.092	2.034	1.993	200.25	11136.9	0	3.00/3
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.139	2.936	1.142	1.202	1.771	1.928	1.418	2.110	1.428	1.472	609.85	23479.2	0	3.00/3
p1--CCp3--p2	tetra unstr	0.1	1.613	1.909	1.262	3.163	1.264	1.332	1.889	1.963	1.555	2.437	1.558	1.614	508.43	24409.8	0	3.00/3
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.920	4.003	1.930	1.852	1.920	2.125	1.827	4.114	1.846	1.680	228.53	13640.9	0	3.00/3
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.045	3.992	2.047	2.039	2.066	2.124	2.031	4.094	2.034	1.993	218.65	13990.9	0	3.00/3
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.139	2.837	1.142	1.240	1.771	1.928	1.418	2.066	1.428	1.496	657.70	24392.8	0	3.00/3
p1--CCp3--p3	tetra unstr	0.1	1.613	1.909	1.262	3.084	1.264	1.375	1.889	1.963	1.555	2.395	1.558	1.644	515.30	25375.1	0	3.00/3
p1--GP--p1	hexa	0.0	1.967	2.095	1.992	2.157	1.999	2.164	1.919	2.125	1.997	2.201	2.001	2.201	104.41	7341.0	0	3.00/3
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.002	2.162	2.066	2.124	1.997	2.199	2.005	2.199	109.30	8899.4	0	3.00/3
p1--GP--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	247.77	10396.4	0	3.00/3
p1--GP--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	222.16	11931.7	0	3.00/3
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.920	2.125	1.993	2.129	1.997	2.128	190.37	8084.3	0	3.00/3
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.093	1.991	2.096	2.066	2.124	1.993	2.128	1.998	2.126	187.99	9545.7	0	3.00/3
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.771	1.928	1.953	1.877	1.777	1.880	288.75	10853.6	0	3.00/3
p1--GP--p2	tetra unstr	0.1	1.613	1.909	1.920	1.897	1.790	1.900	1.889	1.963	1.955	1.921	1.848	1.922	250.54	12308.8	0	3.00/3
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.920	2.125	1.993	2.126	1.996	2.125	374.96	10772.3	0	3.00/3
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.066	2.124	1.993	2.124	1.998	2.123	363.94	11158.1	0	3.00/3
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.874	1.771	1.928	1.951	1.889	1.775	1.892	341.86	11836.5	0	3.00/3
p1--GP--p3	tetra unstr	0.1	1.613	1.909	1.916	1.900	1.786	1.903	1.889	1.963	1.953	1.927	1.846	1.929	311.35	13265.5	0	3.00/3
p2--na--NO	hexa	0.0	1.947	3.025	1.944	1.944	1.948	1.948	2.001	2.766	2.003	2.003	2.005	2.005	159.49	25851.3	0	3.00/3
p2--na--NO	hexa	0.1	1.946	3.053	1.943	1.943	1.947	1.947	2.000	2.807	2.002	2.002	2.005	2.005	153.99	23778.1	0	3.00/3
p2--na--NO	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	410.45	18237.1	0	3.00/3
p2--na--NO	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	380.78	21603.7	0	3.00/3
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.995	2.168	2.003	2.183	2.035	2.884	2.035	2.209	2.040	2.147	169.04	13059.4	0	3.00/3
p2--CCp1--p1	hexa	0.1	2.008	3.133	2.000	2.167	2.008	2.193	2.037	2.925	2.037	2.209	2.042	2.153	209.15	15810.5	0	3.00/3
p2--CCp1--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	623.79	19036.2	0	3.00/3
p2--CCp1--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	448.44	21893.6	0	3.00/3
p2--CCp1--p2	hexa	0.0	2.040	3.142	2.028	2.098	2.039	2.087	2.059	2.936	2.057	2.131	2.064	2.063	172.60	13704.2	0	3.00/3
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.037	2.098	2.049	2.088	2.063	2.975	2.062	2.131	2.069	2.063	185.63	16455.6	0	3.00/3
p2--CCp1--p2	tetra unstr	0.0	2.103	2.703	1.863	1.971	1.864	1.960	2.383	2.718	2.113	1.982	2.121	1.974	614.57	19514.9	0	3.00/4
p2--CCp1--p2	tetra unstr	0.1	2.347	2.823	2.013	1.971	2.016	1.970	2.727	2.903	2.362	1.982	2.372	1.985	575.02	22293.9	0	3.00/4
p2--CCp1--p3	hexa	0.0	2.040	3.142	2.028	2.095	2.039	2.085	2.059	2.936	2.057	2.127	2.064	2.061	204.59	15345.8	0	3.00/3
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.037	2.095	2.049	2.086	2.063	2.975	2.062	2.127	2.069	2.061	214.83	17754.1	0	3.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p2--CCp1--p3	tetra unstr	0.0	2.103	2.703	1.863	1.963	1.864	1.958	2.383	2.718	2.112	1.981	2.121	1.972	616.65	20468.9	0	3.00/4
p2--CCp1--p3	tetra unstr	0.1	2.347	2.823	2.013	1.963	2.016	1.968	2.727	2.903	2.362	1.981	2.372	1.983	557.20	23166.7	0	3.00/4
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.999	3.170	2.004	2.077	2.035	2.884	2.038	3.091	2.040	2.074	196.47	13056.7	0	3.00/3
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.005	3.171	2.009	2.096	2.037	2.925	2.040	3.091	2.043	2.083	175.37	15810.1	0	3.00/3
p2--CCp2--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	632.62	22303.5	0	3.00/3
p2--CCp2--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	636.46	24373.4	0	3.00/3
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.036	3.246	2.040	2.173	2.059	2.936	2.063	3.146	2.065	2.129	209.25	13797.8	0	3.00/3
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.046	3.246	2.050	2.209	2.064	2.975	2.068	3.146	2.070	2.147	206.11	16457.2	0	3.00/3
p2--CCp2--p2	tetra unstr	0.0	2.103	2.704	1.863	2.944	1.864	1.932	2.384	2.719	2.114	2.972	2.121	2.120	659.62	22775.0	0	3.00/4
p2--CCp2--p2	tetra unstr	0.1	2.349	2.824	2.014	2.945	2.016	2.103	2.732	2.904	2.367	2.972	2.373	2.405	574.74	24845.4	0	3.00/4
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.036	3.251	2.040	2.176	2.059	2.936	2.063	3.149	2.065	2.130	232.99	15899.4	0	3.00/3
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.046	3.251	2.050	2.212	2.064	2.975	2.068	3.149	2.070	2.149	215.34	17754.7	0	3.00/3
p2--CCp2--p3	tetra unstr	0.0	2.103	2.704	1.863	2.958	1.864	1.943	2.384	2.719	2.114	2.980	2.121	2.133	670.04	23686.6	0	3.00/4
p2--CCp2--p3	tetra unstr	0.1	2.349	2.824	2.014	2.959	2.016	2.110	2.732	2.904	2.367	2.981	2.373	2.411	579.70	25797.4	0	3.00/4
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.999	4.087	2.004	2.022	2.035	2.884	2.038	4.142	2.040	2.059	217.71	13837.2	0	3.00/3
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.004	4.082	2.009	2.028	2.037	2.925	2.040	4.137	2.043	2.062	220.37	15812.2	0	3.00/3
p2--CCp3--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	810.83	30956.5	0	3.00/3
p2--CCp3--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	778.61	32393.4	0	3.00/3
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.035	4.022	2.040	2.074	2.059	2.936	2.063	4.109	2.065	2.104	243.82	14932.0	0	3.00/3
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.045	4.017	2.050	2.087	2.064	2.975	2.068	4.103	2.070	2.111	245.01	16536.1	0	3.00/3
p2--CCp3--p2	tetra unstr	0.0	2.103	2.704	1.863	3.893	1.864	1.896	2.384	2.719	2.114	3.841	2.121	2.127	812.04	31354.9	0	3.00/4
p2--CCp3--p2	tetra unstr	0.1	2.349	2.824	2.014	3.911	2.016	2.046	2.732	2.904	2.367	3.890	2.373	2.389	790.86	32882.0	0	3.00/4
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.035	4.014	2.040	2.075	2.059	2.936	2.063	4.110	2.065	2.105	256.46	17495.5	0	3.00/3
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.045	4.010	2.050	2.087	2.064	2.975	2.068	4.104	2.070	2.112	259.96	18178.2	0	3.00/3
p2--CCp3--p3	tetra unstr	0.0	2.103	2.704	1.863	3.888	1.864	1.929	2.384	2.719	2.114	3.831	2.121	2.146	944.89	32180.7	0	3.00/4
p2--CCp3--p3	tetra unstr	0.1	2.349	2.824	2.014	3.907	2.016	2.072	2.732	2.904	2.367	3.883	2.373	2.403	819.01	33736.5	0	3.00/4
p2--GP--p1	hexa	0.0	2.003	3.107	2.011	3.133	2.845	3.122	2.035	2.884	2.004	3.005	2.449	2.999	209.03	26058.5	0	3.00/3
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.888	3.131	2.037	2.925	2.004	3.021	2.476	3.015	171.87	23988.3	0	3.00/3
p2--GP--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	419.78	18303.6	0	3.00/3
p2--GP--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	402.79	21700.4	0	3.00/3
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.898	3.215	2.059	2.936	2.001	3.092	2.432	3.083	291.55	26710.8	0	3.00/3
p2--GP--p2	hexa	0.1	2.050	3.166	2.004	3.236	2.920	3.222	2.064	2.975	2.001	3.105	2.440	3.095	265.41	24629.7	0	3.00/3
p2--GP--p2	tetra unstr	0.0	2.103	2.704	1.928	2.794	2.100	2.796	2.384	2.719	1.960	2.779	2.102	2.784	421.04	18693.3	0	3.00/4
p2--GP--p2	tetra unstr	0.1	2.349	2.824	1.928	2.831	2.146	2.833	2.732	2.904	1.960	2.861	2.166	2.866	432.63	21997.2	0	3.00/4
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.897	3.220	2.059	2.936	2.000	3.095	2.432	3.085	524.46	28002.3	0	3.00/3
p2--GP--p3	hexa	0.1	2.050	3.166	2.003	3.242	2.920	3.227	2.064	2.975	2.000	3.108	2.439	3.097	385.37	25931.2	0	3.00/3
p2--GP--p3	tetra unstr	0.0	2.103	2.704	1.924	2.802	2.082	2.802	2.384	2.719	1.958	2.769	2.088	2.773	463.12	19698.3	0	3.00/4
p2--GP--p3	tetra unstr	0.1	2.349	2.824	1.924	2.847	2.128	2.847	2.732	2.904	1.958	2.864	2.152	2.868	486.74	22652.1	0	3.00/4
p3--na--NO	hexa	0.0	2.313	3.204	2.308	2.308	2.312	2.312	2.193	2.830	2.197	2.197	2.200	2.200	784.33	77066.1	0	3.00/3
p3--na--NO	hexa	0.1	2.278	3.207	2.274	2.274	2.278	2.278	2.169	2.826	2.174	2.174	2.176	2.176	737.90	76981.0	0	3.00/3
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1208.8	45193.0	0	3.00/3
p3--na--NO	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1221.0	53420.2	0	3.00/3
p3--CCp1--p1	hexa	0.0	2.761	3.638	2.698	2.166	2.745	2.264	2.550	3.526	2.508	2.207	2.552	2.199	959.09	44827.9	0	3.00/3
p3--CCp1--p1	hexa	0.1	2.717	3.647	2.659	2.166	2.706	2.261	2.501	3.534	2.465	2.207	2.509	2.198	861.81	45950.9	0	3.00/3
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1763.5	45814.3	0	3.00/3
p3--CCp1--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1488.6	53115.7	0	3.00/3
p3--CCp1--p2	hexa	0.0	3.129	3.794	3.046	2.097	3.078	2.094	2.914	3.838	2.857	2.130	2.869	2.063	870.58	45473.1	0	3.00/3
p3--CCp1--p2	hexa	0.1	3.103	3.804	3.033	2.097	3.064	2.093	2.862	3.851	2.828	2.130	2.835	2.063	929.39	46595.2	0	3.00/3
p3--CCp1--p2	tetra unstr	0.0	3.009	3.711	2.897	1.970	2.893	1.968	3.092	3.661	3.022	1.982	3.021	1.961	1680.0	46130.8	0	3.00/3
p3--CCp1--p2	tetra unstr	0.1	3.120	3.786	3.019	1.970	3.015	1.968	3.099	3.741	3.149	1.982	3.147	1.960	1559.0	53372.8	0	3.00/3
p3--CCp1--p3	hexa	0.0	3.129	3.794	3.046	2.094	3.078	2.092	2.914	3.838	2.857	2.127	2.869	2.062	856.84	46769.8	0	3.00/3
p3--CCp1--p3	hexa	0.1	3.103	3.804	3.033	2.094	3.064	2.091	2.862	3.851	2.828	2.127	2.835	2.062	942.46	47890.2	0	3.00/3
p3--CCp1--p3	tetra unstr	0.0	3.009	3.711	2.897	1.963	2.892	1.966	3.091	3.661	3.021	1.981	3.021	1.960	1636.3	46753.4	0	3.00/3
p3--CCp1--p3	tetra unstr	0.1	3.120	3.786	3.018	1.963	3.015	1.966	3.099	3.741	3.149	1.981	3.147	1.960	1566.4	53951.5	0	3.00/3
p3--CCp2--p1	hexa	0.0	2.781	3.643	2.755	3.172	2.763	2.955	2.572	3.538	2.566	3.092	2.573	2.804	852.69	44826.7	0	3.00/3
p3--CCp2--p1	hexa	0.1	2.738	3.653	2.719	3.172	2.726	2.951	2.523	3.546	2.524	3.092	2.530	2.794	884.41	45948.7	0	3.00/3
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1617.2	47060.5	0	3.00/3
p3--CCp2--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1545.2	53594.1	0	3.00/3
p3--CCp2--p2	hexa	0.0	3.332	3.812	3.211	3.247	3.230	3.235	3.260	3.882	3.083	3.147	3.113	3.176	877.41	45474.7	0	3.00/3
p3--CCp2--p2	hexa	0.1	3.350	3.823	3.234	3.247	3.253	3.247	3.275	3.896	3.103	3.147	3.135	3.190	926.79	46596.9	0	3.00/3
p3--CCp2--p2	tetra unstr	0.0	3.196	3.775	2.986	2.942	2.983	2.997	3.461	3.809	3.184	2.970	3.189	3.072	1693.7	47445.3	0	3.00/3
p3--CCp2--p2	tetra unstr	0.1	3.450	3.862	3.163	2.942	3.162	3.045	3.777	3.920	3.434	2.970	3.440	3.120	1589.7	53954.3	0	3.00/3
p3--CCp2--p3	hexa	0.0	3.332	3.812	3.211	3.253	3.230	3.234	3.260	3.882	3.083	3.150	3.113	3.175	923.11	46766.8	0	3.00/3
p3--CCp2--p3	hexa	0.1	3.350	3.823	3.234	3.253	3.253	3.246	3.275	3.896	3.103	3.150	3.135	3.189	974.77	47890.8	0	3.00/3
p3--CCp2--p3	tetra unstr	0.0	3.195	3.776	2.985	2.957	2.983	3.007	3.461	3.809	3.184	2.980	3.188	3.076	1477.0	48215.3	0	3.00/3
p3--CCp2--p3	tetra unstr	0.1	3.450	3.862	3.163	2.957	3.161	3.054	3.777	3.921	3.434	2.980	3.440	3.122	1530.7	54668.5	0	3.00/3
p3--CCp3--p1	hexa	0.0	2.780	3.644	2.753	4.033	2.762	2.775	2.572	3.538	2.566	4.067	2.573	2.597	886.39	44826.9	0	3.00/3
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.716	4.033	2.725	2.746	2.522	3.545	2.523	4.068	2.530	2.555	931.63	45948.9	0	3.00/3
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1735.3	53363.6	0	3.00/3
p3--CCp3--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1703.5	58670.7	0	3.00/3
p3--CCp3--p2	hexa	0.0	3.330	3.812	3.205	3.978	3.228	3.112	3.256	3.881	3.086	4.042	3.110	3.026	875.81	45471.7	0	3.00/3
p3--CCp3--p2	hexa	0.1	3.347	3.823	3.226	3.978	3.250	3.143	3.271	3.895	3.106	4.042	3.131	3.046	1144.0	46591.2	0	3.00/3
p3--CCp3--p2	tetra unstr	0.0	3.196	3.775	2.987	3.943	2.983	3.011	3.461	3.808	3.184	3.959	3.188	3.209	1657.7	53806.1	0	3.00/3
p3--CCp3--p2	tetra unstr	0.1	3.450	3.862	3.164	3.942	3.162	3.181	3.777	3.920	3.434	3.959	3.440	3.451	1699.5	59049.0	0	3.00/3
p3--CCp3--p3	hexa	0.0	3.330	3.812	3.205	3.971	3.228	3.112	3.256	3.881	3.086	4.044	3.110	3.027	945.74	46766.9	0	3.00/3
p3--CCp3--p3	hexa	0.1	3.347	3.823	3.226	3.971	3.250	3.143	3.271	3.895	3.106	4.044	3.131	3.046	1080.8	47885.5	0	3.00/3
p3--CCp3--p3	tetra unstr	0.0	3.195	3.776	2.986	3.944	2.982	3.056	3.460	3.809	3.184	3.961	3.188	3.240	1471.9	54513.0	0	3.00/3
p3--CCp3--p3	tetra unstr	0.1	3.449	3.862	3.163	3.944	3.161	3.224	3.777	3.920	3.434	3.961	3.439	3.481	1691.2	59970.7	0	3.00/3
p3--GP--p1	hexa	0.0	2.780	3.644	2.003	3.792	2.717	3.810	2.572	3.538	2.000	3.687	2.355	3.691	1136.4	77129.5	0	3.00/3
p3--GP--p1	hexa	0.1	2.737	3.653	2.003	3.799	2.707	3.817	2.522	3.546	2.000	3.695	2.347	3.700	826.77	77349.6	0	3.00/3
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1530.2	45303.3	0	3.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p3--GP--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1458.6	53640.9	0	3.00/3
p3--GP--p2	hexa	0.0	3.331	3.812	1.994	3.893	2.454	3.912	3.257	3.881	1.996	3.958	2.171	3.969	1024.7	77926.4	0	3.00/3
p3--GP--p2	hexa	0.1	3.348	3.823	1.994	3.897	2.447	3.916	3.272	3.896	1.996	3.964	2.167	3.975	846.01	77996.9	0	3.00/3
p3--GP--p2	tetra unstr	0.0	3.196	3.775	1.927	3.844	2.032	3.845	3.461	3.808	1.961	3.840	1.977	3.846	1658.5	45560.0	0	3.00/3
p3--GP--p2	tetra unstr	0.1	3.450	3.862	1.927	3.883	2.028	3.884	3.777	3.920	1.961	3.904	1.974	3.909	1726.4	53889.0	0	3.00/3
p3--GP--p3	hexa	0.0	3.331	3.812	1.994	3.886	2.452	3.905	3.257	3.881	1.995	3.960	2.170	3.970	1232.6	79222.2	0	3.00/3
p3--GP--p3	hexa	0.1	3.348	3.823	1.993	3.890	2.446	3.909	3.272	3.896	1.995	3.966	2.167	3.976	1248.1	79294.2	0	3.00/3
p3--GP--p3	tetra unstr	0.0	3.195	3.776	1.923	3.842	2.021	3.844	3.460	3.809	1.958	3.837	1.974	3.844	1679.4	46150.6	0	3.00/3
p3--GP--p3	tetra unstr	0.1	3.450	3.862	1.923	3.884	2.017	3.885	3.777	3.921	1.958	3.904	1.971	3.911	1530.7	54383.0	0	3.00/3

$\Delta t = 10^{-3}$

Table 12: 3D manufactured results for `diagonalIion`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 21 mesh sizes of the row (1050 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--na--NO	hexa	0.0	1.985	1.985	1.985	1.985	1.985	1.985	1.993	1.993	1.993	1.993	1.993	1.993	122.72	4051.0	0	2.00/2
NO--na--NO	hexa	0.1	1.438	1.438	1.293	1.293	1.302	1.302	1.413	1.413	1.211	1.211	1.220	1.220	157.38	6313.0	0	2.00/2
NO--na--NO	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	201.73	5025.8	0	2.00/3
NO--na--NO	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	592.24	12057.2	0	2.00/3
NO--CCp1--p1	hexa	0.0	1.986	1.986	1.989	2.167	1.986	2.246	1.993	1.993	1.995	2.208	1.993	2.188	161.77	4760.6	0	2.00/2
NO--CCp1--p1	hexa	0.1	1.438	1.438	1.298	2.167	1.302	2.230	1.413	1.413	1.212	2.208	1.220	2.158	290.23	6574.4	0	2.00/2
NO--CCp1--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	386.53	6193.2	0	2.00/3
NO--CCp1--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	669.90	12252.5	0	2.00/3
NO--CCp1--p2	hexa	0.0	1.986	1.986	1.991	2.098	1.986	2.090	1.993	1.993	1.995	2.130	1.993	2.062	223.61	6202.6	0	2.00/2
NO--CCp1--p2	hexa	0.1	1.438	1.438	1.300	2.098	1.303	2.086	1.413	1.413	1.212	2.130	1.220	2.055	302.02	7373.6	0	2.00/2
NO--CCp1--p2	tetra unstr	0.0	0.663	0.663	0.377	1.924	0.375	0.459	0.829	0.829	0.617	1.896	0.617	0.670	403.89	6652.1	0	2.00/3
NO--CCp1--p2	tetra unstr	0.1	0.677	0.677	0.400	1.928	0.398	0.484	0.828	0.828	0.629	1.903	0.629	0.684	710.39	12503.9	0	2.00/3
NO--CCp1--p3	hexa	0.0	1.986	1.986	1.991	2.094	1.986	2.088	1.993	1.993	1.995	2.127	1.993	2.060	305.02	9028.5	0	2.00/2
NO--CCp1--p3	hexa	0.1	1.438	1.438	1.300	2.095	1.303	2.085	1.413	1.413	1.212	2.127	1.220	2.054	442.70	10162.7	0	2.00/2
NO--CCp1--p3	tetra unstr	0.0	0.663	0.663	0.377	1.922	0.375	0.462	0.829	0.829	0.617	1.906	0.617	0.671	474.23	7668.9	0	2.00/3
NO--CCp1--p3	tetra unstr	0.1	0.677	0.677	0.400	1.926	0.398	0.487	0.828	0.828	0.629	1.912	0.629	0.685	749.34	13003.3	0	2.00/3
NO--CCp2--p1	hexa	0.0	1.977	1.977	2.002	3.171	1.978	2.610	1.989	1.989	1.999	3.091	1.989	2.376	231.17	5533.9	0	2.00/2
NO--CCp2--p1	hexa	0.1	1.447	1.447	1.258	3.172	1.313	2.174	1.428	1.428	1.175	3.092	1.236	1.620	346.79	6924.0	0	2.00/2
NO--CCp2--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	772.64	9414.5	0	2.00/3
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	960.39	13641.3	0	2.00/3
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.004	3.247	1.970	2.745	1.985	1.985	1.998	3.146	1.985	2.481	293.43	6950.3	0	2.00/2
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.251	3.248	1.326	2.314	1.443	1.443	1.170	3.147	1.253	1.712	405.66	8181.0	0	2.00/2
NO--CCp2--p2	tetra unstr	0.0	0.663	0.663	0.377	2.079	0.375	0.467	0.829	0.829	0.617	1.272	0.617	0.673	833.91	9946.2	0	2.00/3
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.400	2.127	0.398	0.488	0.828	0.828	0.629	1.333	0.629	0.684	1015.5	14096.3	0	2.00/3
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.004	3.252	1.970	2.751	1.985	1.985	1.998	3.149	1.985	2.488	427.37	9779.4	0	2.00/2
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.251	3.253	1.326	2.323	1.443	1.443	1.170	3.150	1.253	1.720	576.86	10941.5	0	2.00/2
NO--CCp2--p3	tetra unstr	0.0	0.663	0.663	0.378	1.991	0.375	0.469	0.829	0.829	0.617	1.240	0.617	0.676	814.59	11030.6	0	2.00/3
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.400	2.038	0.398	0.489	0.828	0.828	0.629	1.295	0.629	0.686	1051.9	14990.0	0	2.00/3
NO--CCp3--p1	hexa	0.0	1.980	1.980	1.993	4.033	1.980	2.052	1.992	1.992	1.995	4.072	1.992	2.002	427.29	7432.8	0	2.00/2
NO--CCp3--p1	hexa	0.1	1.453	1.453	1.241	4.015	1.318	1.271	1.431	1.431	1.167	4.028	1.238	1.184	537.11	8767.6	0	2.00/2
NO--CCp3--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	1766.1	18179.3	0	2.00/3
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	1720.8	22209.3	0	2.00/3
NO--CCp3--p2	hexa	0.0	1.975	1.975	1.992	3.981	1.975	2.098	1.990	1.990	1.995	4.053	1.990	2.007	508.10	8849.4	0	2.00/2
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.230	3.966	1.336	1.300	1.449	1.449	1.160	4.019	1.258	1.185	645.50	9995.3	0	2.00/2
NO--CCp3--p2	tetra unstr	0.0	0.663	0.663	0.378	0.891	0.375	0.420	0.829	0.829	0.617	0.675	0.617	0.658	1800.1	18709.2	0	2.00/3
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.400	0.930	0.398	0.443	0.828	0.828	0.629	0.689	0.629	0.671	1826.2	22660.9	0	2.00/3
NO--CCp3--p3	hexa	0.0	1.975	1.975	1.992	3.973	1.975	2.099	1.991	1.991	1.995	4.055	1.990	2.007	682.75	11674.3	0	2.00/2
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.230	3.959	1.336	1.300	1.449	1.449	1.160	4.021	1.258	1.184	835.59	12809.8	0	2.00/2
NO--CCp3--p3	tetra unstr	0.0	0.663	0.663	0.378	0.888	0.375	0.434	0.829	0.829	0.617	0.684	0.617	0.666	1936.3	19795.1	0	2.00/3
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.400	0.927	0.398	0.456	0.828	0.828	0.629	0.698	0.629	0.678	1866.5	23532.0	0	2.00/3
NO--GP--p1	hexa	0.0	1.985	1.985	1.992	2.071	1.992	2.077	1.993	1.993	1.996	2.038	1.996	2.041	299.85	4511.0	0	2.00/2
NO--GP--p1	hexa	0.1	1.461	1.461	1.992	2.073	1.992	2.079	1.431	1.431	1.996	2.038	1.995	2.041	345.03	6562.4	0	2.00/2
NO--GP--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	254.13	5262.5	0	2.00/3
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	586.17	12250.4	0	2.00/3
NO--GP--p2	hexa	0.0	1.984	1.984	1.984	2.014	1.984	2.018	1.992	1.992	1.992	2.007	1.993	2.009	979.42	5953.0	0	2.00/2
NO--GP--p2	hexa	0.1	1.483	1.483	1.984	2.014	1.985	2.018	1.450	1.450	1.992	2.007	1.992	2.009	849.58	7217.6	0	2.00/2
NO--GP--p2	tetra unstr	0.0	0.663	0.663	1.832	0.862	0.518	0.860	0.829	0.829	1.792	0.867	0.653	0.866	441.41	5718.4	0	2.00/3
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.843	0.883	0.547	0.882	0.828	0.828	1.812	0.884	0.668	0.883	880.58	12497.6	0	2.00/3
NO--GP--p3	hexa	0.0	1.984	1.984	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.992	2.008	1784.3	8779.5	0	2.00/2
NO--GP--p3	hexa	0.1	1.483	1.483	1.984	2.011	1.984	2.015	1.450	1.450	1.992	2.005	1.992	2.008	1811.7	9791.1	0	2.00/2
NO--GP--p3	tetra unstr	0.0	0.663	0.663	1.828	0.864	0.514	0.862	0.829	0.829	1.790	0.869	0.652	0.868	946.10	6720.5	0	2.00/3

Vm-states-Iion	Mesh type	α	Fit N = 10–30						Fit N = 20–30						t_Σ [s]	RSS$_\Sigma$ [MB]	RB	n_{nl}
			V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G				
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.543	0.884	0.828	0.828	1.810	0.886	0.667	0.885	1222.1	12996.5	0	2.00/3
p1--na--NO	hexa	0.0	1.967	2.095	1.924	1.924	1.927	1.927	1.920	2.125	1.833	1.833	1.840	1.840	206.83	7134.2	0	2.00/2
p1--na--NO	hexa	0.1	2.067	2.093	2.044	2.044	2.044	2.044	2.067	2.124	2.028	2.028	2.029	2.029	219.86	8694.8	0	2.00/2
p1--na--NO	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	655.41	10344.2	0	2.00/2
p1--na--NO	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	654.84	11878.0	0	2.00/2
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.924	2.166	1.926	2.192	1.920	2.125	1.833	2.207	1.840	2.114	252.70	7284.9	0	2.00/2
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.043	2.167	2.044	2.224	2.067	2.124	2.028	2.208	2.029	2.167	281.56	8859.1	0	2.00/2
p1--CCp1--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	847.66	11240.4	0	2.00/2
p1--CCp1--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	760.10	12504.4	0	2.00/2
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.924	2.098	1.926	2.083	1.920	2.125	1.833	2.130	1.840	2.049	296.92	8201.4	0	2.00/2
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.043	2.098	2.044	2.090	2.067	2.124	2.028	2.130	2.029	2.061	325.52	9504.8	0	2.00/2
p1--CCp1--p2	tetra unstr	0.0	1.475	1.866	1.142	1.973	1.144	1.493	1.768	1.927	1.422	1.984	1.432	1.592	880.24	11704.0	0	2.00/2
p1--CCp1--p2	tetra unstr	0.1	1.612	1.909	1.265	1.973	1.267	1.622	1.884	1.962	1.558	1.983	1.561	1.723	772.60	12931.7	0	2.00/2
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.924	2.094	1.926	2.081	1.920	2.125	1.833	2.127	1.840	2.048	399.00	11002.1	0	2.00/2
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.043	2.094	2.044	2.088	2.067	2.124	2.028	2.127	2.029	2.059	398.19	11445.5	0	2.00/2
p1--CCp1--p3	tetra unstr	0.0	1.475	1.866	1.142	1.965	1.144	1.506	1.768	1.927	1.422	1.982	1.432	1.599	908.84	12636.4	0	2.00/2
p1--CCp1--p3	tetra unstr	0.1	1.612	1.909	1.265	1.965	1.267	1.633	1.884	1.962	1.558	1.982	1.561	1.729	805.41	13737.5	0	2.00/2
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.924	3.169	1.926	2.040	1.920	2.125	1.827	3.089	1.840	1.840	312.61	7660.9	0	2.00/2
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.048	3.170	2.043	2.196	2.066	2.124	2.028	3.090	2.028	2.085	386.88	8866.5	0	2.00/2
p1--CCp2--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	1202.6	14517.9	0	2.00/2
p1--CCp2--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1051.4	15428.2	0	2.00/2
p1--CCp2--p2	hexa	0.0	1.966	2.095	1.924	3.244	1.925	2.092	1.920	2.125	1.824	3.143	1.840	1.841	373.41	8935.8	0	2.00/2
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.050	3.245	2.042	2.261	2.065	2.124	2.028	3.145	2.028	2.109	392.88	9631.5	0	2.00/2
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.141	2.917	1.145	1.201	1.768	1.927	1.422	2.912	1.432	1.469	1212.9	14965.3	0	2.00/2
p1--CCp2--p2	tetra unstr	0.1	1.612	1.909	1.264	2.928	1.267	1.339	1.884	1.962	1.558	2.938	1.561	1.617	1068.2	15829.5	0	2.00/2
p1--CCp2--p3	hexa	0.0	1.966	2.095	1.924	3.250	1.925	2.095	1.920	2.125	1.824	3.146	1.840	1.843	520.09	11754.7	0	2.00/2
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.050	3.251	2.042	2.265	2.065	2.124	2.028	3.148	2.028	2.111	534.05	12101.7	0	2.00/2
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.141	2.922	1.145	1.213	1.768	1.927	1.422	2.906	1.432	1.479	1211.2	15837.4	0	2.00/2
p1--CCp2--p3	tetra unstr	0.1	1.612	1.909	1.264	2.938	1.267	1.350	1.884	1.962	1.558	2.938	1.561	1.627	1100.2	16782.6	0	2.00/2
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.919	4.067	1.926	1.874	1.920	2.125	1.826	4.139	1.840	1.725	511.31	9564.9	0	2.00/2
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.043	4.057	2.043	2.038	2.067	2.124	2.026	4.119	2.029	2.002	575.29	9935.4	0	2.00/2
p1--CCp3--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	2161.6	23052.8	0	2.00/2
p1--CCp3--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1778.9	24017.3	0	2.00/2
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.917	4.010	1.926	1.850	1.921	2.125	1.823	4.112	1.841	1.677	723.19	10810.6	0	2.00/2
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.042	4.000	2.043	2.035	2.067	2.124	2.025	4.092	2.029	1.988	668.21	11136.3	0	2.00/2
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.141	2.935	1.145	1.205	1.768	1.927	1.422	2.111	1.432	1.475	2163.9	23477.6	0	2.00/2
p1--CCp3--p2	tetra unstr	0.1	1.612	1.909	1.264	3.163	1.267	1.335	1.884	1.962	1.558	2.438	1.561	1.616	1804.9	24406.4	0	2.00/2
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.917	4.003	1.926	1.849	1.921	2.125	1.823	4.114	1.841	1.676	932.98	13646.1	0	2.00/2
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.042	3.992	2.043	2.035	2.067	2.124	2.025	4.094	2.029	1.988	879.35	13988.5	0	2.00/2
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.141	2.836	1.145	1.243	1.768	1.927	1.422	2.067	1.432	1.500	2330.3	24391.1	0	2.00/2
p1--CCp3--p3	tetra unstr	0.1	1.612	1.909	1.264	3.084	1.267	1.378	1.884	1.962	1.558	2.395	1.561	1.647	1858.2	25370.1	0	2.00/2
p1--GP--p1	hexa	0.0	1.968	2.095	1.992	2.157	1.999	2.164	1.920	2.125	1.997	2.201	2.001	2.200	357.41	7338.5	0	2.00/2
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.001	2.162	2.067	2.124	1.997	2.199	2.004	2.199	375.47	8901.2	0	2.00/2
p1--GP--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	610.55	10400.1	0	2.00/2
p1--GP--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	644.92	11931.5	0	2.00/2
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.921	2.125	1.993	2.129	1.996	2.128	863.68	8084.6	0	2.00/2
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.092	1.991	2.096	2.067	2.124	1.993	2.128	1.998	2.126	852.43	9548.3	0	2.00/2
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.768	1.927	1.953	1.877	1.777	1.880	810.89	10858.2	0	2.00/2
p1--GP--p2	tetra unstr	0.1	1.612	1.909	1.920	1.897	1.790	1.900	1.884	1.962	1.955	1.921	1.849	1.923	831.86	12353.3	0	2.00/2
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.921	2.125	1.993	2.126	1.996	2.124	1939.9	10771.9	0	2.00/2
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.067	2.124	1.993	2.124	1.998	2.123	1752.9	11162.2	0	2.00/2
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.873	1.768	1.927	1.951	1.889	1.775	1.892	1182.8	11857.3	0	2.00/2
p1--GP--p3	tetra unstr	0.1	1.612	1.909	1.916	1.900	1.786	1.903	1.884	1.962	1.953	1.927	1.847	1.929	1142.1	13225.7	0	2.00/2
p2--na--NO	hexa	0.0	1.947	3.025	1.942	1.942	1.946	1.946	2.001	2.766	2.000	2.000	2.003	2.003	345.81	25850.3	0	2.00/2
p2--na--NO	hexa	0.1	1.946	3.053	1.941	1.941	1.946	1.946	2.000	2.807	1.999	1.999	2.002	2.002	333.19	23777.0	0	2.00/2
p2--na--NO	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1218.3	18239.1	0	2.00/2
p2--na--NO	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1233.7	21605.4	0	2.00/2
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.993	2.168	2.001	2.183	2.035	2.884	2.031	2.209	2.036	2.145	445.26	13056.2	0	2.00/3
p2--CCp1--p1	hexa	0.1	2.008	3.133	1.997	2.167	2.006	2.192	2.037	2.925	2.033	2.209	2.038	2.152	454.48	15812.8	0	2.00/3
p2--CCp1--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1728.9	19037.9	0	2.00/2
p2--CCp1--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1458.2	21897.2	0	2.00/2
p2--CCp1--p2	hexa	0.0	2.040	3.141	2.024	2.098	2.036	2.087	2.059	2.936	2.052	2.131	2.060	2.062	516.68	13704.3	0	2.00/3
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.033	2.098	2.046	2.088	2.064	2.974	2.056	2.131	2.065	2.062	505.95	16457.1	0	2.00/3
p2--CCp1--p2	tetra unstr	0.0	2.105	2.705	1.863	1.971	1.864	1.960	2.382	2.719	2.113	1.982	2.122	1.974	1831.4	19519.9	0	2.00/3
p2--CCp1--p2	tetra unstr	0.1	2.348	2.824	2.014	1.971	2.018	1.970	2.721	2.901	2.364	1.982	2.374	1.986	2025.1	22291.9	0	2.00/3
p2--CCp1--p3	hexa	0.0	2.040	3.141	2.024	2.095	2.036	2.085	2.059	2.936	2.052	2.127	2.060	2.061	646.23	15346.1	0	2.00/3
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.033	2.095	2.046	2.086	2.064	2.974	2.056	2.127	2.065	2.061	625.23	17752.0	0	2.00/3
p2--CCp1--p3	tetra unstr	0.0	2.105	2.705	1.863	1.963	1.864	1.958	2.382	2.719	2.113	1.981	2.122	1.973	1812.3	20472.8	0	2.00/3
p2--CCp1--p3	tetra unstr	0.1	2.347	2.824	2.014	1.963	2.018	1.968	2.721	2.901	2.364	1.981	2.374	1.984	1955.3	23168.1	0	2.00/3
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.998	3.170	2.002	2.075	2.035	2.884	2.035	3.091	2.037	2.071	534.33	13059.3	0	2.00/3
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.003	3.171	2.007	2.094	2.037	2.925	2.037	3.091	2.039	2.079	573.07	15812.8	0	2.00/3
p2--CCp2--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1963.6	22309.5	0	2.00/2
p2--CCp2--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	2075.7	24373.9	0	2.00/2
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.034	3.246	2.037	2.170	2.059	2.936	2.059	3.146	2.061	2.125	604.10	13794.7	0	2.00/3
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.044	3.246	2.047	2.206	2.064	2.975	2.064	3.146	2.066	2.143	549.34	16456.9	0	2.00/3
p2--CCp2--p2	tetra unstr	0.0	2.105	2.705	1.863	2.944	1.864	1.932	2.384	2.720	2.116	2.972	2.122	2.121	2247.5	22778.6	0	2.00/3
p2--CCp2--p2	tetra unstr	0.1	2.350	2.825	2.016	2.945	2.018	2.104	2.727	2.903	2.369	2.972	2.376	2.408	2176.2	24847.1	0	2.00/3
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.034	3.251	2.037	2.173	2.059	2.936	2.059	3.149	2.061	2.126	722.33	15901.9	0	2.00/3
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.044	3.251	2.047	2.209	2.064	2.975	2.064	3.149	2.066	2.144	682.04	17752.8	0	2.00/3
p2--CCp2--p3	tetra unstr	0.0	2.105	2.705	1.863	2.958	1.864	1.943	2.384	2.720	2.116	2.980	2.122	2.134	2286.3	23691.0	0	2.00/3
p2--CCp2--p3	tetra unstr	0.1	2.350	2.825	2.016	2.959	2.018	2.111	2.727	2.903	2.369	2.981	2.376	2.414	2260.1	25797.0	0	2.00/3
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.997	4.087	2.002	2.020	2.035	2.884	2.035	4.142	2.037	2.056	721.83	13838.5	0	2.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.002	4.082	2.007	2.026	2.037	2.925	2.037	4.137	2.039	2.058	666.51	15813.9	0	2.00/3
p2--CCp3--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	2875.2	30961.1	0	2.00/2
p2--CCp3--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	2895.9	32396.6	0	2.00/2
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.033	4.022	2.037	2.072	2.060	2.936	2.059	4.109	2.061	2.100	857.26	14932.9	0	2.00/3
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.043	4.017	2.047	2.084	2.064	2.975	2.064	4.103	2.066	2.107	785.14	16538.8	0	2.00/3
p2--CCp3--p2	tetra unstr	0.0	2.105	2.705	1.863	3.893	1.864	1.897	2.384	2.720	2.116	3.841	2.123	2.129	3178.2	31359.2	0	2.00/3
p2--CCp3--p2	tetra unstr	0.1	2.350	2.825	2.016	3.911	2.018	2.048	2.727	2.903	2.369	3.890	2.376	2.392	3167.7	32884.7	0	2.00/3
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.033	4.014	2.037	2.073	2.060	2.936	2.059	4.110	2.061	2.101	990.02	17499.1	0	2.00/3
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.043	4.010	2.047	2.085	2.064	2.975	2.064	4.104	2.066	2.107	1022.1	18181.9	0	2.00/3
p2--CCp3--p3	tetra unstr	0.0	2.105	2.705	1.863	3.888	1.864	1.929	2.384	2.720	2.116	3.831	2.123	2.148	3323.1	32183.5	0	2.00/3
p2--CCp3--p3	tetra unstr	0.1	2.350	2.825	2.016	3.907	2.018	2.074	2.727	2.903	2.369	3.884	2.376	2.406	3223.8	33736.6	0	2.00/3
p2--GP--p1	hexa	0.0	2.004	3.107	2.011	3.133	2.844	3.122	2.035	2.884	2.004	3.004	2.447	2.998	590.22	26061.8	0	2.00/3
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.886	3.131	2.037	2.925	2.004	3.021	2.473	3.014	514.55	23988.2	0	2.00/3
p2--GP--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1154.9	18312.9	0	2.00/2
p2--GP--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1382.5	21701.0	0	2.00/2
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.897	3.215	2.060	2.936	2.001	3.092	2.431	3.082	1167.8	26712.3	0	2.00/3
p2--GP--p2	hexa	0.1	2.051	3.166	2.004	3.236	2.920	3.221	2.064	2.975	2.001	3.105	2.439	3.094	1039.5	24632.4	0	2.00/3
p2--GP--p2	tetra unstr	0.0	2.105	2.705	1.928	2.794	2.100	2.795	2.384	2.720	1.960	2.779	2.102	2.783	1472.4	18701.6	0	2.00/3
p2--GP--p2	tetra unstr	0.1	2.350	2.825	1.928	2.831	2.146	2.833	2.727	2.903	1.960	2.862	2.167	2.866	1684.2	21997.0	0	2.00/3
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.896	3.220	2.060	2.936	2.000	3.095	2.430	3.085	2138.9	28004.9	0	2.00/3
p2--GP--p3	hexa	0.1	2.051	3.166	2.003	3.241	2.919	3.226	2.064	2.975	2.000	3.108	2.438	3.097	1870.2	25931.7	0	2.00/3
p2--GP--p3	tetra unstr	0.0	2.105	2.705	1.924	2.801	2.082	2.802	2.384	2.720	1.958	2.768	2.089	2.773	1914.1	19696.0	0	2.00/3
p2--GP--p3	tetra unstr	0.1	2.350	2.825	1.924	2.846	2.128	2.847	2.727	2.903	1.958	2.864	2.153	2.868	2017.0	22649.7	0	2.00/3
p3--na--NO	hexa	0.0	2.313	3.204	2.303	2.303	2.307	2.307	2.193	2.830	2.189	2.189	2.192	2.192	886.77	77052.6	0	2.00/2
p3--na--NO	hexa	0.1	2.278	3.207	2.269	2.269	2.273	2.273	2.169	2.827	2.166	2.166	2.168	2.168	879.82	77137.1	0	2.00/2
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2511.9	45194.0	0	2.00/2
p3--na--NO	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2695.5	53563.8	0	2.00/2
p3--CCp1--p1	hexa	0.0	2.758	3.637	2.675	2.166	2.727	2.263	2.547	3.524	2.476	2.207	2.525	2.199	1245.6	44827.9	0	2.00/3
p3--CCp1--p1	hexa	0.1	2.713	3.646	2.635	2.166	2.687	2.261	2.497	3.532	2.433	2.207	2.481	2.197	1364.9	45948.3	0	2.00/3
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3533.2	45819.6	0	2.00/2
p3--CCp1--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	3589.5	53125.0	0	2.00/2
p3--CCp1--p2	hexa	0.0	3.096	3.790	2.987	2.097	3.017	2.094	2.864	3.829	2.773	2.130	2.782	2.063	1235.1	45475.2	0	2.00/3
p3--CCp1--p2	hexa	0.1	3.064	3.800	2.965	2.097	2.994	2.093	2.807	3.842	2.734	2.130	2.739	2.063	1437.7	46594.9	0	2.00/3
p3--CCp1--p2	tetra unstr	0.0	2.979	3.699	2.857	1.970	2.857	1.968	3.035	3.633	2.949	1.982	2.960	1.961	3474.5	46128.5	0	2.00/3
p3--CCp1--p2	tetra unstr	0.1	3.071	3.771	2.960	1.971	2.963	1.968	3.012	3.707	3.038	1.982	3.053	1.960	3711.0	53360.2	0	2.00/3
p3--CCp1--p3	hexa	0.0	3.096	3.790	2.987	2.094	3.017	2.092	2.864	3.829	2.773	2.127	2.782	2.062	1365.3	46770.8	0	2.00/3
p3--CCp1--p3	hexa	0.1	3.064	3.800	2.965	2.094	2.994	2.091	2.807	3.842	2.734	2.127	2.739	2.062	1565.9	47889.1	0	2.00/3
p3--CCp1--p3	tetra unstr	0.0	2.978	3.699	2.856	1.963	2.857	1.966	3.034	3.633	2.949	1.981	2.960	1.960	3499.2	46754.4	0	2.00/3
p3--CCp1--p3	tetra unstr	0.1	3.071	3.772	2.959	1.963	2.963	1.966	3.011	3.707	3.037	1.981	3.053	1.960	3720.5	53945.5	0	2.00/3
p3--CCp2--p1	hexa	0.0	2.782	3.644	2.745	3.172	2.752	2.948	2.574	3.538	2.546	3.092	2.553	2.790	1235.5	44827.1	0	2.00/3
p3--CCp2--p1	hexa	0.1	2.739	3.653	2.708	3.172	2.715	2.943	2.524	3.546	2.503	3.092	2.509	2.778	1453.5	45947.7	0	2.00/3
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3723.2	47059.4	0	2.00/2
p3--CCp2--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	3912.0	53592.4	0	2.00/2
p3--CCp2--p2	hexa	0.0	3.335	3.812	3.202	3.247	3.220	3.232	3.264	3.882	3.065	3.147	3.095	3.171	1292.4	45473.8	0	2.00/3
p3--CCp2--p2	hexa	0.1	3.352	3.823	3.223	3.247	3.241	3.244	3.280	3.896	3.083	3.147	3.114	3.186	1418.8	46596.0	0	2.00/3
p3--CCp2--p2	tetra unstr	0.0	3.198	3.777	2.984	2.942	2.981	2.996	3.463	3.810	3.182	2.970	3.187	3.073	3489.9	47450.9	0	2.00/3
p3--CCp2--p2	tetra unstr	0.1	3.452	3.863	3.162	2.942	3.161	3.045	3.775	3.921	3.433	2.970	3.439	3.121	3872.8	53952.5	0	2.00/3
p3--CCp2--p3	hexa	0.0	3.335	3.812	3.202	3.253	3.220	3.231	3.264	3.882	3.065	3.150	3.095	3.171	1464.5	46767.1	0	2.00/3
p3--CCp2--p3	hexa	0.1	3.352	3.823	3.223	3.253	3.241	3.243	3.279	3.896	3.083	3.150	3.114	3.185	1614.2	47891.1	0	2.00/3
p3--CCp2--p3	tetra unstr	0.0	3.198	3.777	2.984	2.957	2.981	3.007	3.463	3.810	3.182	2.980	3.187	3.076	3546.9	48213.3	0	2.00/3
p3--CCp2--p3	tetra unstr	0.1	3.451	3.863	3.162	2.957	3.160	3.055	3.776	3.921	3.433	2.980	3.439	3.123	3932.9	54665.7	0	2.00/3
p3--CCp3--p1	hexa	0.0	2.781	3.644	2.742	4.034	2.751	2.764	2.573	3.538	2.546	4.068	2.552	2.578	1433.7	44827.5	0	2.00/3
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.705	4.034	2.713	2.735	2.523	3.546	2.502	4.068	2.508	2.535	1461.2	45807.1	0	2.00/3
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	4234.3	53364.4	0	2.00/2
p3--CCp3--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	4611.4	58671.2	0	2.00/2
p3--CCp3--p2	hexa	0.0	3.332	3.813	3.194	3.978	3.216	3.106	3.260	3.882	3.069	4.042	3.091	3.018	1557.5	45473.6	0	2.00/3
p3--CCp3--p2	hexa	0.1	3.349	3.823	3.214	3.978	3.236	3.135	3.275	3.896	3.087	4.042	3.109	3.036	1885.2	46590.2	0	2.00/3
p3--CCp3--p2	tetra unstr	0.0	3.198	3.777	2.985	3.943	2.981	3.010	3.463	3.810	3.182	3.960	3.187	3.210	4203.9	53808.9	0	2.00/3
p3--CCp3--p2	tetra unstr	0.1	3.452	3.863	3.163	3.943	3.161	3.182	3.775	3.921	3.433	3.960	3.438	3.453	4665.1	59130.7	0	2.00/3
p3--CCp3--p3	hexa	0.0	3.332	3.813	3.194	3.971	3.216	3.106	3.260	3.882	3.069	4.044	3.091	3.018	2220.1	46767.6	0	2.00/3
p3--CCp3--p3	hexa	0.1	3.349	3.823	3.214	3.971	3.236	3.136	3.275	3.896	3.087	4.044	3.109	3.037	2072.9	47885.4	0	2.00/3
p3--CCp3--p3	tetra unstr	0.0	3.197	3.777	2.984	3.944	2.980	3.056	3.462	3.810	3.182	3.962	3.186	3.240	3881.6	54514.6	0	2.00/3
p3--CCp3--p3	tetra unstr	0.1	3.451	3.863	3.162	3.944	3.160	3.224	3.775	3.921	3.433	3.962	3.438	3.483	4625.4	59967.2	0	2.00/3
p3--GP--p1	hexa	0.0	2.781	3.644	2.003	3.789	2.716	3.807	2.573	3.538	2.000	3.679	2.354	3.683	1524.3	77272.0	0	2.00/3
p3--GP--p1	hexa	0.1	2.738	3.653	2.003	3.796	2.706	3.814	2.523	3.546	2.000	3.687	2.345	3.692	1268.9	77349.1	0	2.00/3
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3571.4	45305.4	0	2.00/2
p3--GP--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	4078.1	53641.4	0	2.00/2
p3--GP--p2	hexa	0.0	3.332	3.813	1.994	3.892	2.454	3.910	3.260	3.882	1.996	3.955	2.171	3.965	2174.1	77922.8	0	2.00/3
p3--GP--p2	hexa	0.1	3.350	3.823	1.994	3.896	2.447	3.914	3.275	3.896	1.996	3.961	2.167	3.971	1884.3	77844.3	0	2.00/3
p3--GP--p2	tetra unstr	0.0	3.198	3.777	1.927	3.839	2.032	3.841	3.463	3.810	1.961	3.829	1.977	3.835	3649.2	45565.9	0	2.00/3
p3--GP--p2	tetra unstr	0.1	3.452	3.863	1.927	3.878	2.028	3.879	3.775	3.921	1.961	3.893	1.974	3.898	4018.1	53890.5	0	2.00/3
p3--GP--p3	hexa	0.0	3.332	3.813	1.994	3.884	2.452	3.904	3.260	3.882	1.995	3.956	2.170	3.967	3319.0	79065.9	0	2.00/3
p3--GP--p3	hexa	0.1	3.349	3.823	1.993	3.889	2.446	3.908	3.275	3.896	1.995	3.962	2.167	3.973	3166.2	79290.2	0	2.00/3
p3--GP--p3	tetra unstr	0.0	3.197	3.777	1.923	3.838	2.021	3.840	3.462	3.810	1.958	3.826	1.974	3.833	3682.2	46152.2	0	2.00/3
p3--GP--p3	tetra unstr	0.1	3.451	3.863	1.923	3.879	2.017	3.880	3.775	3.921	1.958	3.893	1.971	3.900	4260.6	54387.8	0	2.00/3

13.4.3 JFNK

$\Delta t = 10^{-2}$

Table 13: 3D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-2}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 21 mesh sizes of the row (105 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–30						Fit $N = 20$–30									
`Vm-states-Iion`	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--na--NO	hexa	0.0	1.985	1.985	1.994	1.994	1.994	1.994	1.993	1.993	2.009	2.009	2.009	2.009	10.903	4081.0	0	2.00/3
NO--na--NO	hexa	0.1	1.436	1.436	1.297	1.297	1.306	1.306	1.411	1.411	1.214	1.214	1.223	1.223	26.673	6311.1	0	2.00/3
NO--na--NO	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	46.900	5184.6	0	3.00/3
NO--na--NO	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	125.86	12050.8	0	3.00/3
NO--CCp1--p1	hexa	0.0	1.985	1.985	1.998	2.166	1.995	2.246	1.992	1.992	2.010	2.208	2.009	2.189	29.636	4914.5	0	2.00/3
NO--CCp1--p1	hexa	0.1	1.436	1.436	1.287	2.167	1.306	2.230	1.411	1.411	1.205	2.208	1.224	2.159	34.670	6578.7	0	2.00/3
NO--CCp1--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	150.50	6443.0	0	3.00/3
NO--CCp1--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	221.53	12246.8	0	3.00/3
NO--CCp1--p2	hexa	0.0	1.985	1.985	1.999	2.098	1.994	2.090	1.992	1.992	2.011	2.130	2.009	2.062	36.531	6570.2	0	2.00/3
NO--CCp1--p2	hexa	0.1	1.436	1.436	1.283	2.098	1.307	2.087	1.411	1.411	1.201	2.130	1.224	2.056	43.424	8032.6	0	2.00/3
NO--CCp1--p2	tetra unstr	0.0	0.664	0.664	0.375	1.924	0.373	0.457	0.826	0.826	0.617	1.896	0.616	0.669	145.87	7077.8	0	3.00/3
NO--CCp1--p2	tetra unstr	0.1	0.677	0.677	0.398	1.928	0.396	0.483	0.823	0.823	0.629	1.903	0.628	0.683	234.85	12496.9	0	3.00/3
NO--CCp1--p3	hexa	0.0	1.985	1.985	1.999	2.094	1.994	2.088	1.992	1.992	2.011	2.127	2.009	2.061	51.184	9863.2	0	2.00/3
NO--CCp1--p3	hexa	0.1	1.436	1.436	1.283	2.095	1.307	2.085	1.411	1.411	1.201	2.127	1.224	2.054	54.930	11329.7	0	2.00/3
NO--CCp1--p3	tetra unstr	0.0	0.664	0.664	0.375	1.923	0.373	0.460	0.826	0.826	0.617	1.906	0.616	0.670	160.88	8341.1	0	3.00/3
NO--CCp1--p3	tetra unstr	0.1	0.677	0.677	0.398	1.926	0.396	0.485	0.823	0.823	0.629	1.912	0.628	0.684	232.20	12995.8	0	3.00/3
NO--CCp2--p1	hexa	0.0	1.977	1.977	2.009	3.171	1.987	2.618	1.989	1.989	2.014	3.091	2.005	2.390	49.078	5690.6	0	2.00/3
NO--CCp2--p1	hexa	0.1	1.447	1.447	1.257	3.172	1.318	2.180	1.426	1.426	1.176	3.092	1.239	1.626	58.805	7158.6	0	2.00/3
NO--CCp2--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	258.79	9806.2	0	3.00/3
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	286.75	14003.1	0	3.00/3
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.010	3.247	1.979	2.753	1.985	1.985	2.013	3.146	2.001	2.494	58.558	7339.2	0	2.00/3
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.248	3.248	1.333	2.320	1.441	1.441	1.170	3.147	1.257	1.718	65.331	8808.0	0	2.00/3
NO--CCp2--p2	tetra unstr	0.0	0.664	0.664	0.375	2.080	0.373	0.465	0.826	0.826	0.617	1.273	0.616	0.673	242.81	10440.8	0	3.00/3
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.398	2.128	0.396	0.486	0.823	0.823	0.629	1.333	0.628	0.683	294.45	14639.7	0	3.00/3
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.010	3.252	1.979	2.758	1.985	1.985	2.013	3.149	2.001	2.501	76.531	10633.9	0	2.00/3
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.248	3.253	1.333	2.328	1.441	1.441	1.170	3.150	1.257	1.726	93.257	12108.2	0	2.00/3
NO--CCp2--p3	tetra unstr	0.0	0.664	0.664	0.375	1.992	0.373	0.466	0.826	0.826	0.617	1.241	0.616	0.675	284.94	11706.3	0	3.00/3
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.398	2.039	0.396	0.487	0.823	0.823	0.629	1.295	0.628	0.686	307.72	15907.4	0	3.00/3
NO--CCp3--p1	hexa	0.0	1.980	1.980	2.001	4.033	1.989	2.062	1.991	1.991	2.011	4.072	2.007	2.017	96.802	7587.6	0	2.00/3
NO--CCp3--p1	hexa	0.1	1.452	1.452	1.244	4.015	1.323	1.276	1.429	1.429	1.170	4.028	1.241	1.186	112.52	9061.8	0	2.00/3
NO--CCp3--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	550.88	18575.4	0	3.00/3
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	510.66	22778.3	0	3.00/3
NO--CCp3--p2	hexa	0.0	1.976	1.976	2.000	3.980	1.985	2.107	1.990	1.990	2.011	4.053	2.005	2.022	102.30	9238.3	0	2.00/3
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.233	3.966	1.343	1.305	1.447	1.447	1.163	4.019	1.261	1.187	129.37	10707.1	0	2.00/3
NO--CCp3--p2	tetra unstr	0.0	0.664	0.664	0.375	0.890	0.373	0.418	0.826	0.826	0.617	0.674	0.616	0.657	624.23	19203.7	0	3.00/3
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.398	0.930	0.396	0.441	0.823	0.823	0.629	0.689	0.628	0.670	524.97	23405.3	0	3.00/3
NO--CCp3--p3	hexa	0.0	1.976	1.976	2.000	3.973	1.985	2.108	1.990	1.990	2.011	4.054	2.005	2.022	122.31	12535.0	0	2.00/3
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.233	3.959	1.343	1.305	1.447	1.447	1.163	4.021	1.261	1.186	143.65	14002.8	0	2.00/3
NO--CCp3--p3	tetra unstr	0.0	0.664	0.664	0.375	0.888	0.373	0.432	0.826	0.826	0.617	0.684	0.616	0.665	573.46	20474.6	0	3.00/3
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.398	0.926	0.396	0.454	0.823	0.823	0.629	0.697	0.628	0.677	559.54	24674.4	0	3.00/3
NO--GP--p1	hexa	0.0	1.984	1.984	1.992	2.072	1.992	2.077	1.992	1.992	1.996	2.038	1.997	2.041	50.383	4665.5	0	2.00/3
NO--GP--p1	hexa	0.1	1.459	1.459	1.992	2.073	1.993	2.079	1.429	1.429	1.996	2.038	1.996	2.041	45.835	6563.2	0	2.00/3
NO--GP--p1	tetra unstr	0.0	0.664	0.664	0.375	0.375	0.373	0.373	0.826	0.826	0.617	0.617	0.616	0.616	48.848	5486.9	0	3.00/3
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.398	0.398	0.396	0.396	0.823	0.823	0.629	0.629	0.628	0.628	127.67	12249.4	0	3.00/3
NO--GP--p2	hexa	0.0	1.983	1.983	1.984	2.014	1.985	2.018	1.992	1.992	1.992	2.007	1.993	2.009	130.18	6314.9	0	2.00/3
NO--GP--p2	hexa	0.1	1.482	1.482	1.984	2.015	1.985	2.018	1.447	1.447	1.992	2.007	1.993	2.010	110.28	7695.8	0	2.00/3
NO--GP--p2	tetra unstr	0.0	0.664	0.664	1.832	0.861	0.516	0.859	0.826	0.826	1.792	0.867	0.652	0.866	134.81	6127.4	0	3.00/3
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.844	0.883	0.546	0.881	0.823	0.823	1.812	0.884	0.668	0.883	173.59	12494.7	0	3.00/3
NO--GP--p3	hexa	0.0	1.983	1.983	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.993	2.008	326.06	9615.9	0	2.00/3
NO--GP--p3	hexa	0.1	1.482	1.482	1.984	2.011	1.985	2.015	1.447	1.447	1.992	2.006	1.992	2.008	241.31	11005.2	0	2.00/3
NO--GP--p3	tetra unstr	0.0	0.664	0.664	1.828	0.863	0.513	0.862	0.826	0.826	1.790	0.868	0.651	0.868	256.29	7394.5	0	3.00/3
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.542	0.884	0.823	0.823	1.810	0.885	0.667	0.885	252.73	12996.4	0	3.00/3
p1--na--NO	hexa	0.0	1.967	2.095	1.927	1.927	1.930	1.930	1.919	2.125	1.838	1.838	1.845	1.845	73.508	7146.7	0	2.00/3
p1--na--NO	hexa	0.1	2.067	2.093	2.047	2.047	2.048	2.048	2.066	2.124	2.033	2.033	2.035	2.035	66.261	8682.2	0	2.00/3
p1--na--NO	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	201.56	10384.5	0	2.00/3
p1--na--NO	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	201.05	11873.0	0	2.00/3
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.926	2.166	1.930	2.193	1.919	2.125	1.836	2.207	1.845	2.115	78.985	7316.5	0	2.00/3
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.046	2.167	2.048	2.225	2.066	2.124	2.033	2.208	2.035	2.168	95.760	8858.9	0	2.00/3
p1--CCp1--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	320.24	11428.9	0	2.00/3
p1--CCp1--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	274.69	12624.6	0	2.00/3
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.925	2.098	1.930	2.083	1.919	2.125	1.835	2.130	1.845	2.049	104.65	8760.1	0	2.00/3
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.046	2.098	2.047	2.090	2.066	2.124	2.033	2.130	2.035	2.061	121.22	9611.3	0	2.00/3
p1--CCp1--p2	tetra unstr	0.0	1.474	1.866	1.139	1.973	1.142	1.493	1.771	1.928	1.419	1.984	1.428	1.590	306.60	12068.1	0	2.00/3
p1--CCp1--p2	tetra unstr	0.1	1.613	1.909	1.262	1.973	1.264	1.621	1.889	1.963	1.556	1.983	1.558	1.721	294.54	13192.7	0	2.00/3
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.925	2.094	1.930	2.081	1.919	2.125	1.835	2.127	1.845	2.048	118.20	12040.5	0	2.00/3
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.046	2.094	2.047	2.088	2.066	2.124	2.033	2.127	2.035	2.060	107.73	12507.1	0	2.00/3
p1--CCp1--p3	tetra unstr	0.0	1.474	1.866	1.139	1.965	1.142	1.505	1.771	1.928	1.419	1.982	1.428	1.597	329.54	13330.6	0	2.00/3
p1--CCp1--p3	tetra unstr	0.1	1.613	1.909	1.262	1.965	1.264	1.632	1.889	1.963	1.556	1.982	1.558	1.727	289.57	14324.8	0	2.00/3

Vm-states-Iion	Mesh type	α	Fit $N = 10$–30						Fit $N = 20$–30						t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
			V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G				
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.926	3.169	1.929	2.043	1.919	2.125	1.831	3.089	1.845	1.844	113.82	7872.0	0	2.00/3
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.050	3.170	2.046	2.200	2.066	2.124	2.033	3.090	2.034	2.090	107.02	8897.9	0	2.00/3
p1--CCp2--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	396.23	14798.2	0	2.00/3
p1--CCp2--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	343.20	15617.1	0	2.00/3
p1--CCp2--p2	hexa	0.0	1.965	2.095	1.925	3.244	1.929	2.095	1.919	2.125	1.828	3.143	1.845	1.846	107.93	9524.3	0	2.00/3
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.051	3.245	2.045	2.265	2.065	2.124	2.033	3.145	2.034	2.115	116.68	9993.9	0	2.00/3
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.139	2.917	1.142	1.199	1.771	1.928	1.418	2.913	1.428	1.465	424.91	15428.2	0	2.00/3
p1--CCp2--p2	tetra unstr	0.1	1.613	1.909	1.262	2.928	1.264	1.336	1.889	1.963	1.555	2.938	1.558	1.614	385.65	16245.6	0	2.00/3
p1--CCp2--p3	hexa	0.0	1.965	2.095	1.925	3.250	1.929	2.098	1.919	2.125	1.828	3.146	1.845	1.848	125.46	12814.8	0	2.00/3
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.051	3.251	2.045	2.269	2.065	2.124	2.033	3.148	2.034	2.117	116.80	13283.4	0	2.00/3
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.139	2.922	1.142	1.211	1.771	1.928	1.418	2.906	1.428	1.476	436.87	16692.6	0	2.00/3
p1--CCp2--p3	tetra unstr	0.1	1.613	1.909	1.262	2.938	1.264	1.348	1.889	1.963	1.555	2.938	1.558	1.624	358.67	17518.9	0	2.00/3
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.922	4.067	1.929	1.877	1.919	2.125	1.830	4.138	1.845	1.729	151.89	9769.3	0	2.00/3
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.046	4.057	2.047	2.042	2.066	2.124	2.032	4.119	2.034	2.007	144.90	10198.8	0	2.00/3
p1--CCp3--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	604.20	23563.3	0	2.00/3
p1--CCp3--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	550.67	24379.4	0	2.00/3
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.920	4.009	1.930	1.853	1.920	2.125	1.827	4.112	1.846	1.682	158.35	11421.3	0	2.00/3
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.045	4.000	2.047	2.039	2.066	2.124	2.031	4.092	2.034	1.993	155.38	11864.5	0	2.00/3
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.139	2.936	1.142	1.202	1.771	1.928	1.418	2.110	1.428	1.472	647.90	24195.5	0	2.00/3
p1--CCp3--p2	tetra unstr	0.1	1.613	1.909	1.262	3.163	1.264	1.332	1.889	1.963	1.555	2.437	1.558	1.614	545.13	25023.2	0	2.00/3
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.920	4.003	1.930	1.852	1.920	2.125	1.827	4.114	1.846	1.680	216.54	14712.4	0	2.00/3
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.045	3.992	2.047	2.039	2.066	2.124	2.031	4.094	2.034	1.993	188.45	15149.0	0	2.00/3
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.139	2.837	1.142	1.240	1.771	1.928	1.418	2.066	1.428	1.496	650.11	25455.3	0	2.00/3
p1--CCp3--p3	tetra unstr	0.1	1.613	1.909	1.262	3.084	1.264	1.375	1.889	1.963	1.555	2.395	1.558	1.644	534.40	26289.6	0	2.00/3
p1--GP--p1	hexa	0.0	1.967	2.095	1.992	2.157	1.999	2.164	1.919	2.125	1.997	2.201	2.001	2.201	94.746	7340.6	0	2.00/3
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.002	2.162	2.066	2.124	1.997	2.199	2.005	2.199	86.296	8892.3	0	2.00/3
p1--GP--p1	tetra unstr	0.0	1.474	1.866	1.140	1.140	1.142	1.142	1.771	1.928	1.419	1.419	1.428	1.428	207.54	10483.4	0	2.00/3
p1--GP--p1	tetra unstr	0.1	1.613	1.909	1.262	1.262	1.264	1.264	1.889	1.963	1.556	1.556	1.558	1.558	211.38	11964.7	0	2.00/3
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.920	2.125	1.993	2.129	1.997	2.128	170.47	8587.1	0	2.00/3
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.093	1.991	2.096	2.066	2.124	1.993	2.128	1.998	2.126	146.84	9578.2	0	2.00/3
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.771	1.928	1.953	1.877	1.777	1.880	282.21	11109.8	0	2.00/3
p1--GP--p2	tetra unstr	0.1	1.613	1.909	1.920	1.897	1.790	1.900	1.889	1.963	1.955	1.921	1.848	1.922	322.57	12472.4	0	2.00/3
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.920	2.125	1.993	2.126	1.996	2.125	316.14	12162.0	0	2.00/3
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.066	2.124	1.993	2.124	1.998	2.123	275.06	12472.4	0	2.00/3
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.874	1.771	1.928	1.951	1.889	1.775	1.892	354.44	12384.3	0	2.00/3
p1--GP--p3	tetra unstr	0.1	1.613	1.909	1.916	1.900	1.786	1.903	1.889	1.963	1.953	1.927	1.846	1.929	367.74	13559.1	0	2.00/3
p2--na--NO	hexa	0.0	1.947	3.025	1.944	1.944	1.948	1.948	2.001	2.766	2.003	2.003	2.005	2.005	150.14	25853.8	0	2.00/3
p2--na--NO	hexa	0.1	1.946	3.053	1.943	1.943	1.947	1.947	2.000	2.807	2.002	2.002	2.005	2.005	121.96	21776.5	0	2.00/3
p2--na--NO	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	311.22	18260.0	0	2.00/3
p2--na--NO	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	393.20	21620.4	0	2.00/3
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.995	2.168	2.003	2.183	2.035	2.884	2.035	2.209	2.040	2.147	195.99	13057.6	0	2.00/3
p2--CCp1--p1	hexa	0.1	2.008	3.133	2.000	2.167	2.008	2.193	2.037	2.925	2.037	2.209	2.042	2.153	171.78	15809.6	0	2.00/3
p2--CCp1--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	424.47	19182.3	0	2.00/3
p2--CCp1--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	410.00	21990.9	0	2.00/3
p2--CCp1--p2	hexa	0.0	2.040	3.142	2.028	2.098	2.039	2.087	2.059	2.936	2.057	2.131	2.064	2.063	170.77	13778.1	0	2.00/3
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.037	2.098	2.049	2.088	2.063	2.975	2.062	2.131	2.069	2.063	166.93	16457.9	0	2.00/3
p2--CCp1--p2	tetra unstr	0.0	2.103	2.703	1.863	1.971	1.864	1.960	2.383	2.718	2.113	1.982	2.121	1.974	419.96	19827.3	0	2.00/3
p2--CCp1--p2	tetra unstr	0.1	2.347	2.823	2.013	1.971	2.016	1.970	2.727	2.903	2.362	1.982	2.372	1.985	409.58	22505.6	0	2.00/3
p2--CCp1--p3	hexa	0.0	2.040	3.142	2.028	2.095	2.039	2.085	2.059	2.936	2.057	2.127	2.064	2.061	168.67	16260.0	0	2.00/3
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.037	2.095	2.049	2.086	2.063	2.975	2.062	2.127	2.069	2.061	192.63	17828.6	0	2.00/3
p2--CCp1--p3	tetra unstr	0.0	2.103	2.703	1.863	1.963	1.864	1.958	2.383	2.718	2.112	1.981	2.121	1.972	468.73	21087.9	0	2.00/3
p2--CCp1--p3	tetra unstr	0.1	2.347	2.823	2.013	1.963	2.016	1.968	2.727	2.903	2.362	1.981	2.372	1.983	412.36	23563.6	0	2.00/3
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.999	3.170	2.004	2.077	2.035	2.884	2.038	3.091	2.040	2.074	184.42	13081.2	0	2.00/3
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.005	3.171	2.009	2.096	2.037	2.925	2.040	3.091	2.043	2.083	193.86	15814.8	0	2.00/3
p2--CCp2--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	574.74	22548.2	0	2.00/3
p2--CCp2--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	474.93	24551.7	0	2.00/3
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.036	3.246	2.040	2.173	2.059	2.936	2.063	3.146	2.065	2.129	211.08	14044.3	0	2.00/3
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.046	3.246	2.050	2.209	2.064	2.975	2.068	3.146	2.070	2.147	173.89	16455.3	0	2.00/3
p2--CCp2--p2	tetra unstr	0.0	2.103	2.704	1.863	2.944	1.864	1.932	2.384	2.719	2.114	2.972	2.121	2.120	588.74	23176.3	0	2.00/3
p2--CCp2--p2	tetra unstr	0.1	2.349	2.824	2.014	2.945	2.016	2.103	2.732	2.904	2.367	2.972	2.373	2.405	471.63	25125.6	0	2.00/3
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.036	3.251	2.040	2.176	2.059	2.936	2.063	3.149	2.065	2.130	221.97	16914.1	0	2.00/3
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.046	3.251	2.050	2.212	2.064	2.975	2.068	3.149	2.070	2.149	206.85	17986.4	0	2.00/3
p2--CCp2--p3	tetra unstr	0.0	2.103	2.704	1.863	2.958	1.864	1.943	2.384	2.719	2.114	2.980	2.121	2.133	539.90	24442.6	0	2.00/3
p2--CCp2--p3	tetra unstr	0.1	2.349	2.824	2.014	2.959	2.016	2.110	2.732	2.904	2.367	2.981	2.373	2.411	504.22	26263.2	0	2.00/3
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.999	4.087	2.004	2.022	2.035	2.884	2.038	4.142	2.040	2.059	219.13	14040.2	0	2.00/3
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.004	4.082	2.009	2.028	2.037	2.925	2.040	4.137	2.043	2.062	240.33	15818.3	0	2.00/3
p2--CCp3--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	769.58	31316.2	0	2.00/3
p2--CCp3--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	642.00	32700.4	0	2.00/3
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.035	4.022	2.040	2.074	2.059	2.936	2.063	4.109	2.065	2.104	235.26	15527.0	0	2.00/3
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.045	4.017	2.050	2.087	2.064	2.975	2.068	4.103	2.070	2.111	268.32	16648.2	0	2.00/3
p2--CCp3--p2	tetra unstr	0.0	2.103	2.704	1.863	3.893	1.864	1.896	2.384	2.719	2.114	3.841	2.121	2.127	785.01	31949.9	0	2.00/3
p2--CCp3--p2	tetra unstr	0.1	2.349	2.824	2.014	3.911	2.016	2.046	2.732	2.904	2.367	3.890	2.373	2.389	665.64	33324.8	0	2.00/3
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.035	4.014	2.040	2.075	2.059	2.936	2.063	4.110	2.065	2.105	273.20	18513.7	0	2.00/3
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.045	4.010	2.050	2.087	2.064	2.975	2.068	4.104	2.070	2.112	261.59	18886.3	0	2.00/3
p2--CCp3--p3	tetra unstr	0.0	2.103	2.704	1.863	3.888	1.864	1.929	2.384	2.719	2.114	3.831	2.121	2.146	808.64	33209.9	0	2.00/3
p2--CCp3--p3	tetra unstr	0.1	2.349	2.824	2.014	3.907	2.016	2.072	2.732	2.904	2.367	3.883	2.373	2.403	644.26	34591.0	0	2.00/3
p2--GP--p1	hexa	0.0	2.003	3.107	2.011	3.133	2.845	3.122	2.035	2.884	2.004	3.005	2.449	2.999	175.28	26059.4	0	2.00/3
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.888	3.131	2.037	2.925	2.004	3.021	2.476	3.015	162.51	21986.0	0	2.00/3
p2--GP--p1	tetra unstr	0.0	2.043	2.626	1.868	1.868	1.869	1.869	2.248	2.602	2.078	2.078	2.084	2.084	358.27	18397.1	0	2.00/3
p2--GP--p1	tetra unstr	0.1	2.198	2.743	1.984	1.984	1.987	1.987	2.433	2.768	2.271	2.271	2.279	2.279	366.28	21760.9	0	2.00/3
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.898	3.215	2.059	2.936	2.001	3.092	2.432	3.083	259.13	26708.4	0	2.00/3
p2--GP--p2	hexa	0.1	2.050	3.166	2.004	3.236	2.920	3.222	2.064	2.975	2.001	3.105	2.440	3.095	232.31	22636.7	0	2.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS$_\Sigma$ [MB]	RB	n_{nl}
p2--GP--p2	tetra unstr	0.0	2.103	2.704	1.928	2.794	2.100	2.796	2.384	2.719	1.960	2.779	2.102	2.784	382.90	19041.3	0	2.00/3
p2--GP--p2	tetra unstr	0.1	2.349	2.824	1.928	2.831	2.146	2.833	2.732	2.904	1.960	2.861	2.166	2.866	390.35	22132.5	0	2.00/3
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.897	3.220	2.059	2.936	2.000	3.095	2.432	3.085	436.66	28002.5	0	2.00/3
p2--GP--p3	hexa	0.1	2.050	3.166	2.003	3.242	2.920	3.227	2.064	2.975	2.000	3.108	2.439	3.097	357.00	24079.3	0	2.00/3
p2--GP--p3	tetra unstr	0.0	2.103	2.704	1.924	2.802	2.082	2.802	2.384	2.719	1.958	2.769	2.088	2.773	406.97	20303.3	0	2.00/3
p2--GP--p3	tetra unstr	0.1	2.349	2.824	1.924	2.847	2.128	2.847	2.732	2.904	1.958	2.864	2.152	2.868	453.30	23026.1	0	2.00/3
p3--na--NO	hexa	0.0	2.313	3.204	2.308	2.308	2.312	2.312	2.193	2.830	2.197	2.197	2.200	2.200	872.23	77059.3	0	2.00/3
p3--na--NO	hexa	0.1	2.278	3.207	2.274	2.274	2.278	2.278	2.169	2.826	2.174	2.174	2.176	2.176	725.16	72597.5	0	2.00/3
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1102.0	45158.7	0	2.00/3
p3--na--NO	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1239.7	53472.5	0	2.00/3
p3--CCp1--p1	hexa	0.0	2.761	3.638	2.698	2.166	2.745	2.264	2.550	3.526	2.508	2.207	2.552	2.199	938.71	44823.5	0	2.00/3
p3--CCp1--p1	hexa	0.1	2.717	3.647	2.659	2.166	2.706	2.261	2.501	3.534	2.465	2.207	2.509	2.198	936.37	45946.8	0	2.00/3
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1491.7	45872.4	0	2.00/3
p3--CCp1--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1295.9	53128.9	0	2.00/3
p3--CCp1--p2	hexa	0.0	3.129	3.794	3.046	2.097	3.078	2.094	2.914	3.838	2.857	2.130	2.869	2.063	914.47	45474.8	0	2.00/3
p3--CCp1--p2	hexa	0.1	3.103	3.804	3.033	2.097	3.064	2.093	2.862	3.851	2.828	2.130	2.835	2.063	936.18	46596.0	0	2.00/3
p3--CCp1--p2	tetra unstr	0.0	3.009	3.711	2.897	1.970	2.893	1.968	3.092	3.661	3.022	1.982	3.021	1.961	1555.3	46211.2	0	2.00/3
p3--CCp1--p2	tetra unstr	0.1	3.120	3.786	3.019	1.970	3.015	1.968	3.099	3.741	3.149	1.982	3.147	1.960	1320.1	53378.0	0	2.00/3
p3--CCp1--p3	hexa	0.0	3.129	3.794	3.046	2.094	3.078	2.092	2.914	3.838	2.857	2.127	2.869	2.062	843.74	46768.3	0	2.00/3
p3--CCp1--p3	hexa	0.1	3.103	3.804	3.033	2.094	3.064	2.091	2.862	3.851	2.828	2.127	2.835	2.062	877.74	47889.0	0	2.00/3
p3--CCp1--p3	tetra unstr	0.0	3.009	3.711	2.897	1.963	2.892	1.966	3.091	3.661	3.021	1.981	3.021	1.960	1563.1	46935.0	0	2.00/3
p3--CCp1--p3	tetra unstr	0.1	3.120	3.786	3.018	1.963	3.015	1.966	3.099	3.741	3.149	1.981	3.147	1.960	1265.7	53955.0	0	2.00/3
p3--CCp2--p1	hexa	0.0	2.781	3.643	2.755	3.172	2.763	2.955	2.572	3.538	2.566	3.092	2.573	2.804	857.17	44820.2	0	2.00/3
p3--CCp2--p1	hexa	0.1	2.738	3.653	2.719	3.172	2.726	2.951	2.523	3.546	2.524	3.092	2.530	2.794	909.48	45950.9	0	2.00/3
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1585.8	46884.2	0	2.00/3
p3--CCp2--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1324.3	53666.8	0	2.00/3
p3--CCp2--p2	hexa	0.0	3.332	3.812	3.211	3.247	3.230	3.235	3.260	3.882	3.083	3.147	3.113	3.176	872.52	45473.9	0	2.00/3
p3--CCp2--p2	hexa	0.1	3.350	3.823	3.234	3.247	3.253	3.247	3.275	3.896	3.103	3.147	3.135	3.190	914.55	46594.9	0	2.00/3
p3--CCp2--p2	tetra unstr	0.0	3.196	3.775	2.986	2.942	2.983	2.997	3.461	3.809	3.184	2.970	3.189	3.072	1570.5	47660.3	0	2.00/3
p3--CCp2--p2	tetra unstr	0.1	3.450	3.862	3.163	2.942	3.162	3.045	3.777	3.920	3.434	2.970	3.440	3.120	1383.0	54049.4	0	2.00/3
p3--CCp2--p3	hexa	0.0	3.332	3.812	3.211	3.253	3.230	3.234	3.260	3.882	3.083	3.150	3.113	3.175	870.79	46768.6	0	2.00/3
p3--CCp2--p3	hexa	0.1	3.350	3.823	3.234	3.253	3.253	3.246	3.275	3.896	3.103	3.150	3.135	3.189	942.03	47892.3	0	2.00/3
p3--CCp2--p3	tetra unstr	0.0	3.195	3.776	2.985	2.957	2.983	3.007	3.461	3.809	3.184	2.980	3.188	3.076	1477.5	48574.3	0	2.00/3
p3--CCp2--p3	tetra unstr	0.1	3.450	3.862	3.163	2.957	3.161	3.054	3.777	3.921	3.434	2.980	3.440	3.122	1312.3	54827.0	0	2.00/3
p3--CCp3--p1	hexa	0.0	2.780	3.644	2.753	4.033	2.762	2.775	2.572	3.538	2.566	4.067	2.573	2.597	848.74	44827.5	0	2.00/3
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.716	4.033	2.725	2.746	2.522	3.545	2.523	4.068	2.530	2.555	924.30	45949.8	0	2.00/3
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1771.8	53595.6	0	2.00/3
p3--CCp3--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1473.3	58804.8	0	2.00/3
p3--CCp3--p2	hexa	0.0	3.330	3.812	3.205	3.978	3.228	3.112	3.256	3.881	3.086	4.042	3.110	3.026	838.54	45472.8	0	2.00/3
p3--CCp3--p2	hexa	0.1	3.347	3.823	3.226	3.978	3.250	3.143	3.271	3.895	3.106	4.042	3.131	3.046	962.87	46597.6	0	2.00/3
p3--CCp3--p2	tetra unstr	0.0	3.196	3.775	2.987	3.943	2.983	3.011	3.461	3.808	3.184	3.959	3.188	3.209	1352.6	54152.0	0	2.00/3
p3--CCp3--p2	tetra unstr	0.1	3.450	3.862	3.164	3.942	3.162	3.181	3.777	3.920	3.434	3.959	3.440	3.451	1423.9	59374.8	0	2.00/3
p3--CCp3--p3	hexa	0.0	3.330	3.812	3.205	3.971	3.228	3.112	3.256	3.881	3.086	4.044	3.110	3.027	887.10	46768.3	0	2.00/3
p3--CCp3--p3	hexa	0.1	3.347	3.823	3.226	3.971	3.250	3.143	3.271	3.895	3.106	4.044	3.131	3.046	959.95	47890.5	0	2.00/3
p3--CCp3--p3	tetra unstr	0.0	3.195	3.776	2.986	3.944	2.982	3.056	3.460	3.809	3.184	3.961	3.188	3.240	1351.6	55284.1	0	2.00/3
p3--CCp3--p3	tetra unstr	0.1	3.449	3.862	3.163	3.944	3.161	3.224	3.777	3.920	3.434	3.961	3.439	3.481	1400.7	60458.9	0	2.00/3
p3--GP--p1	hexa	0.0	2.780	3.644	2.003	3.792	2.717	3.810	2.572	3.538	2.000	3.687	2.355	3.691	1126.1	77279.7	0	2.00/3
p3--GP--p1	hexa	0.1	2.737	3.653	2.003	3.799	2.707	3.817	2.522	3.546	2.000	3.695	2.347	3.700	777.00	72810.8	0	2.00/3
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.216	2.216	2.217	2.217	2.121	2.431	2.176	2.176	2.179	2.179	1407.6	45295.2	0	2.00/3
p3--GP--p1	tetra unstr	0.1	2.155	2.754	2.192	2.192	2.193	2.193	2.080	2.410	2.142	2.142	2.144	2.144	1255.5	53554.9	0	2.00/3
p3--GP--p2	hexa	0.0	3.331	3.812	1.994	3.893	2.454	3.912	3.257	3.881	1.996	3.958	2.171	3.969	1014.5	77925.3	0	2.00/3
p3--GP--p2	hexa	0.1	3.348	3.823	1.994	3.897	2.447	3.916	3.272	3.896	1.996	3.964	2.167	3.975	750.17	69196.2	0	2.00/3
p3--GP--p2	tetra unstr	0.0	3.196	3.775	1.927	3.844	2.032	3.845	3.461	3.808	1.961	3.840	1.977	3.846	1367.1	45574.8	0	2.00/3
p3--GP--p2	tetra unstr	0.1	3.450	3.862	1.927	3.883	2.028	3.884	3.777	3.920	1.961	3.904	1.974	3.909	1245.5	53807.3	0	2.00/3
p3--GP--p3	hexa	0.0	3.331	3.812	1.994	3.886	2.452	3.905	3.257	3.881	1.995	3.960	2.170	3.970	1172.7	79220.7	0	2.00/3
p3--GP--p3	hexa	0.1	3.348	3.823	1.993	3.890	2.446	3.909	3.272	3.896	1.995	3.966	2.167	3.976	870.62	70486.9	0	2.00/3
p3--GP--p3	tetra unstr	0.0	3.195	3.776	1.923	3.842	2.021	3.844	3.460	3.809	1.958	3.837	1.974	3.844	1258.6	46320.8	0	2.00/3
p3--GP--p3	tetra unstr	0.1	3.450	3.862	1.923	3.884	2.017	3.885	3.777	3.921	1.958	3.904	1.971	3.911	1397.1	54335.8	0	2.00/3

$\Delta t = 10^{-3}$

Table 14: 3D manufactured results for `JFNK`, `crankNicolson`, `consistent` mass, `RKF45` states, and $\Delta t = 10^{-3}$. The first convergence block is fitted over $N = 10$–30; the second over $N = 20$–30. The first column records the actual `Vm-states-Iion` triplet used by the run. **RB** is the number of time steps that exhausted the nonlinear iteration budget, accumulated over the 21 mesh sizes of the row (1050 time steps per row in total). With `nonlinearAcceptUnconverged=false` an unconverged step is reverted to its start-of-step values while the clock still advances. n_{nl} reports the median and the maximum number of nonlinear iterations per step over those same steps; the configured budget is `nonlinearIterations`= 2000.

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS$_\Sigma$ [MB]	RB	n_{nl}
NO--na--NO	hexa	0.0	1.985	1.985	1.985	1.985	1.985	1.985	1.993	1.993	1.993	1.993	1.993	1.993	72.358	4083.5	0	2.00/3
NO--na--NO	hexa	0.1	1.438	1.438	1.293	1.293	1.302	1.302	1.413	1.413	1.211	1.211	1.220	1.220	103.30	6309.8	0	2.00/2

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
NO--na--NO	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	166.32	5142.7	0	2.00/3
NO--na--NO	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	351.54	12049.4	0	2.00/3
NO--CCp1--p1	hexa	0.0	1.986	1.986	1.989	2.167	1.986	2.246	1.993	1.993	1.995	2.208	1.993	2.188	146.88	4921.8	0	2.00/3
NO--CCp1--p1	hexa	0.1	1.438	1.438	1.298	2.167	1.302	2.230	1.413	1.413	1.212	2.208	1.220	2.158	141.64	6578.8	0	2.00/2
NO--CCp1--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	397.40	6398.6	0	2.00/3
NO--CCp1--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	587.73	12245.9	0	2.00/3
NO--CCp1--p2	hexa	0.0	1.986	1.986	1.991	2.098	1.986	2.090	1.993	1.993	1.995	2.130	1.993	2.062	207.40	6567.7	0	2.00/3
NO--CCp1--p2	hexa	0.1	1.438	1.438	1.300	2.098	1.303	2.086	1.413	1.413	1.212	2.130	1.220	2.055	200.35	8038.6	0	2.00/2
NO--CCp1--p2	tetra unstr	0.0	0.663	0.663	0.377	1.924	0.375	0.459	0.829	0.829	0.617	1.896	0.617	0.670	423.96	7036.9	0	2.00/3
NO--CCp1--p2	tetra unstr	0.1	0.677	0.677	0.400	1.928	0.398	0.484	0.828	0.828	0.629	1.903	0.629	0.684	613.00	12498.1	0	2.00/3
NO--CCp1--p3	hexa	0.0	1.986	1.986	1.991	2.094	1.986	2.088	1.993	1.993	1.995	2.127	1.993	2.060	288.40	9876.9	0	2.00/3
NO--CCp1--p3	hexa	0.1	1.438	1.438	1.300	2.095	1.303	2.085	1.413	1.413	1.212	2.127	1.220	2.054	282.07	11332.3	0	2.00/2
NO--CCp1--p3	tetra unstr	0.0	0.663	0.663	0.377	1.922	0.375	0.462	0.829	0.829	0.617	1.906	0.617	0.671	469.85	8302.9	0	2.00/3
NO--CCp1--p3	tetra unstr	0.1	0.677	0.677	0.400	1.926	0.398	0.487	0.828	0.828	0.629	1.912	0.629	0.685	615.47	13000.4	0	2.00/3
NO--CCp2--p1	hexa	0.0	1.977	1.977	2.002	3.171	1.978	2.610	1.989	1.989	1.999	3.091	1.989	2.376	216.25	5692.4	0	2.00/3
NO--CCp2--p1	hexa	0.1	1.447	1.447	1.258	3.172	1.313	2.174	1.428	1.428	1.175	3.092	1.236	1.620	226.43	7165.9	0	2.00/2
NO--CCp2--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	773.69	9761.4	0	2.00/3
NO--CCp2--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	936.72	14003.0	0	2.00/3
NO--CCp2--p2	hexa	0.0	1.969	1.969	2.004	3.247	1.970	2.745	1.985	1.985	1.998	3.146	1.985	2.481	275.43	7341.0	0	2.00/3
NO--CCp2--p2	hexa	0.1	1.457	1.457	1.251	3.248	1.326	2.314	1.443	1.443	1.170	3.147	1.253	1.712	331.22	8813.8	0	2.00/2
NO--CCp2--p2	tetra unstr	0.0	0.663	0.663	0.377	2.079	0.375	0.467	0.829	0.829	0.617	1.272	0.617	0.673	824.29	10398.8	0	2.00/3
NO--CCp2--p2	tetra unstr	0.1	0.677	0.677	0.400	2.127	0.398	0.488	0.828	0.828	0.629	1.333	0.629	0.684	919.93	14641.5	0	2.00/3
NO--CCp2--p3	hexa	0.0	1.969	1.969	2.004	3.252	1.970	2.751	1.985	1.985	1.998	3.149	1.985	2.488	407.86	10640.1	0	2.00/3
NO--CCp2--p3	hexa	0.1	1.457	1.457	1.251	3.253	1.326	2.323	1.443	1.443	1.170	3.150	1.253	1.720	489.56	12109.3	0	2.00/2
NO--CCp2--p3	tetra unstr	0.0	0.663	0.663	0.377	1.991	0.375	0.469	0.829	0.829	0.617	1.240	0.617	0.676	921.20	11665.9	0	2.00/3
NO--CCp2--p3	tetra unstr	0.1	0.677	0.677	0.400	2.038	0.398	0.489	0.828	0.828	0.629	1.295	0.629	0.686	992.95	15908.5	0	2.00/3
NO--CCp3--p1	hexa	0.0	1.980	1.980	1.993	4.033	1.980	2.052	1.992	1.992	1.995	4.072	1.992	2.002	420.83	7592.3	0	2.00/3
NO--CCp3--p1	hexa	0.1	1.453	1.453	1.241	4.015	1.318	1.271	1.431	1.431	1.167	4.028	1.238	1.184	548.92	9065.3	0	2.00/2
NO--CCp3--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	2082.3	18530.6	0	2.00/3
NO--CCp3--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	1750.0	22782.3	0	2.00/3
NO--CCp3--p2	hexa	0.0	1.975	1.975	1.992	3.981	1.975	2.098	1.990	1.990	1.995	4.053	1.990	2.007	505.78	9240.4	0	2.00/3
NO--CCp3--p2	hexa	0.1	1.469	1.469	1.230	3.966	1.336	1.300	1.449	1.449	1.160	4.019	1.258	1.185	655.78	10710.5	0	2.00/2
NO--CCp3--p2	tetra unstr	0.0	0.663	0.663	0.378	0.891	0.375	0.420	0.829	0.829	0.617	0.675	0.617	0.658	2065.6	19164.9	0	2.00/3
NO--CCp3--p2	tetra unstr	0.1	0.677	0.677	0.400	0.930	0.398	0.443	0.828	0.828	0.629	0.689	0.629	0.671	1777.1	23407.9	0	2.00/3
NO--CCp3--p3	hexa	0.0	1.975	1.975	1.992	3.973	1.975	2.099	1.991	1.991	1.995	4.055	1.990	2.007	668.97	12534.2	0	2.00/3
NO--CCp3--p3	hexa	0.1	1.469	1.469	1.230	3.959	1.336	1.300	1.449	1.449	1.160	4.021	1.258	1.184	846.23	14010.0	0	2.00/2
NO--CCp3--p3	tetra unstr	0.0	0.663	0.663	0.378	0.888	0.375	0.434	0.829	0.829	0.617	0.684	0.617	0.666	2134.2	20432.8	0	2.00/3
NO--CCp3--p3	tetra unstr	0.1	0.677	0.677	0.400	0.927	0.398	0.456	0.828	0.828	0.629	0.698	0.629	0.678	1875.4	24675.8	0	2.00/3
NO--GP--p1	hexa	0.0	1.985	1.985	1.992	2.071	1.992	2.077	1.993	1.993	1.996	2.038	1.996	2.041	274.71	4669.2	0	2.00/3
NO--GP--p1	hexa	0.1	1.461	1.461	1.992	2.073	1.992	2.079	1.431	1.431	1.996	2.038	1.995	2.041	286.77	6561.2	0	2.00/2
NO--GP--p1	tetra unstr	0.0	0.663	0.663	0.377	0.377	0.375	0.375	0.829	0.829	0.617	0.617	0.617	0.617	172.50	5446.4	0	2.00/3
NO--GP--p1	tetra unstr	0.1	0.677	0.677	0.400	0.400	0.398	0.398	0.828	0.828	0.629	0.629	0.629	0.629	358.09	12247.7	0	2.00/3
NO--GP--p2	hexa	0.0	1.984	1.984	1.984	2.014	1.984	2.018	1.992	1.992	1.992	2.007	1.993	2.009	916.82	6341.8	0	2.00/3
NO--GP--p2	hexa	0.1	1.483	1.483	1.984	2.014	1.985	2.018	1.450	1.450	1.992	2.007	1.992	2.009	749.00	7704.6	0	2.00/2
NO--GP--p2	tetra unstr	0.0	0.663	0.663	1.832	0.862	0.518	0.860	0.829	0.829	1.792	0.867	0.653	0.866	507.51	6090.2	0	2.00/3
NO--GP--p2	tetra unstr	0.1	0.677	0.677	1.843	0.883	0.547	0.882	0.828	0.828	1.812	0.884	0.668	0.883	549.05	12497.3	0	2.00/3
NO--GP--p3	hexa	0.0	1.984	1.984	1.983	2.010	1.984	2.014	1.992	1.992	1.992	2.005	1.992	2.008	2080.8	9677.3	0	2.00/3
NO--GP--p3	hexa	0.1	1.483	1.483	1.984	2.011	1.984	2.015	1.450	1.450	1.992	2.005	1.992	2.008	1710.3	11009.2	0	2.00/2
NO--GP--p3	tetra unstr	0.0	0.663	0.663	1.828	0.864	0.514	0.862	0.829	0.829	1.790	0.869	0.652	0.868	907.09	7355.9	0	2.00/3
NO--GP--p3	tetra unstr	0.1	0.677	0.677	1.840	0.885	0.543	0.884	0.828	0.828	1.810	0.886	0.667	0.885	915.26	12995.6	0	2.00/3
p1--na--NO	hexa	0.0	1.967	2.095	1.924	1.924	1.927	1.927	1.920	2.125	1.833	1.833	1.840	1.840	156.75	7136.1	0	2.00/2
p1--na--NO	hexa	0.1	2.067	2.093	2.044	2.044	2.044	2.044	2.067	2.124	2.028	2.028	2.029	2.029	152.25	8682.5	0	2.00/2
p1--na--NO	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	382.68	10354.9	0	2.00/3
p1--na--NO	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	450.48	11878.2	0	2.00/3
p1--CCp1--p1	hexa	0.0	1.967	2.095	1.924	2.166	1.926	2.192	1.920	2.125	1.833	2.207	1.840	2.114	195.46	7316.9	0	2.00/2
p1--CCp1--p1	hexa	0.1	2.067	2.093	2.043	2.167	2.044	2.224	2.067	2.124	2.028	2.208	2.029	2.167	235.13	8860.8	0	2.00/2
p1--CCp1--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	654.04	11401.3	0	2.00/3
p1--CCp1--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	639.72	12625.4	0	2.00/3
p1--CCp1--p2	hexa	0.0	1.967	2.095	1.924	2.098	1.926	2.083	1.920	2.125	1.833	2.130	1.840	2.049	255.15	8764.0	0	2.00/2
p1--CCp1--p2	hexa	0.1	2.067	2.093	2.043	2.098	2.044	2.090	2.067	2.124	2.028	2.130	2.029	2.061	320.59	9608.3	0	2.00/2
p1--CCp1--p2	tetra unstr	0.0	1.475	1.866	1.142	1.973	1.144	1.493	1.768	1.927	1.422	1.984	1.432	1.592	708.35	12038.6	0	2.00/3
p1--CCp1--p2	tetra unstr	0.1	1.612	1.909	1.265	1.973	1.267	1.622	1.884	1.962	1.558	1.983	1.561	1.723	668.04	13190.7	0	2.00/3
p1--CCp1--p3	hexa	0.0	1.967	2.095	1.924	2.094	1.926	2.081	1.920	2.125	1.833	2.127	1.840	2.048	364.01	12046.0	0	2.00/2
p1--CCp1--p3	hexa	0.1	2.067	2.093	2.043	2.094	2.044	2.088	2.067	2.124	2.028	2.127	2.029	2.059	375.87	12503.0	0	2.00/2
p1--CCp1--p3	tetra unstr	0.0	1.475	1.866	1.142	1.965	1.144	1.506	1.768	1.927	1.422	1.982	1.432	1.599	782.50	13300.4	0	2.00/3
p1--CCp1--p3	tetra unstr	0.1	1.612	1.909	1.265	1.965	1.267	1.633	1.884	1.962	1.558	1.982	1.561	1.729	688.38	14322.8	0	2.00/3
p1--CCp2--p1	hexa	0.0	1.966	2.095	1.924	3.169	1.926	2.040	1.920	2.125	1.827	3.089	1.840	1.840	273.24	7875.3	0	2.00/2
p1--CCp2--p1	hexa	0.1	2.065	2.093	2.048	3.170	2.043	2.196	2.066	2.124	2.028	3.090	2.028	2.085	333.30	8900.6	0	2.00/2
p1--CCp2--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	994.37	14770.4	0	2.00/3
p1--CCp2--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	941.43	15615.3	0	2.00/3
p1--CCp2--p2	hexa	0.0	1.966	2.095	1.924	3.244	1.925	2.092	1.920	2.125	1.824	3.143	1.840	1.841	342.59	9527.4	0	2.00/2
p1--CCp2--p2	hexa	0.1	2.064	2.093	2.050	3.245	2.042	2.261	2.065	2.124	2.028	3.145	2.028	2.109	372.72	9995.8	0	2.00/2
p1--CCp2--p2	tetra unstr	0.0	1.475	1.866	1.141	2.917	1.145	1.201	1.768	1.927	1.422	2.912	1.432	1.469	1121.4	15397.2	0	2.00/3
p1--CCp2--p2	tetra unstr	0.1	1.612	1.909	1.264	2.928	1.267	1.339	1.884	1.962	1.558	2.938	1.561	1.617	1006.7	16247.0	0	2.00/3
p1--CCp2--p3	hexa	0.0	1.966	2.095	1.924	3.250	1.925	2.095	1.920	2.125	1.824	3.146	1.840	1.843	468.05	12820.1	0	2.00/2
p1--CCp2--p3	hexa	0.1	2.064	2.093	2.050	3.251	2.042	2.265	2.065	2.124	2.028	3.148	2.028	2.111	477.61	13286.9	0	2.00/2
p1--CCp2--p3	tetra unstr	0.0	1.475	1.866	1.141	2.922	1.145	1.213	1.768	1.927	1.422	2.906	1.432	1.479	1152.0	16666.6	0	2.00/3
p1--CCp2--p3	tetra unstr	0.1	1.612	1.909	1.264	2.938	1.267	1.350	1.884	1.962	1.558	2.938	1.561	1.627	1051.6	17517.1	0	2.00/3
p1--CCp3--p1	hexa	0.0	1.967	2.095	1.919	4.067	1.926	1.874	1.920	2.125	1.826	4.139	1.840	1.725	490.32	9775.7	0	2.00/2
p1--CCp3--p1	hexa	0.1	2.066	2.093	2.043	4.057	2.043	2.038	2.067	2.124	2.026	4.119	2.029	2.002	549.92	10198.1	0	2.00/2
p1--CCp3--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	2065.7	23533.8	0	2.00/3

			Fit $N = 10$–30						Fit $N = 20$–30									
Vm-states-Iion	Mesh type	α	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	t_Σ [s]	RSS_Σ [MB]	RB	n_{nl}
p1--CCp3--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	1927.3	24383.1	0	2.00/3
p1--CCp3--p2	hexa	0.0	1.967	2.095	1.917	4.010	1.926	1.850	1.921	2.125	1.823	4.112	1.841	1.677	651.67	11428.5	0	2.00/2
p1--CCp3--p2	hexa	0.1	2.066	2.093	2.042	4.000	2.043	2.035	2.067	2.124	2.025	4.092	2.029	1.988	614.83	11868.1	0	2.00/2
p1--CCp3--p2	tetra unstr	0.0	1.475	1.866	1.141	2.935	1.145	1.205	1.768	1.927	1.422	2.111	1.432	1.475	2041.6	24166.4	0	2.00/3
p1--CCp3--p2	tetra unstr	0.1	1.612	1.909	1.264	3.163	1.267	1.335	1.884	1.962	1.558	2.438	1.561	1.616	1820.0	25022.9	0	2.00/3
p1--CCp3--p3	hexa	0.0	1.967	2.095	1.917	4.003	1.926	1.849	1.921	2.125	1.823	4.114	1.841	1.676	904.04	14721.7	0	2.00/2
p1--CCp3--p3	hexa	0.1	2.066	2.093	2.042	3.992	2.043	2.035	2.067	2.124	2.025	4.094	2.029	1.988	852.94	15150.3	0	2.00/2
p1--CCp3--p3	tetra unstr	0.0	1.475	1.866	1.141	2.836	1.145	1.243	1.768	1.927	1.422	2.067	1.432	1.500	2162.4	25434.4	0	2.00/3
p1--CCp3--p3	tetra unstr	0.1	1.612	1.909	1.264	3.084	1.267	1.378	1.884	1.962	1.558	2.395	1.561	1.647	1927.8	26288.7	0	2.00/3
p1--GP--p1	hexa	0.0	1.968	2.095	1.992	2.157	1.999	2.164	1.920	2.125	1.997	2.201	2.001	2.200	321.20	7342.0	0	2.00/2
p1--GP--p1	hexa	0.1	2.067	2.093	1.993	2.155	2.001	2.162	2.067	2.124	1.997	2.199	2.004	2.199	312.60	8892.4	0	2.00/2
p1--GP--p1	tetra unstr	0.0	1.475	1.866	1.142	1.142	1.144	1.144	1.768	1.927	1.422	1.422	1.432	1.432	523.84	10459.1	0	2.00/3
p1--GP--p1	tetra unstr	0.1	1.612	1.909	1.265	1.265	1.267	1.267	1.884	1.962	1.558	1.558	1.561	1.561	524.88	11970.4	0	2.00/3
p1--GP--p2	hexa	0.0	1.968	2.095	1.984	2.094	1.989	2.098	1.921	2.125	1.993	2.129	1.996	2.128	829.14	8590.9	0	2.00/2
p1--GP--p2	hexa	0.1	2.067	2.093	1.985	2.092	1.991	2.096	2.067	2.124	1.993	2.128	1.998	2.126	807.62	9577.6	0	2.00/2
p1--GP--p2	tetra unstr	0.0	1.475	1.866	1.918	1.862	1.726	1.866	1.768	1.927	1.953	1.877	1.777	1.880	711.22	11085.8	0	2.00/3
p1--GP--p2	tetra unstr	0.1	1.612	1.909	1.920	1.897	1.790	1.900	1.884	1.962	1.955	1.921	1.849	1.923	907.71	12474.6	0	2.00/3
p1--GP--p3	hexa	0.0	1.968	2.095	1.984	2.091	1.989	2.094	1.921	2.125	1.993	2.126	1.996	2.124	1859.7	12031.6	0	2.00/2
p1--GP--p3	hexa	0.1	2.067	2.093	1.984	2.089	1.990	2.093	2.067	2.124	1.993	2.124	1.998	2.123	1756.2	12472.6	0	2.00/2
p1--GP--p3	tetra unstr	0.0	1.475	1.866	1.914	1.870	1.723	1.873	1.768	1.927	1.951	1.889	1.775	1.892	1090.2	12361.6	0	2.00/3
p1--GP--p3	tetra unstr	0.1	1.612	1.909	1.916	1.900	1.786	1.903	1.884	1.962	1.953	1.927	1.847	1.929	1269.1	13565.9	0	2.00/3
p2--na--NO	hexa	0.0	1.947	3.025	1.942	1.942	1.946	1.946	2.001	2.766	2.000	2.000	2.003	2.003	262.11	25852.2	0	2.00/3
p2--na--NO	hexa	0.1	1.946	3.053	1.941	1.941	1.946	1.946	2.000	2.807	1.999	1.999	2.002	2.002	239.10	21777.7	0	2.00/3
p2--na--NO	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	749.44	18258.0	0	2.00/2
p2--na--NO	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	925.06	21620.3	0	2.00/2
p2--CCp1--p1	hexa	0.0	2.003	3.107	1.993	2.168	2.001	2.183	2.035	2.884	2.031	2.209	2.036	2.145	347.00	13057.2	0	2.00/3
p2--CCp1--p1	hexa	0.1	2.008	3.133	1.997	2.167	2.006	2.192	2.037	2.925	2.033	2.209	2.038	2.152	340.08	15812.1	0	2.00/3
p2--CCp1--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1129.9	19181.2	0	2.00/2
p2--CCp1--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1164.1	21994.5	0	2.00/2
p2--CCp1--p2	hexa	0.0	2.040	3.141	2.024	2.098	2.036	2.087	2.059	2.936	2.052	2.131	2.060	2.062	444.36	13775.5	0	2.00/3
p2--CCp1--p2	hexa	0.1	2.049	3.165	2.033	2.098	2.046	2.088	2.064	2.974	2.056	2.131	2.065	2.062	452.39	16460.4	0	2.00/3
p2--CCp1--p2	tetra unstr	0.0	2.105	2.705	1.863	1.971	1.864	1.960	2.382	2.719	2.113	1.982	2.122	1.974	1185.2	19827.8	0	2.00/2
p2--CCp1--p2	tetra unstr	0.1	2.348	2.824	2.014	1.971	2.018	1.970	2.721	2.901	2.364	1.982	2.374	1.986	1139.6	22504.4	0	2.00/2
p2--CCp1--p3	hexa	0.0	2.040	3.141	2.024	2.095	2.036	2.085	2.059	2.936	2.052	2.127	2.060	2.061	558.69	16265.4	0	2.00/3
p2--CCp1--p3	hexa	0.1	2.049	3.165	2.033	2.095	2.046	2.086	2.064	2.974	2.056	2.127	2.065	2.061	527.65	17829.2	0	2.00/3
p2--CCp1--p3	tetra unstr	0.0	2.105	2.705	1.863	1.963	1.864	1.958	2.382	2.719	2.113	1.981	2.122	1.973	1232.7	21091.2	0	2.00/2
p2--CCp1--p3	tetra unstr	0.1	2.347	2.824	2.014	1.963	2.018	1.968	2.721	2.901	2.364	1.981	2.374	1.984	1258.2	23563.2	0	2.00/2
p2--CCp2--p1	hexa	0.0	2.003	3.107	1.998	3.170	2.002	2.075	2.035	2.884	2.035	3.091	2.037	2.071	458.92	13080.6	0	2.00/3
p2--CCp2--p1	hexa	0.1	2.008	3.133	2.003	3.171	2.007	2.094	2.037	2.925	2.037	3.091	2.039	2.079	456.09	15807.8	0	2.00/3
p2--CCp2--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	1770.4	22550.0	0	2.00/2
p2--CCp2--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1534.1	24551.7	0	2.00/2
p2--CCp2--p2	hexa	0.0	2.040	3.142	2.034	3.246	2.037	2.170	2.059	2.936	2.059	3.146	2.061	2.125	522.93	14047.8	0	2.00/3
p2--CCp2--p2	hexa	0.1	2.050	3.166	2.044	3.246	2.047	2.206	2.064	2.975	2.064	3.146	2.066	2.143	550.47	16461.9	0	2.00/3
p2--CCp2--p2	tetra unstr	0.0	2.105	2.705	1.863	2.944	1.864	1.932	2.384	2.720	2.116	2.972	2.122	2.121	1788.3	23183.3	0	2.00/2
p2--CCp2--p2	tetra unstr	0.1	2.350	2.825	2.016	2.945	2.018	2.104	2.727	2.903	2.369	2.972	2.376	2.408	1509.6	25124.1	0	2.00/2
p2--CCp2--p3	hexa	0.0	2.040	3.142	2.034	3.251	2.037	2.173	2.059	2.936	2.059	3.149	2.061	2.126	678.82	16916.0	0	2.00/3
p2--CCp2--p3	hexa	0.1	2.050	3.166	2.044	3.251	2.047	2.209	2.064	2.975	2.064	3.149	2.066	2.144	636.70	17987.3	0	2.00/3
p2--CCp2--p3	tetra unstr	0.0	2.105	2.705	1.863	2.958	1.864	1.943	2.384	2.720	2.116	2.980	2.122	2.134	1767.1	24450.4	0	2.00/2
p2--CCp2--p3	tetra unstr	0.1	2.350	2.825	2.016	2.959	2.018	2.111	2.727	2.903	2.369	2.981	2.376	2.414	1630.4	26269.5	0	2.00/2
p2--CCp3--p1	hexa	0.0	2.003	3.107	1.997	4.087	2.002	2.020	2.035	2.884	2.035	4.142	2.037	2.056	658.77	14040.1	0	2.00/3
p2--CCp3--p1	hexa	0.1	2.008	3.133	2.002	4.082	2.007	2.026	2.037	2.925	2.037	4.137	2.039	2.058	633.51	15815.6	0	2.00/3
p2--CCp3--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	2738.8	31317.0	0	2.00/2
p2--CCp3--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	2433.5	32698.6	0	2.00/2
p2--CCp3--p2	hexa	0.0	2.041	3.142	2.033	4.022	2.037	2.072	2.060	2.936	2.059	4.109	2.061	2.100	755.42	15529.7	0	2.00/3
p2--CCp3--p2	hexa	0.1	2.050	3.166	2.043	4.017	2.047	2.084	2.064	2.975	2.064	4.103	2.066	2.107	757.51	16648.0	0	2.00/3
p2--CCp3--p2	tetra unstr	0.0	2.105	2.705	1.863	3.893	1.864	1.897	2.384	2.720	2.116	3.841	2.123	2.129	2923.3	31951.7	0	2.00/2
p2--CCp3--p2	tetra unstr	0.1	2.350	2.825	2.016	3.911	2.018	2.048	2.727	2.903	2.369	3.890	2.376	2.392	2520.1	33327.7	0	2.00/2
p2--CCp3--p3	hexa	0.0	2.041	3.142	2.033	4.014	2.037	2.073	2.060	2.936	2.059	4.110	2.061	2.101	996.82	18508.9	0	2.00/3
p2--CCp3--p3	hexa	0.1	2.050	3.166	2.043	4.010	2.047	2.085	2.064	2.975	2.064	4.104	2.066	2.107	983.63	18888.2	0	2.00/3
p2--CCp3--p3	tetra unstr	0.0	2.105	2.705	1.863	3.888	1.864	1.929	2.384	2.720	2.116	3.831	2.123	2.148	2915.4	33215.1	0	2.00/2
p2--CCp3--p3	tetra unstr	0.1	2.350	2.825	2.016	3.907	2.018	2.074	2.727	2.903	2.369	3.884	2.376	2.406	2557.2	34591.3	0	2.00/2
p2--GP--p1	hexa	0.0	2.004	3.107	2.011	3.133	2.844	3.122	2.035	2.884	2.004	3.004	2.447	2.998	475.17	26058.0	0	2.00/3
p2--GP--p1	hexa	0.1	2.008	3.133	2.011	3.142	2.886	3.131	2.037	2.925	2.004	3.021	2.473	3.014	434.84	21987.3	0	2.00/3
p2--GP--p1	tetra unstr	0.0	2.044	2.627	1.867	1.867	1.868	1.868	2.248	2.603	2.077	2.077	2.084	2.084	923.33	18403.9	0	2.00/2
p2--GP--p1	tetra unstr	0.1	2.198	2.744	1.985	1.985	1.987	1.987	2.429	2.767	2.271	2.271	2.279	2.279	1077.7	21759.6	0	2.00/2
p2--GP--p2	hexa	0.0	2.041	3.142	2.004	3.230	2.897	3.215	2.060	2.936	2.001	3.092	2.431	3.082	1125.6	26708.6	0	2.00/3
p2--GP--p2	hexa	0.1	2.051	3.166	2.004	3.236	2.920	3.221	2.064	2.975	2.001	3.105	2.439	3.094	919.07	22635.8	0	2.00/3
p2--GP--p2	tetra unstr	0.0	2.105	2.705	1.928	2.794	2.100	2.795	2.384	2.720	1.960	2.779	2.102	2.783	1131.3	19048.7	0	2.00/2
p2--GP--p2	tetra unstr	0.1	2.350	2.825	1.928	2.831	2.146	2.833	2.727	2.903	1.960	2.862	2.167	2.866	1319.3	22134.4	0	2.00/2
p2--GP--p3	hexa	0.0	2.041	3.142	2.003	3.235	2.896	3.220	2.060	2.936	2.000	3.095	2.430	3.085	2346.5	28002.5	0	2.00/3
p2--GP--p3	hexa	0.1	2.051	3.166	2.003	3.241	2.919	3.226	2.064	2.975	2.000	3.108	2.438	3.097	1865.3	24076.1	0	2.00/3
p2--GP--p3	tetra unstr	0.0	2.105	2.705	1.924	2.801	2.082	2.802	2.384	2.720	1.958	2.768	2.089	2.773	1507.3	20304.2	0	2.00/2
p2--GP--p3	tetra unstr	0.1	2.350	2.825	1.924	2.846	2.128	2.847	2.727	2.903	1.958	2.864	2.153	2.868	1828.5	23037.4	0	2.00/2
p3--na--NO	hexa	0.0	2.313	3.204	2.303	2.303	2.307	2.307	2.193	2.830	2.189	2.189	2.192	2.192	816.96	68118.1	0	2.00/2
p3--na--NO	hexa	0.1	2.278	3.207	2.269	2.269	2.273	2.273	2.169	2.827	2.166	2.166	2.168	2.168	892.18	72598.4	0	2.00/3
p3--na--NO	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	1833.8	45160.4	0	2.00/2
p3--na--NO	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2087.5	53471.6	0	2.00/2
p3--CCp1--p1	hexa	0.0	2.758	3.637	2.675	2.166	2.727	2.263	2.547	3.524	2.476	2.207	2.525	2.199	1199.2	44824.7	0	2.00/2
p3--CCp1--p1	hexa	0.1	2.713	3.646	2.635	2.166	2.687	2.261	2.497	3.532	2.433	2.207	2.481	2.197	1167.3	45948.4	0	2.00/3
p3--CCp1--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2843.6	45874.6	0	2.00/2
p3--CCp1--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2627.8	53130.2	0	2.00/2
p3--CCp1--p2	hexa	0.0	3.096	3.790	2.987	2.097	3.017	2.094	2.864	3.829	2.773	2.130	2.782	2.063	1176.3	45473.1	0	2.00/2

Vm-states-Iion	Mesh type	α	Fit N = 10–30						Fit N = 20–30						t_Σ [s]	RSS$_\Sigma$ [MB]	RB	n_{nl}
			V_m	V_m^G	u_1	u_1^G	u_2	u_2^G	V_m	V_m^G	u_1	u_1^G	u_2	u_2^G				
p3--CCp1--p2	hexa	0.1	3.064	3.800	2.965	2.097	2.994	2.093	2.807	3.842	2.734	2.130	2.739	2.063	1260.0	46595.7	0	2.00/3
p3--CCp1--p2	tetra unstr	0.0	2.979	3.699	2.857	1.970	2.857	1.968	3.035	3.633	2.949	1.982	2.960	1.961	3055.9	46211.8	0	2.00/2
p3--CCp1--p2	tetra unstr	0.1	3.071	3.771	2.960	1.971	2.963	1.968	3.012	3.707	3.038	1.982	3.053	1.960	2685.8	53376.4	0	2.00/2
p3--CCp1--p3	hexa	0.0	3.096	3.790	2.987	2.094	3.017	2.092	2.864	3.829	2.773	2.127	2.782	2.062	1264.3	46766.6	0	2.00/2
p3--CCp1--p3	hexa	0.1	3.064	3.800	2.965	2.094	2.994	2.091	2.807	3.842	2.734	2.127	2.739	2.062	1457.9	47888.7	0	2.00/3
p3--CCp1--p3	tetra unstr	0.0	2.978	3.699	2.856	1.963	2.857	1.966	3.034	3.633	2.949	1.981	2.960	1.960	2976.5	46930.7	0	2.00/2
p3--CCp1--p3	tetra unstr	0.1	3.071	3.772	2.959	1.963	2.963	1.966	3.011	3.707	3.037	1.981	3.053	1.960	2727.5	53953.5	0	2.00/2
p3--CCp2--p1	hexa	0.0	2.782	3.644	2.745	3.172	2.752	2.948	2.574	3.538	2.546	3.092	2.553	2.790	1126.2	44825.4	0	2.00/2
p3--CCp2--p1	hexa	0.1	2.739	3.653	2.708	3.172	2.715	2.943	2.524	3.546	2.503	3.092	2.509	2.778	1332.6	45949.2	0	2.00/3
p3--CCp2--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3195.5	47169.4	0	2.00/2
p3--CCp2--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2955.2	53666.3	0	2.00/2
p3--CCp2--p2	hexa	0.0	3.335	3.812	3.202	3.247	3.220	3.232	3.264	3.882	3.065	3.147	3.095	3.171	1216.2	45473.7	0	2.00/2
p3--CCp2--p2	hexa	0.1	3.352	3.823	3.223	3.247	3.241	3.244	3.280	3.896	3.083	3.147	3.114	3.186	1389.5	46594.7	0	2.00/3
p3--CCp2--p2	tetra unstr	0.0	3.198	3.777	2.984	2.942	2.981	2.996	3.463	3.810	3.182	2.970	3.187	3.073	3139.1	47660.2	0	2.00/2
p3--CCp2--p2	tetra unstr	0.1	3.452	3.863	3.162	2.942	3.161	3.045	3.775	3.921	3.433	2.970	3.439	3.121	3022.9	54043.4	0	2.00/2
p3--CCp2--p3	hexa	0.0	3.335	3.812	3.202	3.253	3.220	3.231	3.264	3.882	3.065	3.150	3.095	3.171	1407.3	46767.6	0	2.00/2
p3--CCp2--p3	hexa	0.1	3.352	3.823	3.223	3.253	3.241	3.243	3.279	3.896	3.083	3.150	3.114	3.185	1572.6	47888.4	0	2.00/3
p3--CCp2--p3	tetra unstr	0.0	3.198	3.777	2.984	2.957	2.981	3.007	3.463	3.810	3.182	2.980	3.187	3.076	3223.3	48661.1	0	2.00/2
p3--CCp2--p3	tetra unstr	0.1	3.451	3.863	3.162	2.957	3.160	3.055	3.776	3.921	3.433	2.980	3.439	3.123	3093.8	54824.9	0	2.00/2
p3--CCp3--p1	hexa	0.0	2.781	3.644	2.742	4.034	2.751	2.764	2.573	3.538	2.546	4.068	2.552	2.578	1388.8	44828.8	0	2.00/2
p3--CCp3--p1	hexa	0.1	2.737	3.653	2.705	4.034	2.713	2.735	2.523	3.546	2.502	4.068	2.508	2.535	1433.8	45950.0	0	2.00/3
p3--CCp3--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	3971.2	53601.4	0	2.00/2
p3--CCp3--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	3675.5	58844.0	0	2.00/2
p3--CCp3--p2	hexa	0.0	3.332	3.813	3.194	3.978	3.216	3.106	3.260	3.882	3.069	4.042	3.091	3.018	1500.8	45473.3	0	2.00/2
p3--CCp3--p2	hexa	0.1	3.349	3.823	3.214	3.978	3.236	3.135	3.275	3.896	3.087	4.042	3.109	3.036	1588.8	46598.1	0	2.00/3
p3--CCp3--p2	tetra unstr	0.0	3.198	3.777	2.985	3.943	2.981	3.010	3.463	3.810	3.182	3.960	3.187	3.210	3382.6	54155.5	0	2.00/2
p3--CCp3--p2	tetra unstr	0.1	3.452	3.863	3.163	3.943	3.161	3.182	3.775	3.921	3.433	3.960	3.438	3.453	3690.8	59374.8	0	2.00/2
p3--CCp3--p3	hexa	0.0	3.332	3.813	3.194	3.971	3.216	3.106	3.260	3.882	3.069	4.044	3.091	3.018	1721.3	46770.7	0	2.00/2
p3--CCp3--p3	hexa	0.1	3.349	3.823	3.214	3.971	3.236	3.136	3.275	3.896	3.087	4.044	3.109	3.037	1751.7	47890.0	0	2.00/3
p3--CCp3--p3	tetra unstr	0.0	3.197	3.777	2.984	3.944	2.980	3.056	3.462	3.810	3.182	3.962	3.186	3.240	3477.8	55286.0	0	2.00/2
p3--CCp3--p3	tetra unstr	0.1	3.451	3.863	3.162	3.944	3.160	3.224	3.775	3.921	3.433	3.962	3.438	3.483	3867.2	60459.8	0	2.00/2
p3--GP--p1	hexa	0.0	2.781	3.644	2.003	3.789	2.716	3.807	2.573	3.538	2.000	3.679	2.354	3.683	1453.9	77276.2	0	2.00/2
p3--GP--p1	hexa	0.1	2.738	3.653	2.003	3.796	2.706	3.814	2.523	3.546	2.000	3.687	2.345	3.692	1107.5	68550.1	0	2.00/3
p3--GP--p1	tetra unstr	0.0	2.189	2.753	2.208	2.208	2.209	2.209	2.121	2.431	2.162	2.162	2.165	2.165	2698.7	45294.4	0	2.00/2
p3--GP--p1	tetra unstr	0.1	2.155	2.755	2.184	2.184	2.185	2.185	2.080	2.410	2.128	2.128	2.130	2.130	2581.7	53555.5	0	2.00/2
p3--GP--p2	hexa	0.0	3.332	3.813	1.994	3.892	2.454	3.910	3.260	3.882	1.996	3.955	2.171	3.965	2161.7	77923.7	0	2.00/2
p3--GP--p2	hexa	0.1	3.350	3.823	1.994	3.896	2.447	3.914	3.275	3.896	1.996	3.961	2.167	3.971	1552.4	69194.7	0	2.00/3
p3--GP--p2	tetra unstr	0.0	3.198	3.777	1.927	3.839	2.032	3.841	3.463	3.810	1.961	3.829	1.977	3.835	2892.2	45571.9	0	2.00/2
p3--GP--p2	tetra unstr	0.1	3.452	3.863	1.927	3.878	2.028	3.879	3.775	3.921	1.961	3.893	1.974	3.898	2807.7	53808.9	0	2.00/2
p3--GP--p3	hexa	0.0	3.332	3.813	1.994	3.884	2.452	3.904	3.260	3.882	1.995	3.956	2.170	3.967	3441.1	79067.7	0	2.00/2
p3--GP--p3	hexa	0.1	3.349	3.823	1.993	3.889	2.446	3.908	3.275	3.896	1.995	3.962	2.167	3.973	2516.6	70489.4	0	2.00/3
p3--GP--p3	tetra unstr	0.0	3.197	3.777	1.923	3.838	2.021	3.840	3.462	3.810	1.958	3.826	1.974	3.833	2931.1	46322.2	0	2.00/2
p3--GP--p3	tetra unstr	0.1	3.451	3.863	1.923	3.879	2.017	3.880	3.775	3.921	1.958	3.893	1.971	3.900	3400.9	54328.9	0	2.00/2

13.5 Discussion of the 3D results

The qualitative structure repeats, but the 3D sweep is pre-asymptotic. The conclusions on method independence, joint unlocking of the triplet, and the superiority of `cellCentredReconstruct` carry over unchanged to 3D. What does not carry over are the absolute values of the rates: `p3` delivers 3.26 on the hexahedral mesh and 3.46 on the unstructured tetrahedral one, against 4.32 and 4.48 in 2D. This does not indicate a degradation of the method but that the 3D sweep never reached the asymptotic regime. The pairwise rates show it unambiguously: for `p3--CCp2--p2` on hexa, p_{N_1,N_2} grows *monotonically* from 3.154 between $N = 18$ and 19 up to 3.414 between $N = 29$ and 30, with no sign of settling. In 2D, by contrast, the same sequence varies only between 4.346 and 4.381 over the whole $N = 88$–100 range, i.e. it has already converged. The ceiling of the 3D sweep is $N = 30$, that is 27 000 cells, whereas in 2D it is $N = 100$ with 10 000 cells but a three times smaller h. **The 3D rates of this section should be read as lower bounds on the asymptotic order.**

The temporal error does not intervene in 3D. Unlike in 2D, going from $\Delta t = 10^{-2}$ to 10^{-3} hardly changes any 3D rate: the median of $|\Delta p|$ over the 2 496 comparisons is 0.0014, the 95th percentile is 0.021, and the maximum is 0.112 (against a median of 0.0148 and a 95th percentile of 2.31 in 2D). This is consistent with the previous point: because the 3D meshes are coarse, the spatial error stays well above the temporal floor and never touches it. Both Δt tables are therefore readable in 3D.

Stabilisation does matter in 3D, and its sign is case dependent. This is where 2D and 3D differ most. On the hexahedral mesh with the standard operator, $\alpha = 0.1$ *degrades* the order from

1.99 to 1.43; on `p1` it *improves* it, from 1.92 to 2.07 on hexa and from 1.77 to 1.89 on tetrahedra; and on `p2` on the tetrahedral mesh it improves it substantially, from 2.25 to 2.72. In other words, in 3D α is not a harmless parameter that can be fixed once and for all: its effect changes sign with the operator and the mesh family, and it should be verified case by case before being enabled in production.

The unstructured tetrahedral mesh beats the hexahedral one at high order. Against the usual intuition, with `p2` and `p3` the unstructured tetrahedral mesh yields higher rates than the hexahedral one (2.38 against 2.06 for `p2`; 3.46 against 3.26 for `p3`). A reasonable reading is that the LRE stencil on tetrahedra, being less symmetric, saturates later and is therefore further from the floor set by other error sources in this range of N; it should not be extrapolated as an asymptotic advantage. What does repeat is the collapse of the standard operator: `NO` on tetrahedra gives 0.83, practically the same first order observed in 2D.

High order in 3D is limited by memory, not by the solver. For `diagonalIion`, hexa mesh, $\alpha = 0$, and $\Delta t = 10^{-3}$, the median accumulated peak RSS over the 21 mesh sizes goes from 6.95 GB (`NO`) to 8.94 GB (`p1`), 15.3 GB (`p2`), and 46.8 GB (`p3`), while the median time goes from 305 s to 399, 646, and 1434 s. That is, `p3` costs 4.7× in time but 6.7× in memory relative to the standard operator. Since the Krylov iteration count actually *decreases* as the order rises (from 5 to 3 on hexa, from 11 to 5 on tetrahedra; §13.1), the critical resource for scaling the `p3` path to realistic 3D meshes is the memory of the **K** assembly and of the ILU(T) factor, which is exactly what the chunked assembly of §6.2 and the memory layout of §6.3 address.

Configuration recommendation. On the basis of this sweep, and for the manufactured problem, the configuration with the best order/cost ratio is `p3--CCp2--p2` with `crankNicolson`, `consistent` mass, `RKF45`, $\alpha = 0$, and a Δt small enough that the temporal floor stays below the spatial error of the finest mesh one intends to use. The nonlinear method can be chosen on cost grounds (`Picard` in this regime), with the caveat that migrating to a stiff physiological ionic model should prompt a re-evaluation in favour of `diagonalIion` or `JFNK`.